Physics-Informed Neural Quantum States for Quantum Dots: Tangent-Space Geometry, Message Passing, and Wigner Molecules

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Abstract

This thesis studies how far a compact, well-conditioned neural network can be taken as a variational wavefunction for interacting electrons, what a relational architecture contributes that a separable one does not, and whether such a state can be trained without a Markov chain. The system is the two-dimensional harmonic quantum dot, from the kinetic-energy-dominated droplet to the dilute Wigner molecule, for $N \in \{2, 6, 12, 20\}$ electrons and ω from 1 down to 10^{-3}.

The ansatz is a Slater determinant dressed by a neural Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Both are built to be safe for Laplacian-based training: trap-unit coordinates, soft-core pair features, an analytic electron–electron cusp. Each exists in a separable form and a message-passing one, and the trained states are read through the geometry of their log-derivatives. Training is by residual pretraining followed by stochastic reconfiguration (SR), and, separately, by importance-sampled collocation.

For $N = 2$, the available two-electron references are reproduced, including the exact analytic point at $\omega = 1$; for $N \in \{6, 12, 20\}$ the energies match published fixed-node diffusion Monte Carlo wherever it exists: the message-passing-backflow ansatz to 0.0008–0.04% at all three sizes, the conventional-backflow one, run at two of them, to $0.008 - 0.05\%$. Where a reference exists it is also the pretraining target: these are independently evaluated upper bounds agreeing with the reference, not two blind calculations meeting. Below $\omega = 0.1$ no external benchmark exists for $N \geq 6$, and that boundary is marked wherever it occurs.

Message passing's relational channel shows in the tangent geometry before it shows in the energy. At strong confinement it reaches reference-quality solutions along half as many effective tangent directions as a separable network of matched or larger capacity; its tangent space is better explained by a compact dictionary of the Hamiltonian's collective modes at every confinement, increasingly so toward the Wigner crystal; and its local-energy variance is lower throughout, by factors of 1.2 to 3.4—while the two networks agree in energy to four figures. In the nodes it shows as a collapse averted: a conventional backflow falls onto a single collective displacement mode and loses half a percent of the energy, where the message-passing field keeps its rank. Both give way at the same confinement, which points to a common relational bottleneck rather than two independent deficits.

The evidence points to stochastic reconfiguration earning its cost where useful tangent-metric anisotropy remains statistically estimable. Under $|\Psi|^2$ sampling the metric is badly conditioned but well estimated, and SR changes nothing Adam does not already do (tested at $N \leq 6$); under a fixed collocation proposal it is what makes the optimisation work at all, though with proposal adaptation and gradient stabilisation in place Adam is sufficient and more robust; and once the sampling collapses, inverting the metric does harm. Collocation matches SR–VMC while its sampling stays healthy and reaches a radially localised Wigner-regime state at $\omega = 0.0035$ with no Markov chain in the gradient—certified there on energy continuity and radial geometry only, the angular diagnostics not having been run there. Two limits bound the route: a statistical one, pushed back a factor of three in ω by a proposal built from the measured shell geometry, and beyond it a failure of transfer between confinements, from a fine-tuned Slater–Jastrow cancellation that no proposal repairs.

The trained states then map the Fermi-liquid-to-Wigner crossover from the inside, through topology-resolved shell, bond-orientational and stiffness diagnostics. The crossover is staged: the shells expand and stiffen radially well before the electrons acquire angular order within them. A polygon shell model fitted on held-out frames reproduces the measured pair distribution to cosine similarity $0.987 - 0.999$, against $0.936 - 0.996$ for an angle-randomised null. At $\omega = 10^{-3}$ the $N = 6$ dot shows persistent mixing between the $(1, 5)$ and $(0, 6)$ sectors, the $N = 12$ dot a two-shell molecule, and the $N = 20$ dot a three-shell molecule that collapses to a single ring about a central electron within one decade in ω. The $N = 6$ shell radius approaches the classical Wigner value $1.334\,\omega^{-2/3}$ monotonically, to within 1.7%.

Two limits on scope apply throughout. All states reported here are the lowest found within a fixed

spin-projection sector; total spin is not scanned. And the architecture comparison is seeded at $N = 6$, rests on fewer trained models at $N = 12$, and on a single production state at $N = 20$.

To go wrong in one's own way is better than to go right in someone else's.

Fyodor Dostoevsky
Crime and Punishment

Life must be understood backwards. But they forget the other proposition, that it must be lived forwards.

Søren Kierkegaard
Journals

Preface

When I came to UiO, Morten read my background and said: "You know PINNs, you know many-body quantum systems. I want you to do PINNs on quantum dots." I said "sure" and then asked the only sensible follow-up question I had: "Where do I start?"

I did not realise then that this was the question I had been assigned to answer. This thesis is the long answer. The short answer is: start by arranging to be wrong in a way that can be measured.

That is what variational methods make possible. The variational principle does not say how to improve a trial wavefunction, but it does guarantee that its energy lies above the ground-state energy. Improvement therefore has a direction: lower energy is better. For a project with few fixed landmarks, that was enough to begin. The harder task was learning what a disappointing result meant.

It was a solitary problem, not for want of literature—neural quantum states, FermiNet, PauliNet, and neural backflow offered much to learn from—but because the particular route I was trying to take had little precedent. When an experiment failed, it was rarely clear whether the method was unsound, the architecture inadequate, the optimiser poorly matched to the problem, or the code quietly broken. For long stretches, I did not know. The work became an attempt to construct tests that could tell these explanations apart.

The most important lessons came from failures. An error in the importance-sampling code went undetected for months because its signature—loss of accuracy at larger particle numbers and weaker confinement—was exactly what an inadequate architecture might have produced. A small per-particle error in a log-density accumulates over the N particles and is then exponentiated into the importance weights; the bug was invisible where experiments were cheap and ruinous where they mattered. Discovering it damaged my confidence in everything else. Re-establishing that confidence, rather than assuming it, became a substantial part of the later work.

Other failures taught the same lesson from different directions. A Langevin refinement step introduced bias into the gradient. A second-order optimiser became least reliable when its estimated metric was too noisy to resolve. Neither result was useful merely because it was negative. Each exposed an assumption that had been invisible until the computation violated it. Negative results of this kind are not a concession or an embarrassment; they are part of the method, and I have written them up as such.

That diagnostic stance also changed what I thought the network was for. I began by asking whether a machine-learning construction could solve the many-body Schrödinger equation. I ended up using the trained state as an apparatus: something to examine for the structure it had captured and the directions of variation that mattered. That shift, from treating a network as an answer to treating it as an object of study, is the thread running through this thesis.

The work also made a connection harder to ignore. Its obstacles have a double character: each is at once a physics problem and an optimisation problem. Stochastic reconfiguration, developed for variational wavefunctions, is natural gradient descent on the variational manifold. The REINFORCE estimator from reinforcement learning is the score-function gradient of the variational energy. The Fisher information matrix of statistics is, for a real amplitude and up to a factor of four, the quantum geometric tensor of physics. The same mathematics appears under different names in different communities. Recognising that felt less like discovering a bridge between fields than like realising that the two fields had been approaching the same problem from opposite sides.

Looking back, it is easy to make a catalogue of what I would now do differently. Most of that

knowledge, however, was purchased by the failures and could not have been read off in advance. The regret and the result are the same object seen from two sides.

The most consistent source of inspiration, motivation, and steadiness through all of it has been my girlfriend, Elida. Her support made it possible to begin this work and to stay with it when it became difficult. I am more grateful than I can put down here.

I owe a warm and sincere thanks to my family, who have supported me in every undertaking I have had and have simply always been there; and to my friends, inside CCSE and out, who supplied the joy, the laughter, and a good deal of the sanity this project required.

To Morten, again, and properly: I thank you for the assignment, for the two years we've had together, and above all for a genuine and undisguised interest in what I was finding at every stage—including the stages where what I was finding was that I had been wrong. That interest is most of what encouragement actually consists of.

Which is, at last, the answer to the question I asked on the first day. You start by being willing to be wrong on the record, and then you keep the record.

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Part I

Introduction

The world was to me a secret which I desired to divine.

Mary Shelley

Frankenstein; or, The Modern Prometheus

Introduction

A system of many interacting electrons is described completely by its wavefunction. The same object puts the system almost out of reach, because a complete description of N interacting particles is a function of all their coordinates at once, and the space of those coordinates grows with every particle added. The equation that fixes this wavefunction is known exactly; the entire difficulty is solving it.

The competition that makes it hard is physical. Kinetic energy tends to delocalise the particles; interactions and external potentials drive localisation and correlation; and the balance between them plays out in a configuration space far too large to enumerate. Yet the quantities we need—energies, correlation functions, phase boundaries—are precisely the ones that depend on getting that balance right. This is the *many-body problem*, the point at which quantum mechanics turns from a compact set of principles into one of the hardest computational challenges in physics.

A wide range of electronic-structure methods has been developed to address this. Mean-field approaches such as Hartree–Fock replace the full many-body problem with an effective single-particle picture, where each electron moves in an average field from the others. This captures Fermi statistics and some exchange effects but leaves most correlation physics untreated. More systematic methods such as Full Configuration Interaction (FCI) are, in principle, exact within a finite basis, but their cost grows combinatorially with system size. Coupled Cluster (CC) theory provides a more efficient and size-extensive hierarchy of approximations, but it remains computationally demanding and can struggle in strongly correlated regimes or near degeneracies.

Quantum Monte Carlo (QMC) methods, including Variational Monte Carlo (VMC) and Diffusion Monte Carlo (DMC), take a different approach: they represent the wavefunction implicitly through stochastic sampling. For suitable trial states, they can produce highly accurate benchmark energies and correlations for continuum systems. However, QMC is not free of difficulties. The fermion sign problem limits direct simulations, fixed-node approximations inherit bias from the chosen trial nodes, and constructing high-quality trial wavefunctions is increasingly difficult as systems become more strongly correlated or inhomogeneous. In practice, then, much of the many-body problem reduces to a narrower one: choosing an expressive, well-conditioned ansatz and optimising it reliably.

Machine learning and many-body wavefunctions. In parallel with these developments, machine learning—especially deep learning—has become a standard tool for representing complex, high-dimensional functions and fitting them to data. Neural networks now play a central role in domains such as computer vision, language modelling and reinforcement learning. They combine flexible parametric families with inductive biases (e.g. symmetry and locality), normalisation, and optimisation procedures that scale well with dimension.

In physics, this has led to two broad uses of neural networks. In the first, they act as *surrogates*: they learn mappings from parameters or boundary conditions to observables, or approximate solution operators of partial differential equations. Physics-informed neural networks (PINNs) and operator-learning approaches fall into this category by embedding differential operators and constraints directly into the training objective. In the second, neural networks are used as *variational ansätze* for quantum states (neural quantum states, NQS), where the network outputs amplitudes, phases or log-probabilities for many-body configurations and is trained by variational Monte Carlo.

Universal approximation establishes representational capacity, not trainability. The theorems guarantee that a sufficiently large network can *represent* any suitably regular function to arbitrary accuracy. They say nothing about whether gradient-based optimisation will *find* that representation, nor at what sample cost. This thesis measures that gap directly: the same ansatz, demonstrably able to express the ground state, is placed in training schemes that do and do not allow it to get there. Representability is cheap; trainability is the difficulty.

A weaker but sounder case for a neural ansatz survives that objection. Established methods pay for accuracy in one of two currencies. FCI is exact within a finite basis, but its cost grows combinatorially with that basis, so a modest two-dimensional dot of twenty electrons is already far beyond reach. Cheaper methods such as coupled cluster buy their favourable scaling instead by deciding in advance which correlations may be described at all.

A variational neural ansatz avoids both of those particular costs. It requires no explicit enumeration of the determinant space, and it fixes no prior truncation on which correlations may be described; its parameter count is set empirically by the structure of the target rather than by the size of a basis. That is not freedom from scaling—capacity can still have to grow with N or with the accuracy demanded, as the twenty-electron results below show—but it is freedom to concentrate capacity wherever the solution actually varies. Whether it does so in practice is an empirical question, and this thesis treats it as one.

That question also suggests a different way to use a solver. Nearly every established method for the many-body Schrödinger equation is constructed outward from physical principles: an approximation is chosen, justified physically, and then implemented. A trained network inverts the order. Physical structure is still imposed on it—antisymmetry, the cusp, trap scaling, permutation symmetry—but the *specific correlations* it develops within those constraints are not prescribed, and are interrogated only afterwards—and unlike an implicit stochastic representation, it is a concrete parameterised object whose internal structure can be examined directly. This motivates a question that the size of the problem does not by itself settle. A twenty-electron dot is a function of forty coordinates, and any basis expansion of it grows combinatorially—but it does not follow that the solution itself is forty-dimensional in any useful sense. How many active tangent directions does a given parameterisation actually use to represent the solved state—a question about the solution and its representation rather than about the equation. It is worth saying at once that this is not an intrinsic dimension of the wavefunction: the count depends on how the state is parameterised, as §2.2.5 makes precise.

We make that question quantitative through the geometry of the wavefunction's log-derivatives. Perturbing the parameters of a trained ansatz over an ensemble of configurations and collecting the resulting sensitivities into a covariance matrix yields, in the simplest reading, a principal-component analysis of the state: the eigenvalues report how much variation each independent direction carries, and their distribution counts how many directions are doing genuine work. The physically appropriate version of this covariance is the quantum geometric tensor (equivalently, the Fisher metric), and the corresponding count is the participation ratio of its spectrum—an effective dimension rather than a hard rank, insensitive to the raw parameter count but not to the choice of parameterisation. Chapter 7 develops this instrument and applies it to architecture, optimiser, and training paradigm alike; the questions it answers are how this effective dimension behaves as particles are added and as the trap is opened, and whether different architectures arrive at the same solution by different routes.

In principle, then, neural networks provide highly expressive ansätze for quantum problems. In practice, several issues arise. Objectives based on local energies involve second derivatives and can be numerically delicate. Coordinate parameterisations that do not respect known structure can lead to large gradients and Laplacian spikes near electron coalescence. Generic architectures can waste capacity on degrees of freedom that are irrelevant from a physical point of view. These challenges motivate hybrid approaches in which as much known structure as possible—antisymmetry, cusp conditions, scaling, and symmetries—is built into the ansatz and its coordinates, and the network is used to learn the remaining correlation corrections.

A further ingredient concerns not *what* structure is imposed but *how* the network represents the electrons. Correlation is intrinsically relational—what one electron should do depends on where the others are—so architectures that let particles exchange information (message passing), rather than processing each coordinate independently, are a natural candidate for the correlator and for the backflow that deforms the nodes. How much this relational capacity actually buys, and why, is one of the questions this thesis takes up.

Where this work sits. Representing a quantum state by a neural network and optimising it variationally was introduced by Carleo and Troyer [1]; for continuum fermions the idea has since matured into highly accurate *ab initio* solvers such as FermiNet [2] and PauliNet [3], and into explicit neural *backflow* transformations [4]—the direct ancestor of the coordinate backflow used here. Message passing, the relational mechanism at the centre of this thesis, has recently been built into neural quantum states for extended fermion systems [5, 6]. Against that backdrop the present work is deliberately narrow and complementary. Rather than pushing a new architecture to ever larger systems, it asks *why* the relational channel helps, and uses the tangent-kernel geometry to make the answer precise; it focuses on the two-dimensional quantum-dot Wigner regime, where correlation is strongest and where topology-resolved structural diagnostics remain sparse; and it treats conditioning, the optimiser, and Markov-chain-free training as objects of study in their own right rather than fixed choices. To our knowledge these questions have not previously been addressed together for parabolic quantum dots.

Quantum dots as a controlled testbed. Two-dimensional parabolic quantum dots provide a clean and tunable setting in which to explore such ideas. They offer a simple route from Fermi-liquid–like behaviour to Wigner-molecule physics. As the trap frequency ω decreases, the single-particle level spacing collapses while Coulomb repulsion becomes dominant. Electrons then tend to localize into ring or shell patterns. In the lab frame, rotational symmetry and quantum fluctuations smear these into smooth densities; in a co-rotating frame, one recovers crystalline-like structures whose topology depends on particle number. For $N{=}6$ and $N{=}12$, classical and semiclassical studies have identified preferred shellings such as $(1,5)$ and $(3,9)$ and mapped parts of the crossover [7–10].

From the perspective of this thesis, quantum dots are attractive for two reasons. First, they are genuinely non-trivial: Coulomb singularities, strong correlations and an actual Fermi-liquid–to–Wigner crossover appear already for moderate N. Second, they are still simple enough that accurate DMC references exist in many regimes and that detailed structural diagnostics (shell occupancies, bond order, virial ratios) are computationally feasible. For fully quantum dots, however, quantitative topology-resolved diagnostics (occupancy fractions, bond-orientational order, shell "stiffness") and a systematic picture of the Wigner regime remain relatively sparse. In addition, it is not well understood how modern neural ansätze internally represent these states.

Scope and questions. In this thesis we revisit two-dimensional harmonic quantum dots with a Slater–Jastrow(+backflow) neural quantum state, pretrained on a physics-informed residual objective and refined variationally. The design is meant to be *safe by construction* for Laplacian-based objectives: coordinates are scaled to trap units ($\tilde{R} = \sqrt{\omega}R$), learned pair features are soft-core (avoiding spurious $1/r$ behaviour in derivatives), and the electron–electron cusp is enforced analytically rather than learned. On top of a Slater determinant of harmonic-oscillator orbitals, two learnable objects carry the correlation: a Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Each can be built either without message passing—encoding and pooling per-particle and pairwise features in a single pass, as in a DeepSet—or as a *message-passing* network in which the particles exchange and update information through explicit, transported edge state before a quantity is read out.

Once such an ansatz reaches reference accuracy—which, as we will see, a surprisingly compact one does—the remaining questions concern *why* its parts behave as they do, and what it can tell us about the electrons. We address four. The first three share an analytical centre: the geometry of the wavefunction's log-derivatives, captured by the quantum geometric tensor and the neural tangent kernel.

1. **Architecture.** What does a message-passing network learn that a separable one cannot, given that both can be trained to the same energy? We compare the two forms in the correlator and in the backflow, holding everything else fixed.

2. **Optimiser.** When does natural-gradient (stochastic reconfiguration) preconditioning change the result, and when is it unnecessary or harmful?

3. **Paradigm.** Can an accurate wavefunction be trained *without* Markov-chain Monte Carlo, using importance-sampled collocation instead, and how far into the correlated regime does that reach before it fails?

4. **Physics.** Can the trained wavefunction serve as a scientific instrument for Wigner-molecule physics—shell topologies, angular order, and melting—and does its internal representation reflect that physics?

Diagnostics. Total energies alone cannot separate the models compared here, since both architectures can be trained to the same energy. Alongside the DMC benchmarks we therefore use three further classes of measurement. For the physics: automatic shell detection with shell-resolved reconstructions of the pair distribution $g(r)$, bond-orientational order parameters $|\Phi_m|$ and angular Lindemann-like ratios adapted from classical Wigner and Yukawa layers [11], and the density parameter r_s estimated from the first peak of $g(r)$ following Egger *et al.* [7]. For the learned representation: entropy effective ranks of correlator and backflow features, principal-component ablations, and linear probes relating latent axes to physical observables. And for the ansatz itself: the quantum geometric tensor and neural tangent kernel, the effective dimension of the variational tangent space, symmetric state overlaps, the local-energy variance $\mathrm{Var}(E_L)$, and—for the Markov-chain-free route—the effective sample size and the Pareto shape parameter $\hat{k}$ of the importance weights. All are applied as post-processing to $|\Psi|^2$ samples, so the same analysis could be applied to classical, quantum-chemical or other neural wavefunctions on the same footing.

Main contributions. The main contributions of this thesis are:

1. **Compact, conditioned accuracy.** A small Slater–Jastrow(+backflow) ansatz in trap units reaches DMC-level energies for N=2 and holds $\sim 10^{-2}$–10^{-1}% relative errors for $N \in \{6, 12, 20\}$ across $\omega \in \{0.1, 0.5, 1.0\}$, extrapolating smoothly to ultra-weak traps ($\omega = 10^{-3}$). Conditioning—trap units, analytic cusp, safe pair features—matters at least as much as network size for stability and accuracy in these dots.

2. **Message passing as a relational channel, made precise by the tangent kernel.** Read through the quantum geometric tensor, a message-passing correlator represents the ground state in roughly half as many effective tangent directions as a separable one at strong confinement—and keeps its leading directions inside a compact, physically motivated operator dictionary at *every* confinement, where the separable network does not—while a message-passing backflow retains full rank exactly where a conventional, single-round-message backflow collapses to a single collective mode at the Wigner crystal and loses accuracy. The two effects point to a common relational mechanism—a channel the physics requires—and both are invisible in the total energy until weak confinement turns the geometric deficit into an energetic one.

3. **Optimiser and paradigm, governed by the same geometry.** Natural-gradient (SR) preconditioning helps precisely when the tangent metric is anisotropic *and* still estimable. Under $|\Psi|^2$ sampling the empirical metric is badly conditioned yet SR buys nothing, so conditioning alone cannot be the criterion: what matters is whether the batch can resolve it. SR is therefore negligible for $|\Psi|^2$-VMC (tested at $N \leq 6$) and decisive for collocation under a fixed proposal—though once the proposal adapts, Adam recovers and is more robust in the deep-Wigner regime. And an MCMC-free, importance-sampled collocation objective matches VMC when the sampling is healthy and, with a physics-informed proposal built from the Wigner shell, reaches a radially localised Wigner-regime state at $\omega = 0.0035$ with no Markov chain in the gradient training (a Metropolis probe is used only for checkpoint selection and final certification).

4. **Topology-resolved Wigner diagnostics.** Across $\omega \in [1, 10^{-3}]$ we quantify the crossover using two-shell probabilities and inner-ring occupancy histograms, a polygon shell model fitted on held-out frames that reproduces $g(r)$ to cosine similarity 0.987–0.999 against 0.936–0.996 for an angle-randomised null, and sector-resolved bond order and Lindemann indices read against their random-angle baselines. At $\omega = 10^{-3}$ the N=6 dot shows persistent mixing between the $(1, 5)$ and $(0, 6)$ sectors, while the N=12 dot forms a two-shell molecule with a commensurate $(3, 9)$ split as the dominant topology, complementing classical and quantum studies of parabolic dots [7–9, 12].

One further observation is recorded, at a lower evidential level and not as a contribution of the same rank:

- **Representation insight, with its gauge explicitly drawn.** The correlator features occupy a low-dimensional manifold (one to three principal directions) whose leading axes act as collective coordinates for global size and radial fluctuations, with a $\phi \leftrightarrow \psi$ recomposition across regimes, while backflow spans many local modes. We take care to separate the parts of this picture that are *gauge-like*—fixed by how correlation is divided between the Jastrow and the backflow, two substitutable channels—from the state-level invariants (energy, the local-energy variance, and the overlap between two trained states) and the controlled interventions (the energetic price of ablating a component, reproducible for a fixed model and protocol but not itself gauge-invariant) on which the architectural conclusions actually rest.

Thesis structure. Chapters 1 and 2 set out the physics and the machine-learning background the later chapters require. Chapters 3 to 5 give the ansatz, the optimisation pipeline, and the diagnostics.

Chapter 6 reports the energies and the Wigner-molecule structure; Chapter 7 analyses architecture, optimisation and sampling through the tangent geometry, and reports the Markov-chain-free training programme. Part IV opens with a short guide to how the two fit together. A discussion and a conclusion follow, and the appendices develop the conditioning theory, the collocation–backflow obstruction that motivated the weak-form objective, and the full diagnostic tables.

Part II

Theory

Men do not know how what is at variance agrees with itself. It is an attunement of opposite tensions, like that of the bow and the lyre.

HERACLITUS
Fragments

Chapter 1

The many-body problem and its numerical difficulty

The system is N electrons in a two-dimensional harmonic trap, repelling one another through the Coulomb interaction. Its Hamiltonian is elementary, and three of its properties make the numerical solution difficult.

The electrons are identical fermions, so the wavefunction changes sign under exchange, and that sign structure is the expensive part of the problem. The interaction diverges where two electrons meet, so the local energy $\hat{H}\Psi/\Psi$ diverges with it unless the wavefunction is constructed to cancel the divergence. And the kinetic term is a second derivative, which exposes any method that evaluates the energy pointwise to the curvature of the function it is given.

1.1 The many-body problem

Quantum mechanics is formulated in a complex Hilbert space: a complete inner-product space, so that limits and basis expansions are well defined. A state is a normalised vector $|\psi\rangle$, observables are Hermitian operators, and expanding in an orthonormal basis $\{|i\rangle\}$ turns operators into matrices $O_{ij} = \langle i|\hat{\mathcal{O}}|j\rangle$ and states into coefficient vectors $c_i = \langle i|\psi\rangle$. In the coordinate representation $\psi(x) = \langle x|\psi\rangle$ is the wavefunction. Choosing the basis to match the physics—harmonic-oscillator orbitals for confined charges, plane waves for periodic systems—is what makes a representation sparse, and it is the reason the orbitals of Chapter 3 are those of the trap itself.

For N particles the space is the N-fold tensor product $\mathcal{H}_1^{\otimes N}$, and a convenient fermionic basis is the set of Slater determinants $\{|\Phi_I\rangle\}$ built from N distinct spin-orbitals, in which any state expands as $|\Psi\rangle = \sum_I C_I |\Phi_I\rangle$. That expansion is exact and, as §1.2 makes precise, unaffordable.

Time evolution is unitary and generated by the Hamiltonian, $i\partial_t |\Psi\rangle = H|\Psi\rangle$, so for time-independent H the problem reduces to its stationary eigenpairs. We work throughout in atomic units ($\hbar = m_e = e = 1$) and with the many-body Hamiltonian

$$H = -\frac{1}{2}\sum_{i=1}^{N}\nabla^2_{\mathbf{r}_i} + \sum_{i=1}^{N} V_{\text{ext}}(\mathbf{r}_i) + \sum_{i<j} V_{\text{int}}(|\mathbf{r}_i - \mathbf{r}_j|), \qquad H\Psi(R) = E\Psi(R), \tag{1.1}$$

where $R = (\mathbf{r}_1, \ldots, \mathbf{r}_N)$ and Ψ is normalised. Identical particles restrict Ψ to a symmetry sector: symmetric for bosons, antisymmetric for fermions. For the quantum dots studied here $V_{\text{ext}}(\mathbf{r}) = \frac{1}{2}\omega^2 r^2$ and $V_{\text{int}}(r) = 1/r$ in two dimensions. Everything that follows is an attempt to approximate the lowest eigenpair of Eq. (1.1).

1.2 Concepts from electronic structure

Three concepts from quantum chemistry are used throughout this thesis and computed nowhere in it: correlation energy, closed-shell structure, and exactness within a basis. The accuracy claims of later chapters are calibrated against them.

Second quantisation. A first-quantised N-fermion state is an antisymmetric function expanded in Slater determinants of M one-particle orbitals, and the number of such determinants is $\binom{M}{N}$. The occupation-number formalism replaces the bookkeeping with algebra. Over an orthonormal set of spin-orbitals $\{|p\rangle\}$ one defines creation and annihilation operators obeying the canonical anticommutation relations

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \qquad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0,$$

so that antisymmetry is carried by the operators rather than by the wavefunction, and a determinant is simply $|\Phi_I\rangle = a_{p_1}^\dagger \cdots a_{p_N}^\dagger |0\rangle$. One- and two-body operators then take forms independent of N,

$$O = \sum_{pq} O_{pq}\, a_p^\dagger a_q, \qquad V = \tfrac{1}{4} \sum_{pqrs} \langle pq\|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r,$$

with $\langle pq\|rs\rangle$ the antisymmetrised two-electron integral. This is the language in which the next two ideas are usually stated. Our own ansatz is first-quantised throughout—it is a function of real-space coordinates—so the formalism appears here as vocabulary and not as machinery.

Hartree–Fock, and what correlation means. Hartree–Fock approximates the ground state by a *single* determinant, with the orbitals themselves optimised: stationarity of the determinant energy under orbital rotations gives the eigenproblem $F|\phi_p\rangle = \varepsilon_p |\phi_p\rangle$ for the Fock operator $F = \hat{h}_0 + \sum_{j\in\mathrm{occ}}(J_j - K_j)$, in which each electron moves in the mean field of the others. Because F depends on its own solutions, it is solved self-consistently.

The *correlation energy* is whatever a single determinant cannot reach, $\Delta E = E_{\mathrm{exact}} - E_{\mathrm{HF}}$: a small fraction of the total and essentially all of the difficulty, which is why a method can look converged in total energy while representing the correlation badly, a distinction Chapter 7 makes quantitative. A structural fact comes with it. When the occupied orbitals fill complete shells the determinant is a spin singlet and is unique, with no degenerate manifold to choose from, and those *closed-shell* particle numbers are the ones studied here.

Full configuration interaction, and the cost of being exact. Expanding in *all* N-electron determinants of a given basis and diagonalising, $|\Psi\rangle = \sum_I C_I |\Phi_I\rangle$ with $H\mathbf{C} = E\mathbf{C}$, gives the exact ground state *within that basis*. The Slater–Condon rules make H extremely sparse—determinants differing by more than two spin-orbitals do not couple—so the diagonalisation is done iteratively and symmetry sectors can be treated separately. What cannot be avoided is the $\binom{M}{N}$ growth of the space itself. For the twenty-electron dot studied here, an accurate one-particle basis puts the determinant count far beyond reach. This is the combinatorial wall that motivates variational methods in the first place.

Why none of it is computed here. Our determinant is built from harmonic-oscillator orbitals rather than self-consistent HF orbitals, and our external benchmark is diffusion Monte Carlo [13], which for these systems is both more accurate than an affordable FCI and available at the particle numbers we care about.

1.3 Quantum Dots

1.3.1 Model and units

We consider N spin-$\frac{1}{2}$ fermions in d=2 with harmonic confinement and Coulomb repulsion,

$$H = \sum_{i=1}^{N} \Big(-\tfrac{1}{2}\nabla_{\mathbf{r}_i}^2 + \tfrac{1}{2}\omega^2 r_i^2 \Big) + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1).$$

Spin does not appear in H, so the eigenstates can be classified by S and S_z and the spatial problem solved at fixed spin assignment. For more than two identical fermions a spin-adapted eigenstate of S^2 is in general a sum of spatial–spin products rather than a single one; what we use computationally is the weaker and sufficient statement that with the numbers of up- and down-spin electrons held fixed, the antisymmetriser factorises and the spatial problem may be treated on its own.

1.3.2 One–particle structure and shell filling

We use two-dimensional harmonic-oscillator spin–orbitals. In the implementation these are the *Cartesian* 2D-HO eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\chi_\sigma\}$ with energies $E_{n_x n_y} = \omega(n_x+n_y+1)$ and spin $\sigma \in \{\uparrow,\downarrow\}$. The principal index $k = n_x + n_y$ labels a shell of $(k+1)$ degenerate spatial states, each twofold spin-degenerate; for the closed shells studied here this Cartesian basis spans the same filled subspace as any complete basis of those shells, so the reference determinant is basis-independent. Filling from the bottom yields the *closed-shell* electron numbers

$$N_{\text{closed}} = (K+1)(K+2), \quad K = 0, 1, 2, \ldots \quad (2, 6, 12, 20, 30, \ldots).$$

All other N correspond to a *partially filled* top shell (open shell), with a degenerate noninteracting manifold.

1.3.3 Spin structure and Slater form

Write single-particle spin-orbitals as $\varepsilon_\alpha(x) = \phi_\alpha(\mathbf{r})\,\chi_\sigma(s)$, with $x = (\mathbf{r}, s)$. Because $\chi_\uparrow$ and $\chi_\downarrow$ are orthogonal and H is spin-independent, the Slater determinant simplifies:

Closed shells. For even N with doubly occupied spatial orbitals $\{\phi_1, \ldots, \phi_{N/2}\}$, the determinant factorizes into two identical spatial determinants, one for each spin:

$$\Psi_{\text{closed}}(X) = \frac{1}{(N/2)!}\ \det[\phi_a(\mathbf{r}_i)]_{i\in\uparrow, a=1..N/2}\ \det[\phi_a(\mathbf{r}_j)]_{j\in\downarrow, a=1..N/2}\ \times\ \Xi_{S=0,S_z=0},$$

where Ξ is the singlet spin function. In the computational representation this last factor is not carried: fixing which electrons are spin-up and which spin-down makes the full spin-orbital determinant block diagonal, and the product of the two spatial determinants *is* the ansatz. This has the *form* of a restricted closed-shell determinant, but its orbitals are the harmonic-oscillator eigenstates of the trap, not orbitals optimised self-consistently in the mean field of the other electrons. It is therefore the closed-shell *non-interacting* reference, not an RHF one, and no self-consistent-field calculation is performed anywhere in this thesis. The distinction matters for how the later results should be read: everything the correlator and the backflow achieve is measured against a determinant that has not been given the benefit of a mean field, so what they recover includes the orbital relaxation an HF calculation would have supplied as well as the correlation beyond it.

1.3.4 Kato's Cusp Conditions (2D Coulomb systems)

For Coulomb interactions the potential diverges as $1/r$ at coalescence. Finiteness of the local energy

$$E_L(R) = \frac{(H\Psi)(R)}{\Psi(R)}$$

requires that the kinetic singularity cancels the $1/r$ divergence. Kato's theorem, first proved in [14], gives the necessary short–distance behaviour of the (spherically averaged) wavefunction. Let r_{ij} be the interparticle distance, $\mu_{ij} = m_i m_j/(m_i + m_j)$ the reduced mass (in a.u.), Z_i the charge number (electron $Z = -1$, nucleus $Z = +Z_A$), and D the spatial dimension. Then

$$\frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\,Z_i Z_j}{D-1} \quad \text{(s-wave / opposite-spin channel)}.$$

For *identical fermions* the s-wave is excluded; the leading channel is p-wave (relative angular momentum $k = 1$), and the wavefunction carries a linear node at coalescence. The cusp condition then applies to the *regular part* left after that node is factored out, and this distinction is easy to lose. Writing

$\overline{\Psi}(r) = r\,\overline{u}(r)$ with $\overline{u}$ regular and nonzero at contact, the logarithmic derivative of the full wavefunction diverges, $\partial_r \ln \overline{\Psi} = 1/r + \partial_r \ln \overline{u}$, so it is $\overline{u}$ and not $\overline{\Psi}$ that obeys a finite cusp condition:

$$\frac{\partial}{\partial r_{ij}} \ln \overline{u}\bigg|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\,Z_i Z_j}{2k + D - 1} = \frac{2\,\mu_{ij}\,Z_i Z_j}{D+1} \quad \text{(like-spin, } k = 1\text{)}.$$

In a Slater–Jastrow form the antisymmetric factor supplies the linear zero and the Jastrow supplies $\overline{u}$, so this is exactly a condition on the Jastrow slope—which is how it is imposed in Chapter 3.

Specializing to 2D electrons (atomic units). With $m_e = 1$ and $\mu_{ee} = 1/2$:

$$\boxed{a_{ee}^{\uparrow\downarrow} = \frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_0 = 1\ , \qquad a_{ee}^{\uparrow\uparrow} = \frac{\partial}{\partial r_{ij}} \ln \overline{u}\bigg|_0 = \tfrac{1}{3}\ .}$$

The two derivatives are taken on different objects: the full spherically averaged wavefunction in the antiparallel case, where it is finite at contact, and its regular part in the parallel case, where the antisymmetric node has been removed. The like-spin value follows from the p-wave (linear-node) analysis carried out in full in Appendix B.2, the canonical derivation used throughout this thesis; it gives the familiar $\frac{1}{4}$ in three dimensions and $\frac{1}{3}$ here in two. The 2D like-spin cusp is therefore *not* simply half the opposite-spin value—a coincidence that holds only in 3D—and the implemented value is $1/(D+1) = \frac{1}{3}$. For electron–nucleus coalescence ($\mu_{eA} \approx 1$, $Z_i Z_j = -Z_A$):

$$\boxed{a_{eA} = \frac{\partial}{\partial r_{iA}} \ln \overline{\Psi}\bigg|_0 = -2Z_A\ .}$$

Why this matters. These conditions ensure the kinetic energy's r^{-1} singularity cancels the Coulomb r^{-1} term so that E_L remains finite as particles coalesce. Enforcing the cusp explicitly improves conditioning.

1.4 Structural diagnostics and Wigner markers

A Wigner molecule is not defined by a single number, so the crossover has to be read from several markers at once. This section says what each one measures and why it responds to Wigner order; the estimators, thresholds and stability protocol are defined once, in the Analysis chapter (§5.1), and the measurements themselves are in Chapter 6.

Radial localisation. The pair distribution $g(r)$ and the radial probability $P(r)$ say how far apart the electrons sit and how tightly that distance is fixed. The informative quantity is not the typical radius—which simply grows as the trap weakens—but the *ratio* of the radial width to it. A distribution that becomes relatively narrower around a growing radius is a shell stiffening; one that merely dilates is a droplet expanding. This is the finite-system analogue of a Lindemann criterion.

Shell topology. In a parabolic trap the classical minimum energy configuration is a set of concentric rings with definite occupancies—$(1,5)$ at N=6, $(3,9)$ at N=12 [8, 9]. Detecting those occupancies frame by frame turns a qualitative picture into a distribution over topologies, which is what makes it possible to say that a state is *mostly* $(3,9)$ with real weight on $(2,10)$—a rotating molecule fluctuating between near-degenerate shellings rather than a single frozen crystal.

Orientational order. Radial structure alone cannot distinguish a ring of electrons at fixed angular spacing from a ring free to slide. The bond-orientational order parameter Φ_m measures m-fold angular order on a detected ring after removing global rotation, and an angular Lindemann number measures how stiff those spacings are. Both are standard in two-dimensional Coulomb and Yukawa systems [11, 12], and together with the radial markers they separate angular from radial melting—the two need not happen at the same ω.

Classical scaling. Finally, the strong-coupling limit makes a parameter-free prediction: inter-particle distances should approach $r \propto \omega^{-2/3}$. Fitting the measured exponent and watching it approach $-2/3$ from above is an external check on the whole picture, since nothing in the ansatz knows about that number.

How to read them together. As ω falls, the width ratios and angular Lindemann numbers shrink, shell gaps sharpen, the shell-resolved $|\Phi_m|$ rise above their finite-ring random-angle baseline, and the radial exponent approaches its classical value. No one of these is decisive—a narrow $P(r)$ can be an artefact of poor sampling, and a large $|\Phi_m|$ can be a fluctuation on a short chain—but they fail in different ways, which is why Chapter 6 reports them together and Chapter 5 states the stability checks each one is subjected to. Quantum Wigner markers in parabolic dots are reviewed in Refs. [7, 10–12].

1.5 Quantum Monte Carlo and the status of its benchmarks

Two stochastic methods supply the numbers this thesis is measured against, and the difference between what they guarantee is the whole of its external evidence. Variational Monte Carlo optimises a trial state under the variational principle. Diffusion Monte Carlo projects toward the ground state in imaginary time. Only the first is what we do; only the second is what we are scored against; and neither is exact for fermions.

1.5.1 Variational Monte Carlo

For any normalised trial state $\Psi_T(\mathbf{R};\boldsymbol{\alpha})$ the Rayleigh quotient

$$E[\Psi_T] = \frac{\langle \Psi_T|\hat{H}|\Psi_T\rangle}{\langle \Psi_T|\Psi_T\rangle} \;\geq\; E_0$$

bounds the exact ground-state energy from above, so minimising it over $\boldsymbol{\alpha}$ can only improve the estimate. Writing the expectation as an average of the *local energy*

$$E_L(\mathbf{R}) \;=\; \frac{\hat{H}\Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})} \tag{1.2}$$

over configurations drawn from $P(\mathbf{R}) = |\Psi_T(\mathbf{R})|^2/\int|\Psi_T|^2$ turns the integral into a sampling problem,

$$E[\Psi_T] \;\approx\; \frac{1}{N_{\mathrm{MC}}}\sum_i E_L(\mathbf{R}_i),$$

with the configurations generated by Metropolis–Hastings [15] at acceptance

$$A(\mathbf{R}\to\mathbf{R}') = \min\left(1,\; \frac{|\Psi_T(\mathbf{R}')|^2}{|\Psi_T(\mathbf{R})|^2}\,\frac{T(\mathbf{R}'\to\mathbf{R})}{T(\mathbf{R}\to\mathbf{R}')}\right),$$

optionally drifted by the quantum force $\mathbf{F} = 2\nabla\ln|\Psi_T|$ toward regions of high density [16].

The zero-variance principle. One property of Eq. (1.2) does more work in this thesis than the energy itself. If Ψ_T were an exact eigenstate, $\hat{H}\Psi_T = E\Psi_T$ pointwise, and E_L would be *constant* over the whole of configuration space. Its variance

$$\mathrm{Var}(E_L) \;=\; \langle E_L^2\rangle - \langle E_L\rangle^2 \tag{1.3}$$

therefore vanishes only at an exact eigenstate, and decreases as the trial state improves. This makes it a measure of wavefunction quality that is independent of any external reference—which matters here for two reasons. It keeps saying something after the energy has saturated, when two ansätze agree to four figures and the energy can no longer tell them apart. And it is defined on the state rather than on any component of it, so it survives changes of parameterisation that quantities such as a component's rank do not. Chapter 7 uses it in both roles.

1.5.2 Diffusion Monte Carlo, and the fixed-node constraint

DMC removes the dependence on a trial state by projection. In imaginary time $\tau = it$ the Schrödinger equation becomes a diffusion equation,

$$-\frac{\partial \Psi(\mathbf{R}, \tau)}{\partial \tau} = (\hat{H} - E_T)\, \Psi(\mathbf{R}, \tau),$$

whose long-time limit is the ground state, provided the initial state overlaps it. The evolution is simulated by an ensemble of walkers that diffuse, drift along the quantum force, and are branched or killed with weight $w = \exp[-(E_L(\mathbf{R}) - E_T)\Delta\tau]$, so that the walker density converges on the ground-state density.

That argument works because a diffusion equation propagates a *probability*. For bosons the ground state is nodeless and can be treated as one directly. For fermions it cannot: antisymmetry forces Ψ to change sign, and a signed density is not a probability. Splitting it into positive and negative populations is formally possible but their difference is swamped by their sum at a rate exponential in $N\tau$—the *fermion sign problem* [17].

The standard escape is to impose the sign structure rather than sample it. The *fixed-node* approximation confines the walkers to the nodal pockets of a trial state Ψ_T, killing any walker that crosses a node. Within each pocket the wavefunction has a definite sign, the projection is well posed, and the method returns the exact ground-state energy *of the problem with those nodes imposed as a boundary condition.* That energy is a rigorous upper bound on E_0, and it is exact only if the nodes are. Otherwise it carries a systematic *nodal bias* that no amount of sampling removes. The bound is a statement about the fixed-node eigenvalue in the limit of zero time step and infinite walker population; a published DMC number is an estimate of it and carries time-step, population-control and finite-statistics errors of its own, which is part of why comparing a variational energy against one at the fifth decimal place requires the matched uncertainty analysis we did not undertake (§8.1).

Three consequences follow. A published DMC energy is not ground truth; it is the best energy attainable inside the nodal surface of the trial state used to produce it. A variational state with better nodes than that surface may therefore lie *below* a fixed-node DMC number without violating anything, since the variational principle bounds the energy from below by E_0 and not by another approximation (§8.1). And improving nodes becomes the thing to attempt, which is what the coordinate backflow of Chapter 3 is for.

1.5.3 Requirements on the trial state

Both methods take a trial state as given. VMC is only as good as the state it optimises, and fixed-node DMC only as good as the nodes it is handed, so the quality of the ansatz stops being an implementation detail and becomes the object of study. Three requirements bear on it at once.

Antisymmetry has to hold exactly and by construction, since the sign structure is what makes the problem hard in the first place. In practice that means a determinant, multiplied by a symmetric correlation factor of the kind Jastrow introduced for strongly interacting systems [18]. The nodes have to be placed well, since that is where the remaining systematic error lives once the amplitude is right. And the state has to be smooth enough that $E_L = \hat{H}\Psi/\Psi$, containing as it does a Laplacian and a potential that diverges at coalescence, can be evaluated without the two singularities amplifying one another. The cusp conditions of §1.3.4 say that those two divergences can be made to cancel exactly, and hard-wiring that cancellation is the first design decision of Chapter 3.

The remaining design choice is the functional form of the smooth correlation factor. The next chapter takes it from a literature that grew up asking different questions.

Chapter 2

Neural networks as variational ansätze

2.1 The variational learning objective

Machine learning is usually introduced as *empirical risk minimisation*: an unknown joint distribution $P(X,Y)$ generates data, a loss ℓ scores predictions, and one minimises the empirical average $\hat{\mathcal{R}}(\theta) = \frac{1}{n}\sum_i \ell(f_\theta(x_i), y_i)$ over a finite sample as a proxy for the population risk $\mathcal{R}(\theta) = \mathbb{E}_P[\ell(f_\theta(X), Y)]$. The gap between the two is the subject of the field: overfitting, the bias–variance tradeoff, regularisation, held-out validation [19, 20].

The variational problem considered here differs from that setting in two respects. There is no fixed labelled dataset, and so no conventional train/test generalisation problem. What replaces it is not the absence of a finite-sample difficulty but a different one: the objective is a population expectation that must be estimated afresh at every step, and an optimiser can certainly overreact to a particular finite batch—which is exactly what the effective-sample-size and weight-tail diagnostics of Chapter 7 measure. The objective is an expectation,

$$E[\Psi_\theta] \;=\; \mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[E_L(\mathbf{R})\big], \tag{2.1}$$

under a distribution the model itself defines, re-estimated from fresh samples at every step. The optimisation target is the physical state rather than predictive performance on unseen data.

What replaces generalisation as the difficulty is already visible in the form of Eq. (2.1). Its measure $|\Psi_\theta|^2$ moves: it changes with every update, so the gradient, the samples used to estimate it, and the metric used to precondition it are all functions of the current parameters. In supervised learning the data distribution sits still and none of this arises. Meanwhile $E_L = \hat{H}\Psi_\theta/\Psi_\theta$ contains a Laplacian and a potential that diverges, so the objective reads the network's second derivatives rather than its values, and reads them near a singularity. Smoothness stops being a stylistic preference and becomes a requirement of the loss.

One term from the usual taxonomy recurs below. The objectives here are *unsupervised*: no labelled targets exist, and the loss comes entirely from the physics the network is asked to satisfy. The exception is the second phase of residual pretraining, where a published reference energy is used as an explicit target, and that is why those runs are later described as trained without a Markov chain rather than as unsupervised solutions of the Schrödinger equation. Reinforcement learning enters obliquely, through the REINFORCE estimator borrowed for the collocation gradient.

2.2 Optimisation

Whatever the objective, minimising it over a nonlinear, high-dimensional hypothesis class admits no closed form, so the parameters $\theta \in \mathbb{R}^p$ are updated iteratively according to the local geometry of the loss.

2.2.1 Gradient descent

The most basic algorithm is *gradient descent*. Starting from an initial guess $\theta^{(0)}$, parameters are updated by

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}), \tag{2.2}$$

where $\eta > 0$ is the *learning rate*. Intuitively, the gradient points in the direction of steepest increase of the loss, so subtracting it decreases the loss most rapidly in a local sense.

If η is too small convergence is slow; if too large the iteration diverges.

2.2.2 Stochastic gradient descent (SGD)

For large datasets, evaluating the full gradient at every step is prohibitively expensive. Instead, one uses *stochastic gradient descent (SGD)*, which approximates the gradient on a randomly sampled mini-batch $B \subset D$:

$$\nabla_\theta \hat{\mathcal{R}}_B(\theta) = \frac{1}{|B|} \sum_{(x_i, y_i) \in B} \nabla_\theta \ell(f_\theta(x_i), y_i). \tag{2.3}$$

The update rule is then

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)}). \tag{2.4}$$

The batch estimate is unbiased, and its variance sets the scale of the noise in the trajectory [21].

2.2.3 Momentum and adaptive methods

Plain gradient descent treats all directions in parameter space equally. To accelerate convergence, one may use *momentum*, maintaining a velocity vector v_t:

$$v_{t+1} = \beta v_t + (1 - \beta)\, \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)}), \tag{2.5}$$

$$\theta^{(t+1)} = \theta^{(t)} - \eta v_{t+1}, \tag{2.6}$$

where $\beta \in (0, 1)$ controls the contribution of past gradients. Momentum damps oscillation across narrow directions and accumulates displacement along consistent ones.

Beyond momentum, *adaptive methods* such as RMSProp and Adam rescale learning rates per parameter based on historical gradient magnitudes [22]. For instance, Adam combines momentum with an adaptive normalisation:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \tag{2.7}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \tag{2.8}$$

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{2.9}$$

where $g_t = \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)})$ and $\hat{m}_t, \hat{v}_t$ are bias-corrected versions of m_t, v_t. Adam is the default optimiser in this work; where it is replaced by SR, and why, is the subject of §7.2 [23].

2.2.4 Second-order methods and natural gradient

First-order methods only use gradient information. In principle, convergence can be accelerated by incorporating curvature via the Hessian $H(\theta) = \nabla_\theta^2 \hat{\mathcal{R}}(\theta)$. Newton's method updates parameters as

$$\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1}\, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.10}$$

However, computing and inverting the Hessian is usually infeasible for large models.

An attractive alternative is the *natural gradient* [24, 25], which replaces the Euclidean metric with the geometry of probability distributions. The key object is the *Fisher information matrix* (FIM),

$$F(\theta) = \mathbb{E}\big[\nabla_\theta \log p_\theta(x)\, \nabla_\theta \log p_\theta(x)^\top\big], \tag{2.11}$$

where p_θ denotes the model distribution. The natural gradient update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, F(\theta^{(t)})^{-1}\, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.12}$$

This preconditioning makes updates invariant to reparameterizations of θ and often improves conditioning of the optimisation. In practice, computing F^{-1} exactly is intractable, but approximations (e.g. stochastic reconfiguration in variational Monte Carlo, Kronecker-factored approximations in deep learning) make the method viable [2].

2.2.5 The variational tangent space: geometric tensor, tangent kernel, and effective dimension

For a variational quantum state the Fisher metric takes a particularly concrete form, and that form is the analytical lens used throughout the results of this thesis. Write the per-sample log-derivatives—the *score*—of an ansatz Ψ_θ as

$$O_{ki} = \frac{\partial \log |\Psi_\theta(\mathbf{x}_k)|}{\partial \theta_i}, \qquad k = 1, \ldots, B, \;\; i = 1, \ldots, P, \tag{2.13}$$

for a batch of B configurations drawn from $|\Psi_\theta|^2$ and P parameters θ (here θ collects *all* variational parameters). Its centred Gram matrices define two dual objects. Throughout, O is taken *column-centred*, $O_{ki} \mapsto O_{ki} - \frac{1}{B}\sum_{k'} O_{k'i}$, which is what makes the Gram matrix a covariance rather than a second moment and is how it is formed in the implementation (§4.4). The *quantum geometric tensor* (QGT) is then

$$S = \tfrac{1}{B}\, O^\top O \in \mathbb{R}^{P\times P}, \tag{2.14}$$

is the Fisher–Rao metric of the state written in terms of the amplitude score. Every state studied here is real and positive up to the determinant sign, at zero magnetic field, so the amplitude score is the whole story; for a general complex wavefunction the quantum geometric tensor also carries phase sensitivity, and the identification with the classical Fisher matrix is then only of its real part. It is proportional to the Fisher matrix $F(\theta)$ of the previous subsection specialised to $p_\theta = |\Psi_\theta|^2$: because $\nabla_\theta \log |\Psi_\theta|^2 = 2\,\nabla_\theta \log |\Psi_\theta|$, one has $F = 4S$, the factor of four being a convention absorbed into the learning rate. Stochastic reconfiguration is natural-gradient descent under this metric, $\dot\theta = -S^{-1}\nabla_\theta E$ [26, 27]. Its Gram dual, the *neural tangent kernel* (NTK) [28],

$$K = OO^\top \in \mathbb{R}^{B\times B}, \tag{2.15}$$

shares the nonzero spectrum of S up to the overall factor B (its nonzero eigenvalues are $B\lambda_a$), and is cheaper to diagonalise when $B < P$.

The analysis is organised around two scalars read from that spectrum. The *condition number* $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measures how anisotropic the tangent metric is, and therefore how far a natural-gradient step departs from a Euclidean one. It is the quantity one would expect to decide when SR is worth its cost; Chapter 7 measures it and finds that it does not, which is how the estimability criterion of that chapter arises. The *effective dimension*

$$d_{\text{eff}}(S) = \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2} \tag{2.16}$$

is the participation ratio of the spectrum: it counts how many independent tangent directions carry appreciable Fisher weight locally around the solution, regardless of the raw parameter count P. A small d_{eff} means the local tangent sensitivity is concentrated in a few collective directions; a large one means it is spread across many. Because these are properties of the tangent geometry rather than of the raw parameter count, they permit architectures, optimisers and training paradigms to be compared on a common footing—provided the comparison is made under matched conventions. In the comparisons of §7.1.1 the measured d_{eff} proved stable across seeds and across the choice of probe measure; that is an empirical finding at those settings, not a general property.

The participation ratio d_{eff} is parameterisation-dependent. It is computed from the QGT eigenvalues in the ordinary Euclidean parameter coordinates, and those eigenvalues are *not* invariant under an arbitrary reparameterization: rescaling a single parameter $\theta_1 \mapsto c\,\theta_1$ leaves the physical state untouched but changes d_{eff} (for a diagonal metric it moves from 2 toward 1 as $c \to \infty$). d_{eff} is therefore *not* a coordinate-free intrinsic dimension of the wavefunction; it is a property of the (architecture, parameterisation, normalisation) triple. We use it accordingly—as a comparison between fixed architectures under matched conventions and a common probe measure—and read a smaller d_{eff} as "this architecture concentrates its Fisher weight into fewer active directions," not as an absolute manifold dimension.

2.2.6 Tangent-space diagnostics

Convergence speed is not what matters about the optimiser here. A natural-gradient step is only as good as the metric it inverts, so the estimator of S becomes part of the method rather than an implementation detail, and Chapter 7 turns that into a three-regime result. Separately, $\kappa(S)$ and d_{eff} can be read off a trained state, which makes them diagnostics as much as convergence aids, and they are used that way from Chapter 6 onward.

2.3 Networks that can be differentiated twice

A feed-forward network composes affine maps with elementwise nonlinearities,

$$h^{(0)} = x, \qquad h^{(\ell)} = \sigma\big(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}\big), \qquad f_\theta(x) = W^{(L)}h^{(L-1)} + b^{(L)},$$

and without the nonlinearity σ the composition collapses to a single affine map. For most applications the choice of σ is settled by convention or by a sweep. Here the objective settles it, because the local energy $E_L = \hat{H}\Psi_\theta/\Psi_\theta$ contains a Laplacian and training therefore backpropagates through second derivatives of the network.

The activation. $\mathrm{ReLU}(x) = \max(0, x)$ has a second derivative that vanishes almost everywhere and is distributional at its kinks, so it cannot represent the pointwise curvature a Laplacian-based local energy asks for; it is excluded at once. Less obviously, being C^∞ does not suffice either. The relevant property is the reach of the higher derivatives: an activation whose second and third derivatives decay quickly away from the origin contributes little curvature over most of its input range, so the Laplacian of a deep composition stays bounded and its variance finite. Tanh and the logistic sigmoid are perfectly smooth, yet their higher derivatives are sharply peaked and change sign, and curvature accumulates unevenly through depth. We use the $\mathrm{SiLU}(x) = x\,\sigma(x)$ family instead. Figure 2.1 shows GELU and Mish, which are the members of that family whose derivative shapes we compared and which SiLU closely resembles; their second and third derivatives are broader and decay more slowly than those of tanh or the logistic sigmoid. This is a design heuristic rather than a theorem: the figure establishes the shapes of the derivatives, not a bound on the Laplacian variance of a deep composition, and we did not run an activation ablation.

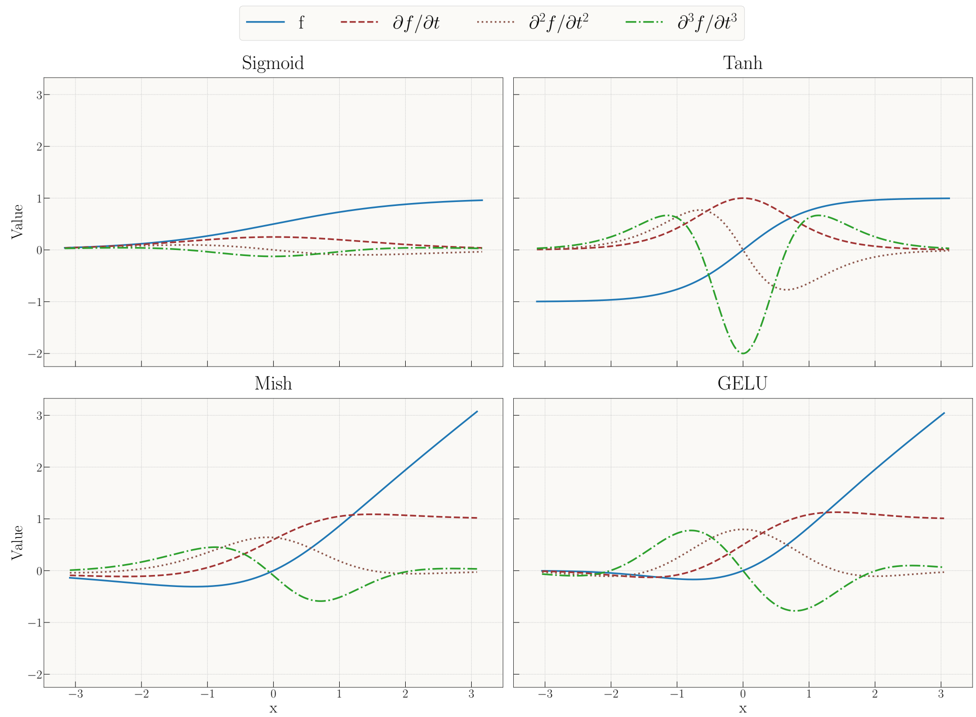


Figure 2.1: Common activation functions and their first three derivatives. All of GELU, Mish, tanh and the logistic sigmoid are C^∞, so the distinction is not smoothness but the *reach* of the higher derivatives: for GELU and Mish the second and third derivatives are broad and decay slowly to zero, while for tanh and sigmoid they are sharply peaked near the origin and change sign, so curvature accumulates unevenly when such units are composed through depth. ReLU is shown for contrast; its second derivative is a distribution and it cannot be used in a Laplacian-based objective at all.

Initialisation. The standard schemes—Xavier/Glorot, scaling weight variance as $1/(n_{\mathrm{in}} + n_{\mathrm{out}})$, and Kaiming/He, scaling as $2/n_{\mathrm{in}}$ for rectified units—are designed to keep forward activations and backward gradients from shrinking or exploding through depth [29, 30]. That is necessary but not sufficient when the loss contains second derivatives, because a scheme that stabilises $\partial f/\partial x$ says nothing directly about $\partial^2 f/\partial x^2$. We use Xavier-style initialisation with the smooth activations above, and—more consequentially than any initialisation choice—zero-initialise the final layer of the backflow, so that the ansatz begins exactly at the Slater nodes and the learned nodal deformation grows from nothing rather than from noise (§3.5.7).

None of this makes the Laplacian well conditioned; it only stops the network from making it worse. The conditioning itself comes from the coordinate scaling, the soft-core pair features and the analytic cusp of Chapter 3, and is analysed in Appendices A and B.

2.4 Architectures for many-body wavefunctions

The feed-forward network of the previous section maps a fixed-size input vector to an output. A many-body wavefunction is not a generic function of such a vector: identical particles impose a permutation symmetry, and correlation is intrinsically *relational*—what one particle should do depends on where the others are. A plain multilayer perceptron acting on the concatenated coordinates $(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ respects neither property; it would have to learn the symmetry from data, and its parameter count would grow with N. This section introduces the two architecture classes that build the required structure in—permutation-invariant pooling and message passing—and situates the specific message-passing network used in this thesis as one instance of the latter.

2.4.1 Permutation symmetry and equivariance

For identical particles the modulus $|\Psi|$ is invariant under exchange (the sign is carried by the antisymmetric Slater reference of Chapter 3). The two learnable objects of the ansatz therefore have distinct symmetry requirements. A scalar *correlator* f must be permutation-*invariant*,

$$f(P_\sigma R) = f(R),$$

while a per-particle *backflow* displacement field $\mathbf{g} = (\mathbf{g}_1, \ldots, \mathbf{g}_N)$ must be permutation-*equivariant*,

$$\mathbf{g}(P_\sigma R) = P_\sigma\, \mathbf{g}(R),$$

for any permutation σ (here P_σ relabels the particles). Enforcing these by construction is both more data-efficient than learning them and yields a parameter count independent of N, because the same shared maps are applied to every particle and pair.

2.4.2 Permutation-invariant pooling: DeepSets

The simplest way to build an invariant function is to embed each particle independently and pool the embeddings symmetrically. This is the DeepSets construction [31]: any suitably regular permutation-invariant function admits the form

$$f(R) = \rho\Big(\textstyle\sum_i \phi(\mathbf{x}_i)\Big),$$

with a per-particle embedding ϕ, a symmetric pool $\sum_i$, and a readout ρ. A two-body extension encodes pairs, $\phi(\mathbf{x}_i, \mathbf{x}_j)$, and pools over them, capturing pairwise correlation. What characterises this class is that each particle—or pair—is embedded *independently* and only then pooled: no information is exchanged between particles before the pool. We refer to such networks as *separable*, and they serve as the non-message-passing baseline for the correlator.

2.4.3 Message passing and graph networks

To let a particle's representation depend on the others' in an iterated, higher-order way, one places the particles on a graph—nodes are particles, edges are pairs—and passes learned messages along the edges [32, 33]. In a generic round ℓ, each edge gathers information from its endpoint nodes and each node aggregates its incoming edge messages,

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e\big(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}^{(\ell)}\big), \qquad \mathbf{h}_i^{(\ell+1)} = \phi_v\Big(\mathbf{h}_i^{(\ell)},\ \mathrm{Agg}_{j\neq i}\, \mathbf{m}_{ij}^{(\ell)}\Big),$$

with shared maps ϕ_e, ϕ_v and a symmetric aggregation Agg, so that permutation equivariance holds by construction; a final readout produces an invariant scalar (correlator) or an equivariant field (backflow). Both DeepSets and message passing use pairwise information, but only message passing lets that information *propagate*: after several rounds a particle's representation reflects not just its neighbours but their neighbours' responses. This iterated propagation is what we call the *relational channel*, and asking what it buys—in the correlator and in the backflow—is one of the central questions of this thesis. Message passing has been used in exactly this spirit for neural quantum states of extended fermion systems [5, 6].

2.4.4 The CTNN: one message-passing instance

The message-passing correlator and backflow studied here are realised by a particular network we refer to as the *CTNN*: a copresheaf-style message-passing network that maintains explicit *node and edge* features and transports information between them through learned maps, rather than recomputing and discarding one-shot messages. The correlator iterates this exchange over a down/up "V-cycle"; the backflow applies a single such copresheaf round. Its concrete update equations, hidden dimensions, and parameter counts are specified in the Methods chapter (§3.5.2, Eqs. (3.13)–(3.15)). The framing matters in two respects. First, the CTNN is *one* concrete realisation of the message-passing class defined above, and the comparison run in this thesis is between two specific inductive biases—the CTNN against a DeepSets-type separable baseline—not between the classes they belong to. Conclusions are stated for message passing *as instantiated by the CTNN, relative to the separable baseline studied here*; whether they generalise to other message-passing neural quantum states is a plausible extrapolation and not something these experiments test. Wherever we contrast a "message-passing" network with a "separable" one, the separable member is a DeepSets-type network and the message-passing member is the CTNN. Second, we write "copresheaf-*style*" deliberately: the name reflects the node/edge transport structure of the updates, not a claim to a formal sheaf-theoretic construction.

2.4.5 Relevance to the ansatz

Both learnable objects of the ansatz—the Jastrow correlator and the coordinate backflow—can be built in either form: as a separable DeepSets-type network, or as the message-passing CTNN. Holding everything else fixed and varying only this choice isolates the effect of the relational channel, which Chapter 7 analyses through the geometry of the tangent kernel introduced in §2.2.5.

2.5 Approximation theory and its limits

The usual justification for using a neural network as a variational ansatz is that it can represent anything. That argument needs disposing of early, since this thesis is about what happens after it stops being relevant.

The universal approximation theorems say that a single hidden layer with a non-polynomial activation is dense in $C(K)$ on compacta: for any continuous f and any $\varepsilon > 0$ there exist a width and a parameter vector achieving $\sup_{x\in K} |f(x) - f_\theta(x)| < \varepsilon$ [34–36]. Sharper versions bound the width needed for a given accuracy, and show that depth can be exponentially more efficient than width for compositional targets. For our purposes they establish exactly one thing, and it is best put negatively: there is no *fundamental* representability obstruction, because a sufficiently expressive network of this form exists. They say nothing about the particular 20k–236k networks used here. Whether those are expressive enough is an empirical question, and it is answered later by what they achieve variationally, not by any theorem.

What they do not establish is anything about whether gradient-based training will find that representation, at what sample cost, or how the answer depends on how the function is parameterised. Those questions are the subject of this thesis, and the theorems are silent on all three. The existence proofs are non-constructive; the widths they guarantee are typically astronomical; and the same function can be represented by two networks whose optimisation behaviour is entirely different—which is, in miniature, the comparison Chapter 7 makes.

One further result does more than assert possibility. In high dimension a generic continuous target requires a network size growing exponentially in d, so approximation alone cannot explain why these methods work at all in Nd dimensions. The escapes are all statements about *structure* rather than capacity: targets of finite Barron norm admit dimension-independent rates [37]; compositional targets admit efficient representation by deep networks when the composition matches the architecture; and imposing a symmetry removes the redundant copies the network would otherwise have to learn, reducing the burden on the representation rather than the dimension of the domain. The last of these is the one we exploit directly, and the middle one is why the message-passing architectures of §2.4 deserve attention: their usefulness comes from a compositional form matching the relational structure of the physics, not from added capacity.

The rest of the thesis therefore assumes representability and measures the other things: trainability, conditioning, and the geometry of the parameterisation.

2.6 Physics-Informed Neural Networks

In scientific applications the governing equations already encode most of what is known about a system, and a model fitted only to data throws that information away. *Physics-informed neural networks* (PINNs) [38] put it back, by penalising the residual of the equation itself alongside whatever data exists. Training a network against a differential operator predates the name by two decades [39]; what [38] added was automatic differentiation, which makes the residual cheap enough to evaluate that it can drive the optimisation directly.

2.6.1 Formulation of PINNs

Let $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ denote the unknown solution of a partial differential equation (PDE) of the form

$$\mathcal{N}[u](x) = 0, \qquad x \in \Omega, \tag{2.17}$$

subject to boundary conditions $u(x) = g(x)$ for $x \in \partial\Omega$. Here $\mathcal{N}$ is a (possibly nonlinear) differential operator.

In a PINN, $u_\theta(x)$ is represented by a neural network, and the loss function enforces both data fitting and PDE residual minimization:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \sum_{i=1}^{n_d} |u_\theta(x_i) - y_i|^2 + \lambda_{\text{PDE}} \sum_{j=1}^{n_r} |\mathcal{N}[u_\theta](\xi_j)|^2 + \lambda_{\text{BC}} \sum_{k=1}^{n_b} |u_\theta(\zeta_k) - g(\zeta_k)|^2. \tag{2.18}$$

Here $\{x_i, y_i\}$ are observed data, $\{\xi_j\}$ are collocation points in the domain for enforcing the PDE, and $\{\zeta_k\}$ are boundary points. The weights $\lambda_{\text{data}}, \lambda_{\text{PDE}}, \lambda_{\text{BC}}$ balance the different contributions.

2.6.2 Automatic differentiation

A key ingredient of PINNs is the use of *automatic differentiation* (AD) to evaluate ∇u_θ and Δu_θ. Unlike finite-difference approximations, AD computes derivatives exactly (up to floating-point error), making it feasible to incorporate higher-order differential operators directly into the loss. This enables end-to-end training of networks constrained by physical laws.

2.6.3 Theoretical difficulties

Although PINNs are conceptually straightforward, making them reliable and scalable raises several non-trivial challenges that go beyond classical universal approximation results.

Approximation in Sobolev spaces. Universal approximation theorems control function error in L^2/L^∞, but strong-form PINN losses depend on *derivatives* through $L[u_\theta]$. To make residuals small one must approximate in matching Sobolev norms $W^{\ell,p}$ (with ℓ the PDE order), which imposes smooth activations (e.g., tanh/SiLU for $\ell \geq 2$) and architecture choices that control higher derivatives. Low-regularity targets (discontinuities, cusps) break global $W^{\ell,p}$ control and favor weak/variational or entropy-based formulations.

Stability (residual-to-error). A small residual does not imply a small solution error unless the forward problem is (conditionally) stable. For well-posed elliptic/parabolic problems one can bound $\|u - u_\theta\|$ by $\|L[u_\theta] - f\|$, but inverse/ill-posed settings may require regularization, priors, or restricted hypothesis classes to prevent "residual overfitting" with large state error.

Sampling and generalisation. Collocation-based training minimizes a sampled surrogate of a continuum loss. Generalisation requires capacity control (e.g., bounded weights/Lipschitz loss) and sufficient sampling of interior vs. boundary/IC points; otherwise the empirical residual can be small while the population residual remains large.

Optimisation and conditioning. PINN losses couple multiple physics terms (interior, boundary, data) with disparate scales, producing ill-conditioned gradient dynamics. In the linearized/NTK view, convergence speed is governed by the spectrum of a problem-dependent kernel; large condition numbers lead to slow or stalled training.

Spectral bias. Gradient-based training tends to fit low-frequency components first, delaying convergence for solutions with sharp layers or oscillations unless the sampling, features, or architecture emphasize high-frequency content (e.g., Fourier features, windowed bases, curriculum).

Mitigating these challenges requires careful design of architectures, activations, initialisation, loss weighting, and sampling strategies. These topics are explained and discussed in detail in [40]

2.6.4 Relevance to this work

The four difficulties above are not a list of things that might go wrong. Each of them recurs, by name, in the results. Sobolev approximation is why the activation is chosen for its higher derivatives rather than its values. Residual-to-error stability is why a small residual is never taken as evidence of a good state without an independent energy evaluation. Sampling and generalisation, in this setting, becomes the question of whether an importance-weighted estimator has a variance at all, which is what bounds the Markov-chain-free programme. And optimisation conditioning is the subject of the entire tangent-kernel analysis.

The fourth is the one that determined the design of the collocation programme. A strong-form residual loss, differentiating through the Laplacian, meets a structural obstruction as soon as it is asked to train a nodal surface, because the learning signal lives where E_L diverges. The weak form keeps the local energy as a reward and never differentiates it, and so avoids the obstruction. Appendix C shows why, and it is the reason the collocation programme of Chapter 7 takes the form it does.

2.7 Constraints carried into the method

The two traditions of this chapter and the last have been describing the same objects in different words. The quantum geometric tensor of variational Monte Carlo is the Fisher metric of information geometry; stochastic reconfiguration is natural-gradient descent under it; the neural tangent kernel is that metric's Gram dual; and the sampling measure a physicist chooses for a Metropolis chain is what a statistician calls a proposal distribution. Neither community needs the other's vocabulary in order to do its own work, which is why the identifications are easy to miss. They matter here because the architecture determines the spectrum of that metric, the optimiser inverts it, and the sampling measure determines how well it can be estimated.

Four properties of this problem bear on that metric.

The objective contains a second-order operator. The local energy carries a Laplacian, so training differentiates through second derivatives of the network. This is what dictates smooth activations, controlled initialisation, and a determinant that is never asked to do anything violent (§2.3).

The interaction is singular, and the state carries nodes. The Coulomb term diverges at coalescence, and the trial function vanishes on a surface whose location is itself being learned. Both are places where $E_L = \hat{H}\Psi/\Psi$ can blow up, and the cusp conditions of §1.3.4 say that the first divergence can be cancelled analytically if the short-range form is built in rather than learned.

The measure is part of the method. The objective is an expectation under $|\Psi_\theta|^2$, so the sampling distribution enters the gradient, the metric, and the estimate of both. Whether that metric can be estimated at all turns out to decide whether preconditioning helps or hurts (§2.2.5).

Correlation is relational. What one electron should do depends on where the others are, and the two architecture classes of §2.4 differ precisely in how far that dependence is allowed to propagate.

The first two are met by construction in Chapter 3, which hard-wires antisymmetry and the cusp and conditions the geometry so that the learned part can remain small and smooth. The third is treated in

Chapter 4, where the sampling measure is a design variable rather than a fixed background. The fourth is not settled here: both architectures are constructed, and the comparison between them is reported in Chapter 7.

Part III

Methods

No great thing is created suddenly, any more than a bunch of grapes or a fig.

EPICTETUS

Discourses

Chapter 3

Trial Wavefunction

The trial state is constructed to satisfy fermionic antisymmetry and the electron–electron cusp analytically, and to keep the learned terms smooth under the Laplacian. Antisymmetry requires a determinant. The Coulomb divergence admits an analytic cancellation, so it is imposed rather than learned. Second derivatives enter the loss, so the features fed to the network are bounded and smooth and the coordinates are scaled to keep those derivatives $\mathcal{O}(1)$ across three decades of confinement. The form of the learned part is not fixed by these constraints; both candidate architectures are constructed here and compared in Chapter 7. Unless stated otherwise we work in atomic units.

3.1 Preliminaries

Hamiltonian and units. We study N spin-$\frac{1}{2}$ fermions in two spatial dimensions confined by a harmonic trap and interacting via Coulomb repulsion,

$$H = \sum_{i=1}^{N} \Big(-\tfrac{1}{2}\nabla^2_{\mathbf{r}_i} + \tfrac{1}{2}\,\omega^2 r_i^2 \Big) + \sum_{1\le i<j\le N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1). \tag{3.1}$$

Spin does not appear in H; eigenstates factorize into a spatial function with good L_z and a spin function with good (S, S_z).

Coordinates and scaling. We collect positions as $R = (\mathbf{r}_1, \ldots, \mathbf{r}_N) \in \mathbb{R}^{2N}$ and use trap scaling $\tilde{\mathbf{r}}_i = \sqrt{\omega}\,\mathbf{r}_i$, $\tilde{R} = \sqrt{\omega}\,R$. This aligns typical length scales to $\mathcal{O}(1)$, improves conditioning of second derivatives, and makes the noninteracting ground state ω-independent. When useful, we also use pair distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and their scaled versions $\tilde{r}_{ij} = |\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j|$.

Sectors and focus. We primarily report *closed-shell* cases (unique spin singlet at the noninteracting level). Open shells introduce a degenerate manifold and additional choices for the reference; we keep this to brief remarks only where needed.

Spin–orbitals and basis. One-particle spatial orbitals are the Cartesian two-dimensional harmonic-oscillator eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\}$ with $E_{n_x n_y} = \omega(n_x + n_y + 1)$, $n_x, n_y \in \mathbb{N}_0$ (the basis used in the implementation; for the closed shells studied it spans the same filled shells as any complete basis of those shells). Spin–orbitals are $\varepsilon_\alpha(x) = \phi_{n_x n_y}(\mathbf{r})\chi_\sigma(s)$, $\sigma \in \{\uparrow, \downarrow\}$.

3.2 Overview of the ansatz

Our trial state combines exact antisymmetry, an analytic short-range cusp, and a compact neural residual, while optionally evaluating the Slater determinant on backflowed coordinates:

$$\Psi_{\theta,\beta}(R) = \mathrm{SD}\big(\tilde{R}'\big)\ \exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big), \tag{3.2}$$

$$\tilde{R}' = \tilde{R} + \Delta_\beta(\tilde{R}). \tag{3.3}$$

Here SD enforces antisymmetry; u hard-wires the Kato cusp; W_θ (the NQS correlator) learns smooth medium/long-range structure; and Δ_β (backflow) adjusts the nodes by smoothly warping the coordinates used inside the determinants; its per-particle displacements $\Delta_{\beta,i}$ are written $\Delta\mathbf{x}_i$ in the results chapters. We deliberately feed the NQS with $\tilde{R}$ (not $\tilde{R}'$) to avoid self-reinforcing feedback and keep local-energy variance low.

Design principles. (i) *Separate guaranteed physics from learned details:* antisymmetry and the cusp are hard-wired; the network learns the rest. (ii) *Condition the geometry:* work in trap units and use soft-core pair features so that first/second derivatives are bounded. (iii) *Keep the learned object small and smooth:* once the cusp is explicit, a compact NQS suffices across (N, ω).

3.3 Slater reference

For closed shells we use a restricted product of spin-block determinants of the lowest $N/2$ harmonic oscillator orbitals (trap units):

$$\mathrm{SD}(\tilde{R}) = \det\big[\phi_a(\tilde{\mathbf{r}}_i)\big]_{i\in\uparrow,\ a\le N/2}\ \det\big[\phi_a(\tilde{\mathbf{r}}_j)\big]_{j\in\downarrow,\ a\le N/2}. \tag{3.4}$$

The correlator reweights amplitudes *within* nodal pockets; without backflow, the nodes are those of (3.4).

3.4 Neural correlator W_θ

As with the backflow, the correlator comes in two forms that are compared directly in §7.1: a *separable* network that encodes particle- and pair-features independently and pools them, and a copresheaf *message-passing* (CTNN) network. We describe the separable network first—it is the baseline and also supplies the safe features and analytic cusp shared by both—and then the CTNN.

3.4.1 Separable correlator (DeepSets baseline)

The separable NQS aggregates *particlewise* and *pairwise* embeddings with permutation-invariant pooling and appends two global statistics, all in $\tilde{R}$. A core safety requirement is that pair channels satisfy $ds_k/d\tilde{r} \to 0$ as $\tilde{r} \to 0$ so that W_θ never injects spurious $1/\tilde{r}$ terms into $\nabla \log \Psi$ or $\Delta \log \Psi$.

Particle branch (DeepSets).

This part of the network is typically referred to as Deep Sets [31]. The permutation invariance of quantum many-body mechanics is naturally captured by this architecture. A neural network ϕ creates a latent embedding for each particle individually. All embeddings are pooled together to form a global representation, which is used later in the readout. Formally, we have

$$\mathbf{h}_i^\phi = \phi(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{d_L}, \qquad \overline{\mathbf{h}}^\phi = N^{-1}\sum_i \mathbf{h}_i^\phi \in \mathbb{R}^{d_L}.$$

Pair branch (soft-core, even channels).

Although the particle branch could theoretically produce pair correlations, in practice it struggles to represent sharp near-field structure. We therefore add a dedicated pair branch that builds features from interparticle distances. This network processes all pairs $(i,j), i \neq j$ individually and pools the resulting latent embeddings. The problem is that raw distances $\tilde{r}_{ij}$ have unbounded derivatives at coalescence, which destabilizes Laplacian evaluations. We therefore build *safe* pair features with bounded first and second derivatives as $\tilde{r} \to 0$. With $\tilde{r}^{\text{soft}} = \sqrt{\tilde{r}^2 + \varepsilon^2}$ ($\varepsilon \in [0.15, 0.30]$ in trap units), we build six "safe" features

$$\begin{aligned} s_1 &= \log\big(1 + (\tilde{r}^{\text{soft}}/\varepsilon)^2\big), \quad s_2 = \tfrac{\tilde{r}^2}{\tilde{r}^2+\varepsilon^2}, \quad s_3 = (\tilde{r}^{\text{soft}}/\varepsilon)^2 \exp\big[-(\tilde{r}^{\text{soft}}/\varepsilon)^2\big], \\ \text{rbf}_g &= \exp(-g\, s_1), \quad g \in \{0.25, 1, 4\}, \qquad \mathbf{u}_{ij} = [s_1, s_2, s_3, \text{rbf}_{0.25}, \text{rbf}_1, \text{rbf}_4] \in \mathbb{R}^6. \end{aligned} \tag{3.5}$$

An MLP ψ maps $\mathbf{u}_{ij} \mapsto \mathbf{h}^{\psi}_{ij} \in \mathbb{R}^{d_L}$. A short-range gate

$$\chi(\tilde{r}) = \frac{\tilde{r}^2}{\tilde{r}^2 + r_g^2}, \quad r_g \approx 0.3, \quad \chi(0) = \chi'(0) = 0,$$

suppresses learned responses *at* coalescence; downstream we use $\chi\, \mathbf{h}^{\psi}_{ij}$.

3.4.2 Analytic short-range cusp (shared)

The analytic cusp below is shared by both correlator variants (and is independent of the backflow). Because the safe features are necessary in the pair branch, the NQS cannot represent the Kato cusp condition exactly. We therefore add an analytic cusp term to the exponent in (3.2), written in *physical* coordinates so that its slope at contact is manifestly correct:

$$u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r/a_{\text{ho}}}, \qquad a_{\text{ho}} = 1/\sqrt{\omega}, \qquad \gamma_{\uparrow\downarrow} = \tfrac{1}{d-1}, \quad \gamma_{\uparrow\uparrow} = \tfrac{1}{d+1}. \tag{3.6}$$

Because the linear prefactor is the physical separation r, the slope at contact is $\partial_r u|_0 = \gamma_{\sigma\sigma'}$, exactly the Kato condition in general d; for $d = 2$ this gives $\gamma_{\uparrow\downarrow} = 1$ and $\gamma_{\uparrow\uparrow} = \frac{1}{3}$, the standard two-dimensional quantum-dot values, derived—including the parallel-spin case, where antisymmetry already forces a linear node—in Appendix B.2. Only the exponential taper carries the trap length a_{ho}, suppressing long-range interference so the NQS need not unlearn cusp structure at medium and long range; in trap units ($\tilde{r} = \sqrt{\omega}\, r$) the same term reads $u = (\gamma_{\sigma\sigma'}/\sqrt{\omega})\, \tilde{r}\, e^{-\tilde{r}}$.

3.4.3 Separable readout

With contributions from the particle and pair branches, we form a global representation for the final readout. In addition to the pooled embeddings $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$, we also include two global statistics that help the network gauge overall system size and density. We average pairs by default or apply degree-normalised radial attention

$$w_{ij} = \exp[-(\tilde{r}_{ij}/r_c)^p], \quad \hat{w}_{ij} = w_{ij} \Big/ \sum_{k \neq i} w_{ik}, \quad (r_c, p) = (0.4, 6),$$

then form $g_i = \sum_{j \neq i} \hat{w}_{ij}\, \mathbf{h}^{\psi}_{ij}$ and $\overline{\mathbf{h}}^{\psi} = N^{-1} \sum_i g_i$. Two system-level scalars in trap units,

$$\mathbf{g}(\tilde{R}) = \Big[\tfrac{1}{Nd} \sum_i \|\tilde{\mathbf{x}}_i\|^2,\ \tfrac{2}{N(N-1)} \sum_{i<j} s_1(\tilde{r}_{ij})\Big] \in \mathbb{R}^2,$$

are concatenated with $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$ and mapped to a scalar by a tiny MLP ρ:

$$W_\theta(\tilde{R}) = \rho\Big(\overline{\mathbf{h}}^{\phi} \parallel \overline{\mathbf{h}}^{\psi} \parallel \mathbf{g}(\tilde{R})\Big). \tag{3.7}$$

3.4.4 Message-passing correlator (CTNN)

The message-passing correlator (code: `CTNNJastrowVCycle`) keeps the same node/edge embeddings, safe pair features, and analytic cusp, but replaces the independent pooling of the separable network with the copresheaf transport of §2.4.4: at each round the endpoint nodes are transported into an edge update and the updated edges are transported back and aggregated onto the nodes,

$$\mathbf{h}_{ij} \mathrel{+}= F_e\big(\mathbf{h}_{ij} \,\|\, \rho_{v\to e}\mathbf{h}_i \,\|\, \rho_{v\to e}\mathbf{h}_j\big), \qquad \mathbf{h}_i \mathrel{+}= F_v\Big(\mathbf{h}_i \,\|\, \underset{j\neq i}{\mathrm{Agg}}\, w_{ij}^{\mathrm{spin}}\, \rho_{e\to v}\mathbf{h}_{ij}\Big), \tag{3.8}$$

with residual updates and per-round transport maps. Unlike the single-round backflow, the correlator stacks these rounds in a multiscale "V-cycle": $n_\downarrow$ down-pass rounds, a linear coarsening to a bottleneck width, then $n_\uparrow$ up-pass rounds with skip connections back to the down-pass features (defaults $n_\downarrow = n_\uparrow = 2$). A permutation-invariant readout over the final node and edge statistics returns the scalar $W_\theta(\tilde{R})$, exactly as in (3.7). Holding the Slater reference, cusp, and training fixed and swapping only the separable readout for this V-cycle isolates the relational channel analysed in §7.1.

3.5 Backflow

Backflow deforms the coordinates *used only inside the Slater reference* so that the nodal surface can adapt to interparticle correlations:

$$\tilde{R}' \;=\; \tilde{R} + \Delta_\beta(\tilde{R}), \qquad \Psi_{\theta,\beta}(R) \;=\; \mathrm{SD}(\tilde{R}')\, \exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big). \tag{3.9}$$

The correlator W_θ and cusp u are evaluated on $\tilde{R}$ to avoid feedback loops in the local energy; only the determinant "sees" $\tilde{R}'$. This choice cleanly separates (i) amplitude reweighting within nodal pockets (handled by $u + W_\theta$) from (ii) nodal *geometry* (handled by backflow).

Design goals. (i) *Stability near coalescence:* messages are built from mollified radii and short-range gates so that $\partial\Delta_\beta/\partial\tilde{r}$ stays bounded; (ii) *Permutation symmetry:* permutation equivariance holds by construction (shared maps and symmetric aggregation); (iii) *Close to identity at initialisation:* a zero-initialised last layer and a positive learnable scale keep $\Delta_\beta \approx 0$ at the start. Rotational symmetry, by contrast, is only *approximate*: the displacement head reads out a d-vector from node features that include the raw coordinates, so the map is not rotation-equivariant by construction, and it is encouraged instead by the geometry-based edge features and by rotationally-augmented sampling. A global translation is not a symmetry of the trap in any case; the zero-mean projection (3.19) is a regulariser that removes a spurious centre-of-mass drift, not an imposed physical symmetry.

We study *two* backflow architectures. Both map the configuration to a per-particle displacement field and share the output bounding, spin masking, and centre-of-mass projection below; they differ only in how a particle's displacement is made to depend on the others. The conventional network is the baseline of the architecture comparison in Chapter 7; the copresheaf CTNN is the production ansatz.

3.5.1 Conventional backflow (single-round messages)

The baseline (code: `BackflowNet`) uses a single round of pairwise messages. For each ordered pair (i, j) with $i \neq j$,

$$\mathbf{m}_{ij} = \phi_\beta\big(\tilde{\mathbf{x}}_i,\ \tilde{\mathbf{x}}_j,\ \mathbf{r}_{ij},\ \tilde{r}_{ij}^{\mathrm{soft}},\ (\tilde{r}_{ij}^{\mathrm{soft}})^2\big)\cdot w_{ij}^{\mathrm{spin}}, \qquad \tilde{r}_{ij}^{\mathrm{soft}} = \sqrt{\tilde{r}_{ij}^2 + \varepsilon_{\mathrm{bf}}^2}, \tag{3.10}$$

with ϕ_β a small MLP and $w_{ij}^{\mathrm{spin}} \in \{0, 1\}$ a spin mask (same-spin only or all pairs). Messages are aggregated to a node state and a node network returns the displacement,

$$\mathbf{m}_i = \underset{j\neq i}{\mathrm{Agg}}\, \mathbf{m}_{ij}, \qquad \delta\tilde{\mathbf{x}}_i = \psi_\beta\big(\tilde{\mathbf{x}}_i \,\|\, \mathbf{m}_i\big), \qquad \mathrm{Agg} \in \{\mathrm{sum}, \mathrm{mean}, \mathrm{max}\}. \tag{3.11}$$

There is no persistent edge state: each message is recomputed from the raw geometry and discarded after aggregation.

3.5.2 CTNN backflow (copresheaf)

The message-passing architecture (code: `CTNNBackflowNet`) instead maintains explicit *node and edge* features and transports information between them through learned linear maps—the copresheaf-style step of §2.4.4. Node features are embedded from the position and a spin scalar, edge features from the relative geometry,

$$\mathbf{h}_i^{(0)} = W_v\,[\tilde{\mathbf{x}}_i,\, s_i], \qquad \mathbf{h}_{ij}^{(0)} = \mathrm{MLP}_e\big(\mathbf{r}_{ij},\, |\mathbf{r}_{ij}|,\, |\mathbf{r}_{ij}|^2\big). \tag{3.12}$$

A node-to-edge transport $\rho_{v\to e}$ feeds the endpoint nodes into an edge update; an edge-to-node transport $\rho_{e\to v}$ returns the updated edges to the nodes, which are refined by a residual node update; a linear head reads out the displacement:

$$\mathbf{h}'_{ij} = \mathrm{MLP}'_e\big(\mathbf{h}_{ij}^{(0)} \,\|\, \rho_{v\to e}\mathbf{h}_i^{(0)} \,\|\, \rho_{v\to e}\mathbf{h}_j^{(0)}\big), \tag{3.13}$$

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} w_{ij}^{\mathrm{spin}}\, \rho_{e\to v}\big(\mathbf{h}'_{ij}\big), \tag{3.14}$$

$$\mathbf{h}_i = \mathbf{h}_i^{(0)} + \mathrm{MLP}_v\big(\mathbf{h}_i^{(0)} \,\|\, \mathbf{m}_i\big), \qquad \delta\tilde{\mathbf{x}}_i = W_\Delta\, \mathbf{h}_i. \tag{3.15}$$

The learned transport maps $\rho_{v\to e}, \rho_{e\to v}$ are the channel whose ablation collapses the backflow at the Wigner crystal (Chapter 7). The backflow uses a single copresheaf round; the message-passing *correlator* of the architecture comparison (`CTNNJastrowVCycle`) iterates the same style of update over a down/up "V-cycle".

3.5.3 Shared output, scaling, and initialisation

Both variants take inputs in trap units ($\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$), bound the displacement, and start at the Slater nodes,

$$\Delta_\beta(\tilde{R}) \;=\; \tanh\big(\delta\tilde{X}\big)\cdot s_\beta, \qquad s_\beta = \mathrm{softplus}(s_\beta^{\mathrm{raw}}) > 0, \tag{3.16}$$

with the last linear layer (ψ_β or W_Δ) zero-initialised so that $\Delta_\beta\approx 0$ initially. Empirically this parameterisation stabilises the SR/energy tails and protects the Laplacian.

3.5.4 Short-range behaviour and gates

As $\tilde{r}_{ij}\to 0$, raw $1/\tilde{r}$ factors can make $\nabla\cdot\Delta_\beta$ spiky. We therefore (i) mollify $\tilde{r}_{ij}$ as in (3.10), and optionally (ii) gate the message with

$$\chi_{\mathrm{bf}}(\tilde{r}_{ij}) = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + r_g^2}, \qquad \chi_{\mathrm{bf}}(0) = \chi'_{\mathrm{bf}}(0) = 0, \quad r_g \approx 0.3, \tag{3.17}$$

or with degree-normalised radial attention

$$w_{ij} = \exp\big[-(\tilde{r}_{ij}/r_c)^p\big], \qquad \hat{w}_{ij} = \frac{w_{ij}}{\sum_{k\neq i} w_{ik}}. \tag{3.18}$$

Either choice keeps the learned response smooth at coalescence and limits the BF contribution to the local energy variance.

3.5.5 Symmetries and COM neutrality

Permutation symmetry is automatic (pairwise messages and symmetric aggregation). To avoid a spurious centre-of-mass (COM) drift, we project the flow to zero mean:

$$\Delta_\beta \;\leftarrow\; \Delta_\beta \;-\; \frac{1}{N}\sum_{i=1}^{N} \Delta_{\beta,i}. \tag{3.19}$$

The messages are built from the relative vectors $\mathbf{r}_{ij}$ and the zero-mean projection removes the spurious centre-of-mass drift; as noted under the design goals, the map is not translation-invariant or rotation-equivariant by construction (the displacement head reads a d-vector from the absolute coordinates).[1]

The two backflow variants apply this projection at *different points*, and the difference is invisible in the wavefunction but not in the diagnostics. The copresheaf network (§3.5.2) applies Eq. (3.19) inside its forward pass, so the field it returns is already COM-free; the conventional network returns the raw displacement and the projection is applied by the caller when $\tilde{R}'$ is formed. The shifted coordinates are therefore identical either way, but any diagnostic evaluated on a network's *returned* field sees a COM component in one case and not the other. This is what puts the two columns of the backflow-rank measurement in §7.1.2 under ceilings that differ by two, and it is stated there again where it bites.

[1] Implementation note: the mean is taken over particles *within each configuration*, not across configurations in the batch, so the constraint removes centre-of-mass motion configuration by configuration. It is a one-liner and removes the small residual COM drift seen in ablations.

3.5.6 Effect on the nodes and variational safety

Because only SD is evaluated at $\tilde{R}'$, backflow changes the *nodes* but leaves the correlator exponent on $\tilde{R}$. No Jacobian term is introduced (we are not performing a change of variables of the full wavefunction). This "Feynman–Cohen style" [41] design targets fixed-node error directly while keeping the amplitude model simple and smooth; the same idea in its modern quantum-Monte-Carlo form is due to Holzmann and Ceperley [42].

3.5.7 Regularization, initialisation, and cost

We found the following defaults robust across (N, ω): (i) $\varepsilon_{\text{bf}} = 10^{-6}$ (trap units) for the backflow soft norm of (3.10), fixed rather than tuned. It is a numerical guard, not a modelling choice: $\sqrt{\tilde{r}^2 + \varepsilon_{\text{bf}}^2}$ differs from $\tilde{r}$ by less than ε_{bf} everywhere, so it changes no value the network can resolve and exists only to keep $\nabla\tilde{r}$ finite at exact coincidence, which $\nabla\sqrt{\tilde{r}^2}$ is not. It should not be read as a short-range physical scale: that role belongs to the pair-feature softening $\varepsilon \in [0.15, 0.30]$ of (3.5), four to five orders of magnitude larger, which is what actually sets where the correlator stops resolving structure; (ii) bounded output tanh with a learnable scale s_β initialised in $[0.02, 0.10]$; (iii) optional gate χ_{bf} with $r_g \approx 0.3$ (trap units). Zero-initialising the last layer of ψ_β makes the ansatz start exactly at the Slater nodes and improves optimiser stability. The compute/memory cost scales as $O\big(BN^2(d + H)\big)$ for batch size B and hidden width H.

3.5.8 Mapping to implementation

In the conventional network, ϕ_β and ψ_β correspond to `BackflowNet.phi` (message MLP) and `BackflowNet.psi` (node/update MLP). In the copresheaf network, the embeddings, transport maps, and update MLPs of (3.13)–(3.15) correspond to `CTNNBackflowNet`'s `node_embed/edge_embed`, `rho_v_to_e/rho_e_to_v`, `edge_update/node_update`, and `dx_head`. For both, aggregation is selectable (`sum|mean|max`); spin handling (`use_spin`, `same_spin_only`) produces the mask w_{ij}^{spin}; output control matches (3.16) via `out_bound="tanh"` and s_β = softplus(`bf_scale_raw`); and (3.19) is enforced by subtracting the per-configuration mean over particles of the predicted Δ_β before forming $\tilde{R}'$.

Summary. Backflow provides nodal freedom by moving the Slater coordinates, while also modifying the determinant amplitude inside each nodal pocket. With mollified distances, optional short-range gates/attention, zero-mean projection, and bounded outputs, it integrates cleanly with the analytic cusp and smooth NQS correlator, designed to reduce nodal error while controlling local-energy variance.

3.6 Architecture sizes and hyperparameters

Table 3.1 collects the representative settings. The two comparisons in Chapter 7 stand on different footings in this respect and it is worth saying so before either is read.

The *correlator* comparison is capacity-controlled. The two variants are held at comparable parameter counts, so a difference in outcome cannot be attributed to one simply being larger, and where §7.1.1 departs from matched capacity it does so in the *separable* network's favour, scaling it from 20k to 164k parameters against a fixed 80k message-passing network without closing the gap.

The *backflow* comparison is not. At equal width the copresheaf network is intrinsically the heavier object—it carries a persistent edge state where the conventional network carries only a per-particle aggregate—so at the shared production setting (message width 128, node MLP 128×3) it holds 182k parameters against 51k, a factor of 3.6. No matched backflow pair was trained. What separates architecture from capacity there is therefore not a matched control but the conventional network's own behaviour across ω, which §7.1.2 sets out where the measurement is reported. All settings below are those of the accompanying code (`scripts/run_depth_analysis.py` for the comparison tiers; `scripts/source_state_check.py` for the production widths).

Table 3.1: The three sets of models trained in this thesis. They exist for different purposes and are never compared with one another. Widths are hidden dimensions; “rounds” are message-passing steps ($n_{\downarrow}+n_{\uparrow}$ for the V-cycle). Within the architecture study the correlator variants are not capacity-matched, which is why §7.1.1 also reports the matched 89.4k separable network and the full 20k–164k ladder; the backflow variants are not matched either, and cannot be at equal width, for the reason given above. Settings follow the accompanying code (`scripts/run_depth_analysis.py`, `scripts/source_state_check.py`).

Programme	Correlator	Backflow
Physics production	separable, width 128, 2 layers, 54k	conventional, msg 128, node 128 × 3, 51k
(§6.1)	*(as above)*	message-passing, same widths, 182k
Architecture study	separable, pair-MLP 64 × 4, out 32, 47.7k	*as production; not matched*
(§7.1)	message-passing, node/edge 32, bottleneck 16, 2+2 rounds, 79.8k	
Collocation	message-passing V-cycle, 25k	none
(§7.3)	*(as above)*	copresheaf, h_{bf}=128, +24k

Readout for every correlator: hidden 64, 2–3 layers, SiLU. Optimiser: Adam at $\mathrm{lr} = 5 \times 10^{-4}$ (backflow) and 5×10^{-5} (Jastrow); CG-SR with Tikhonov damping λ, a trust-region cap and CG restarts. Collocation batch $n_{\mathrm{coll}} \sim 3 \times 10^3$ with oversampling $m = 8$–16; the collocation families are described further in §4.5.3.

Chapter 4

Optimisation

4.1 Primary objective and two–stage training

Given a parameterised wavefunction Ψ_θ, the variational ground–state energy is

$$E(\theta) = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \mathbb{E}_{\tilde{R} \sim \pi_\theta}\big[E_L(\tilde{R})\big], \qquad \pi_\theta(\tilde{R}) = \frac{|\Psi_\theta(\tilde{R})|^2}{\int |\Psi_\theta|^2}. \tag{4.1}$$

The local energy is evaluated in *physical* coordinates, consistently with the Hamiltonian (3.1),

$$E_L(R) = -\tfrac{1}{2} \sum_{i=1}^{N} \Big[\Delta_{\mathbf{r}_i} \ln \Psi_\theta + \|\nabla_{\mathbf{r}_i} \ln \Psi_\theta\|^2 \Big] + \tfrac{1}{2}\, \omega^2 \sum_{i=1}^{N} r_i^2 + \sum_{i<j} \frac{1}{r_{ij}}, \tag{4.2}$$

and the zero–variance principle holds: $\mathrm{Var}_{\pi_\theta}[E_L] = 0$ iff Ψ_θ is an eigenstate. The network *features* are computed in trap units $\tilde{\mathbf{r}} = \sqrt{\omega}\, \mathbf{r}$ for conditioning, but the local energy itself is formed in physical units, so no ω factors are absorbed into the kinetic, trap, or Coulomb terms. (In trap units the same operator reads $E_L = \omega\big[-\tfrac{1}{2} \sum_i (\Delta_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta + \|\nabla_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta\|^2) + \tfrac{1}{2} \sum_i \tilde{r}_i^2\big] + \sum_{i<j} \sqrt{\omega}/\tilde{r}_{ij}$; the kinetic and trap terms scale by ω and the Coulomb term by $\sqrt{\omega}$.)

Our optimisation follows two stages that mirror the implementation:

1. **Residual–based pretraining (stage I).** We fit the residual of E_L to a moving scalar target E_{eff} using the loss $\mathcal{L}_{\text{res}}(\theta) = \mathbb{E}\big[(E_L(\tilde{R}) - E_{\text{eff}})^2\big]$. We begin with a *self–target* (stabilizer) $E_{\text{eff}} = \mu$ where $\mu = \mathbb{E}[E_L]$ over the current batch. This minimises the *sample* variance of E_L under the current collocation measure—which is not the same object as $\mathrm{Var}_{|\Psi|^2}(E_L)$ unless the two measures coincide—and prevents the network from chasing a potentially inaccurate external target before it can represent low–variance pockets. We then anneal toward a trusted reference E_{DMC} via

$$E_{\text{eff}}(\alpha) \;=\; \alpha\, E_{\text{DMC}} + (1-\alpha)\, \mu, \qquad \alpha \in [0,1], \tag{4.3}$$

with a smooth cosine schedule $\alpha = \alpha(t)$ that ramps from α_{start} to α_{end} over a prescribed fraction of epochs (code: `alpha_start`, `alpha_end`, `alpha_decay_frac`). Setting the `objective` to `"residual"` uses $E_{\text{eff}} = \mu$; `"energy"` uses $E_{\text{eff}} = E_{\text{DMC}}$; and `"energy_var"` uses the annealed blend above.

2. **Stochastic Reconfiguration (stage II).** After residual pretraining, we refine with SR (natural–gradient VMC). Writing $\bar{O}(\tilde{R}) = \partial_\theta \log \Psi_\theta(\tilde{R}) - \mathbb{E}_\pi[\partial_\theta \log \Psi_\theta]$ for the centred score,

$$S \;=\; \mathbb{E}_\pi\big[\bar{O}^\top \bar{O}\big], \qquad g \;=\; 2\, \mathbb{E}_\pi\big[(E_L - \mathbb{E}_\pi[E_L])\, \bar{O}\big], \tag{4.4}$$

we solve the damped normal equations

$$(S + \lambda I)\, \Delta\theta \;=\; -\, g, \tag{4.5}$$

using (preconditioned) conjugate gradients with restarts, a trust–region scale on the final step, and batchwise centering/whitening of O (code: `damping`, `max_param_step`, CG options).

This pipeline exploits the zero–variance principle for stable shaping of W_θ (and backflow) before applying the geometry–aware SR update.

4.2 Configuration–space sampling

Guiding principle (permutation invariance). In a many–body fermionic system, *which* particle indices are close is irrelevant; what matters is *how many* close pairs and at which length scales. Consequently, we can design samplers that (i) emphasize *configurational motifs* (clusters, shells, dimers, long tails) rather than specific labels, and (ii) deliberately permute particles and randomly rotate configurations to explore the space efficiently without biasing towards any ordering.

4.2.1 Stratified capped–simplex mixture

We draw collocation batches $X \in \mathbb{R}^{B\times N\times d}$ from a five–component mixture,

$$(\text{centre, tails, mixed, shells, dimers}),$$

with nonnegative weights $w \in \Delta^4$ projected to a *capped simplex* $\{\, w : \sum_k w_k = 1,\ 0 \le w_k \le 0.3 \,\}$ to enforce coverage (code: `_project_simplex_with_caps`). Each component proposes x as follows (all in physical units; the code internally scales by $a_{\text{ho}} = 1/\sqrt{\omega}$):

- **Centre:** narrow isotropic Gaussian around the trap centre (samples near the density core).
- **Tails:** wide Gaussian (probes low–density outskirts and large–r behaviour).
- **Mixed:** hybrid batches mixing narrow and moderate spreads per configuration (tests robustness to inhomogeneous spreads).
- **Shells:** points placed on K hyperspherical shells with trap–scaled radii $\{r_k\}$ and occupancy $\{q_k\}$, with small radial/tangential jitter (captures ring/shell order at weak traps).
- **Dimers:** we induce a few close pairs per configuration by sampling short interparticle offsets with random directions and log–uniform distances (enforces near–coalescence coverage without singularities).

We then apply a random in–plane rotation (for $d = 2$), and a random permutation of particle indices; both preserve the target expectations by symmetry.

Adaptive mixture and shell occupancy. Per epoch we compute simple difficulty scores and update w and q by exponentiated–gradient (EG) steps with momentum, temperature, and an exploration term; w is then projected back to the capped simplex (code: `eg_eta`, `eg_temp`, `eg_momentum`, `explore_gamma`, `prob_floor`). Mixture difficulty uses the residual $(E_L - E_{\text{eff}})^2$ per component; shell difficulty uses a proxy

$$\underbrace{V_i}_{\text{harm.+Coulomb per particle}} + \underbrace{\gamma_{g^2}\,\|\nabla_{\mathbf{x}_i}\log\Psi\|^2}_{\text{stiff gradients}} + \underbrace{\gamma_{\text{cusp}}\, r_{\min}^{-1}}_{\text{coalescence pressure}},$$

aggregated by nearest shell index (code: `g2_weight`, `cusp_gamma`, safe clip `rmin_clip`). This shifts probability mass toward underfit regions *without* collapsing coverage thanks to the caps.

Hard–example injection. We maintain a small buffer of previously challenging configurations (based on residuals) and stochastically inject them, with mild multiplicative and additive jitter, into the next batch (code: `sampler_hard_*`). This provides a curriculum–like pressure while avoiding overfitting to outliers.

Robustness tweaks. We trim extreme local–energy outliers by quantiles (default 3%) *before* forming losses/updates; we also micro–batch the evaluation to keep memory bounded, and apply gradient clipping to f_θ (and backflow) parameters.

4.3 Residual–based training (stage I)

For a batch $\{\tilde{R}_b\}_{b=1}^{B}$ we compute $E_L(\tilde{R}_b)$ and the batch mean $\mu = \frac{1}{B}\sum_b E_L(\tilde{R}_b)$. The residual loss is

$$\mathcal{L}_{\text{res}}(\theta;\alpha) = \frac{1}{B}\sum_{b=1}^{B}\big(E_L(\tilde{R}_b) - E_{\text{eff}}(\alpha)\big)^2, \qquad E_{\text{eff}}(\alpha) = \alpha\, E_{\text{DMC}} + (1-\alpha)\,\mu. \tag{4.6}$$

- **Self–target warmup ($\alpha = 0$).** Minimising $\mathbb{E}[(E_L - \mu)^2]$ is variance minimisation *under the sampling measure in use*; μ is treated as a detached constant, so no gradient flows through the batch mean. It teaches the network to produce smooth, low–variance amplitudes inside nodal pockets and reduces sensitivity to early sampler noise.
- **Anneal to reference ($\alpha\uparrow$).** As capacity grows, we blend in the external target, gently steering the mean energy while retaining the variance pressure. The cosine ramp (code: `alpha_start`, `alpha_end`, `alpha_decay_frac`) avoids abrupt objective switches.
- **Reference-target residual ($\alpha = 1$).** Stage I may end with $E_{\text{eff}} = E_{\text{DMC}}$. Despite the code name `"energy"`, this is not variational energy minimisation: it is a squared residual against a fixed scalar, and it can in principle drive $\langle E_L\rangle$ *up* toward a target the ansatz could beat. The variational optimisation is stage II.

In practice we compute $\nabla_\theta \mathcal{L}_{\text{res}}$ by autodiff, use micro–batches of size `micro_batch`, apply optional gradient clipping (code: `grad_clip`), and log per–component usage and shell radii in physical units.

Local–energy derivatives in practice. We support three Laplacian backends:

1. **Exact:** chunked second partials (reference–accurate, heavier).
2. **HVP–Hutchinson:** $\Delta \log\Psi = \mathbb{E}_v[v^\top H_{\log\Psi} v]$ with Rademacher v (fast and stable; fallback to FD if a probe is non–finite).
3. **FD–Hutchinson:** centred finite differences of directional gradients (robust in corner cases; slightly noisier).

We average a small number of probes (2–16 in ablations), and trim quantiles before forming losses.

4.4 Stochastic Reconfiguration (stage II)

Stochastic reconfiguration [26, 43] preconditions the energy gradient with the (inverse) quantum geometric tensor, and is the imaginary-time / natural-gradient step for a variational state [24, 27]. It uses the covariance form of the energy gradient

$$\partial_{\theta_k} E = 2\,\mathrm{Cov}_\pi\big(E_L, O_k\big) = 2\Big(\langle E_L O_k\rangle - \langle E_L\rangle\,\langle O_k\rangle\Big), \tag{4.7}$$

and approximates a natural–gradient step by solving $(S+\lambda I)\Delta\theta = -g$ with the quantum geometric tensor S of §2.2.5. Implementation details:

- **Centering/whitening.** We centre O per batch and whiten by the diagonal of S before CG to improve conditioning.
- **Trust region.** The final step is scaled to satisfy a maximum parameter norm (code: `max_param_step`).
- **Damping.** A small λ (code: `damping`) stabilizes low–variance directions.

Because our backflow only alters the Slater coordinates (and W_θ stays on safe features of $\tilde{R}$), the SR statistics remain well–behaved even when nodes shift.

Optimiser variants compared. Three first-order updates appear in the results, and their comparison is itself a finding (§7.2):

- **Adam** [22]: a diagonal adaptive step, used as the robust default, with separate learning rates for the Jastrow and backflow blocks.
- **Plain SR**: the natural-gradient step above with a fixed diagonal damping λ.
- **CG-SR**: the same step solved matrix-free by preconditioned conjugate gradients, with Tikhonov damping and a trust-region cap. It is most useful where S is anisotropic but still statistically resolved, and becomes unstable at weak confinement, where the sampling that builds S collapses and the empirical metric can no longer be estimated (§7.2).

Solving the SR step without forming S is what makes it affordable at $\mathcal{O}(10^5)$ parameters. The same obstacle is addressed from the other side by the minSR and kernel-formulation reductions, which exploit the fact that the number of samples, rather than the number of parameters, sets the true rank of the problem [44, 45]. In the $N \leq 6$ comparisons of §7.2, Adam and SR reach indistinguishable energies under $|\Psi|^2$ sampling; they differ in conditioning and, decisively, in the collocation regime. Whether that equivalence persists at larger N is not established there.

4.5 Sampling paradigms and the weak-form collocation objective

The alternative pipeline described in this section draws its batches from an analytic proposal rather than a Markov chain—the SR route of §4.4 samples $|\Psi_\theta|^2$ by Metropolis—so the choice of *sampling measure* must be stated explicitly: the results treat it as a variable in its own right. Two paradigms appear in this thesis. In *variational Monte Carlo* (VMC) the batch is drawn from $|\Psi_\theta|^2$ itself, usually by a Metropolis chain, so the estimators of the QGT and of the gradient share the very metric they are meant to precondition. In *collocation* the batch is drawn from a fixed analytic proposal $q(\mathbf{R})$ and reweighted to $|\Psi_\theta|^2$ by importance weights $w_i = |\Psi_\theta(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$, so that the gradient training requires no Markov chain. A Metropolis probe is used only outside the gradient loop—for periodic checkpoint selection and for the final energy certification.

For the collocation objective we use a *weak form*: the local energy E_L enters only as a reward in a score-function (REINFORCE) gradient [46],

$$\nabla_\theta \langle E \rangle = 2\,\mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[(E_L(\mathbf{R}) - b)\,\nabla_\theta \log|\Psi_\theta(\mathbf{R})|\big], \tag{4.8}$$

with a variance-reducing baseline b (a running mean of E_L), and the Laplacian inside E_L is never differentiated through. This is deliberate. Differentiating the Laplacian through a backflow-shifted determinant is the source of the nodal instability analysed in Appendix C; the weak form keeps the full local-energy signal as a reward while sidestepping that pathology. Two robustness controls stabilise the reward: local energies are clipped to a robust median/MAD location–scale band ($\sim 5\sigma$), and the most extreme rewards are quantile-trimmed before the gradient is formed.

4.5.1 Analytic proposal and importance resampling

The collocation points are drawn from an analytic mixture of isotropic Gaussians centred at the trap minimum and reweighted to $|\Psi_\theta|^2$ [16],

$$q(\mathbf{R}) = \frac{1}{K}\sum_{k=1}^{K} \mathcal{N}\big(\mathbf{R};\,0,\,\sigma_{f,k}^2/\omega\; I_{Nd}\big), \quad \sigma_f \in \{0.8, 1.3, 2.0\}, \qquad w_i = \frac{|\Psi_\theta(\mathbf{R}_i)|^2}{q(\mathbf{R}_i)}, \tag{4.9}$$

where the $1/\omega$ scaling ties the proposal width to the oscillator length $\ell = 1/\sqrt{\omega}$ and the three components cover core, bulk, and tails. A batch of n_{coll} points is obtained by multinomial resampling on $p_i = w_i/\sum_j w_j$ from $m\,n_{\mathrm{coll}}$ candidates ($m \approx 8$–16). Reweighting quality is tracked by the effective sample size $\mathrm{ESS} = (\sum_i w_i)^2/\sum_i w_i^2$ and, as a heavier-tailed diagnostic, the Pareto shape parameter $\hat{k}$ of the weights [47]. The second is not a second opinion on the first. Importance weights are heavy-tailed, and $\hat{k}$ estimates the tail index of the generalised Pareto distribution fitted to the largest of them. The thresholds are statements about the moments of the *fitted* tail. For $\hat{k} < 1/2$ the weights behave as though the variance were finite and the reweighted estimator obeys a central limit theorem. Between 1/2 and 1 the variance behaves as infinite though the mean does not, and convergence is slow enough that $\hat{k} \approx 0.7$ is the practical limit of trustworthiness. Above 1 the fitted tail has no finite mean.

That last statement needs care, because in this setting the underlying ratio does have a finite mean by construction: $w = |\Psi_\theta|^2/q$ with q a proper density gives $\mathbb{E}_q[w] = \int |\Psi_\theta|^2 < \infty$ whenever Ψ_θ is square-integrable, so asymptotically $\hat{k} < 1$. A measured $\hat{k} > 1$ is therefore a *finite-sample* diagnostic: it says the sample is deep in a pre-asymptotic regime where the realised estimator behaves as if no finite mean existed, not that the expectation fails to exist [47]. We use it in that sense throughout. An ESS can look survivable while $\hat{k}$ says the estimator has no variance to speak of, which is why we track both. Both degrade when q ceases to overlap $|\Psi_\theta|^2$, and that degradation is the failure mode which bounds the paradigm at large N and weak ω (Chapter 7, §7.3.4). Three controls manage the tails: weight tempering ($w \to w^\alpha$, $\alpha < 1$), log-weight clipping at an upper quantile, and ESS-adaptive oversampling that raises m when the achieved ESS falls below a floor. The first two are deliberate bias–variance trades: they change the distribution the reward is averaged over, so the reweighted energy readout is no longer an estimate of $\langle E_L \rangle_{|\Psi_\theta|^2}$. That is why the final certification is always a separate evaluation under $|\Psi_\theta|^2$ (§7.2). An instability-rollback reverts to the last good checkpoint and decays the learning rate if the energy jumps beyond a threshold.

4.5.2 Physics-informed proposal for the Wigner regime

The origin-centred proposal (4.9) places its mass where a Wigner molecule has almost none: at strong correlation the density collapses onto a shell at the classical ring radius r_0, and $|\Psi_\theta|^2$ has essentially no support near the origin, so the ESS collapses. We therefore replace q by a proposal built from the Wigner shell itself—a radial Gaussian centred at r_0 (set by balancing confinement against Coulomb repulsion) times an angular distribution over the ring, so that proposed configurations resemble electrons spread around the ring rather than clustered at the centre. This restores a usable ESS and carries MCMC-free training to $\omega \approx 0.0035$ (Chapter 7, §7.4), far past where the naive proposal collapses. Angular crystallisation of the Wigner ring sets the next limit and is addressed there by a Dirichlet model of the angular gaps; what bounds the route below $\omega = 0.0035$ is a separate, non-sampling failure of transfer between confinements. Difficult confinements are reached by *cascade warm-starting*: parameters converged at one ω initialise the next, following $\omega : 1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001$. Whether the two paradigms reach the same *state*, and where they diverge, is one of the questions of Chapter 7.

4.5.3 Ansatz families used in the collocation programme

Three ansatz families are trained under this objective in Chapter 7, all on the same Slater–Jastrow core $\Psi(\mathbf{R}) = \det[\phi_i(\mathbf{r}_j + \Delta\mathbf{r}_j)]\, e^{J(\mathbf{R})}$ and differing only in what dresses it. They are deliberately smaller than the production networks of §3.6, because this programme is limited by gradient noise and memory rather than by capacity.

Jastrow-only. The correlator alone at ∼25k parameters, in its message-passing form (`CTNNJastrowVCycle`), acting on node features (single-particle orbital values) and edge features (soft-core pair distances). Fastest to train and the most robust; used both standalone and as the initialiser for backflow runs.

Backflow + Jastrow. Adds the copresheaf displacement network (`CTNNBackflowNet`), ∼49k parameters in total at $h_{\rm bf}$=128. Its edge tensor scales as $\mathcal{O}(N^2 h_{\rm bf})$, which is the memory bottleneck at large N: at N=20 the hidden width must be cut to $h_{\rm bf} \in \{48, 64\}$.

Neural Pfaffian. Replaces the determinant with $\mathrm{Pf}[A_{ij}]$, a more general antisymmetric form admitting richer nodal topology [48].

4.6 Units, scaling, and batch construction

All features are computed in trap units $\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$ so that length scales and gates (e.g. the short–range gate in the NQS and backflow) behave consistently across ω. The sampler internally widens/narrows Gaussians and shell radii using $a_{\rm ho} = 1/\sqrt{\omega}$ in a fixed, explicit way, so that the five components cover comparable *dimensionless* regimes for $\omega \in [0.01, 1]$. Proposal candidates are drawn independently, so the batch carries no serial Markov-chain autocorrelation; the retained batch is not i.i.d. in the strict sense, since multinomial resampling can repeat candidates and the hard-example buffer reinjects earlier ones. What the design buys is the absence of a mixing time, not statistical independence.

4.7 The two workflows

Two training routes have now been described and they are used for different purposes throughout the thesis, so it is worth setting them side by side before the diagnostics chapter. Both start from the same ansatz and end at an energy evaluated under $|\Psi_\theta|^2$; they differ in the measure that defines the loss in between.

SR–VMC (§4.3–§4.4)	**Collocation (§4.5)**
stratified mixture $\to$ residual pretraining	pretrained checkpoint
$\downarrow$	$\downarrow$
$\|\Psi_\theta\|^2$ sampling (Metropolis)	analytic proposal q, importance reweighting
$\downarrow$	$\downarrow$
SR step $(S + \lambda I)\Delta\theta = -g$	weak-form REINFORCE gradient (Adam)
$\downarrow$	$\downarrow$
energy under $\|\Psi_\theta\|^2$	Metropolis probe for checkpoint choice, then energy under $\|\Psi_\theta\|^2$

The right-hand route places no Markov chain in any gradient; the probe that selects its checkpoints sits outside the update loop entirely. The steps of the left-hand route, which carries the ground-state energies of §6.1, are:

1. **Sample collocation set** X from the stratified capped–simplex mixture (centre, tails, mixed, shells, dimers), then apply rotation and permutation; optionally inject hard examples.
2. **Compute** E_L using the selected Laplacian backend; trim outliers.
3. **Stage I (residual).** Minimize $\frac{1}{B}\sum_b (E_L(\tilde{R}_b) - E_{\text{eff}}(\alpha))^2$ with α ramped from 0 to α_{end}; update mixture/shell weights via EG.
4. **Stage II (SR).** Form S and g, solve $(S + \lambda I)\Delta\theta = -g$, apply a trust–region step.

This procedure keeps the early dynamics variance–dominated and sampler–aware, then transitions to a geometry–aware natural–gradient refinement. Empirically it yields stable convergence across (N, ω), including weak traps where shell/dimer coverage and permutation–respecting stratification are essential.

4.8 Statistical uncertainties

Every energy and observable we report carries a statistical error, and two distinct sources contribute. Within a single evaluation, successive MCMC samples are autocorrelated, so a naive standard error over B samples understates the sampling uncertainty; this is what blocking addresses. Across training runs, the optimisation is itself stochastic and lands on different states from different seeds; blocking says nothing about that, and it is assessed separately by repeating runs at independent seeds wherever the claim depends on it (§7.3.4, §7.1.1). We therefore estimate the error on a scalar mean $\bar{A} = \frac{1}{B}\sum_i A(\mathbf{R}_i)$ by *blocking*: successive samples are averaged into blocks of growing size and the plateau of the block standard error as the block size exceeds the autocorrelation time is taken as the uncertainty [49, 50]. For the final ground-state energy this is the dominant reported error; a parenthetical digit, e.g. 155.8738(7), denotes one standard error on the last quoted figure.

For quantities that are *nonlinear* functions of sample statistics—participation ratios, effective ranks, PCA cosines, and the coupling diagnostics of Chapter 6—a blocked standard error is not available in closed form, and we use a *block bootstrap* instead: blocks (of the plateau size above) are resampled with replacement, the statistic is recomputed on each resample, and its spread across resamples is the reported error (§5.2.9). Where a number is quoted as a seed average, we additionally report the spread across independent random seeds, which captures optimisation—not just sampling—variability; the two are distinguished in the captions. Reference energies drawn from the literature carry the uncertainties quoted by their sources.

Chapter 5

Analysis

This chapter describes *what* we measure on fresh $|\Psi_\theta|^2$ samples, *how* we compute the diagnostics, and *why* each quantity is informative. Results and numbers are presented in Chapter 6.

5.1 Structural diagnostics and Wigner markers

Unless stated otherwise, curves are accumulated in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to compare shapes across ω, while all headline distances are reported in *Bohr* (physical units $\mathbf{x}$). Production chains are thinned; we split samples into halves for stability checks (acceptance window, JSD between halves, and reproducibility under restarts). Using trap units to compare shapes across confinements is standard in parabolic dots; see the overview in [10].

5.1.1 Pair distribution and radial statistics

We estimate an *effective* pair distribution—a shell-area-corrected pair-distance density—in trap units. For a finite trapped dot there is no homogeneous-system normalisation, because the one-body density varies strongly across the cloud, so A_{eff} below is a finite-dot convention rather than a thermodynamic volume and $g(r)$ does not tend to unity at large r. Peak positions and relative profiles, which is what the analysis uses, are insensitive to that choice; absolute amplitudes are not.

$$g(r_\alpha) \;=\; \frac{A_{\text{eff}}}{N(N-1)}\,\frac{\big\langle \sum_{i<j} \mathbf{1}\{r_{ij} \in [r_\alpha^-, r_\alpha^+)\}\big\rangle}{2\pi r_\alpha\,\Delta r_\alpha}, \qquad r_{ij} = \|\mathbf{y}_i - \mathbf{y}_j\|_2, \tag{5.1}$$

with $A_{\text{eff}} = \pi R_{0.95}^2$ and $R_{0.95}$ the 95% quantile of single-particle radii in trap units. This $g(r)$-based structural analysis follows established practice in mesoscopic 2D electron systems and quantum dots [7, 10, 12]. The pair-distance probability is therefore denoted $P_{\text{pair}}(r_{ij}) \propto r_{ij}^{d-1} g(r_{ij})$. It is used for pair-structure diagnostics, distinct from the one-body radial probability $P_1(r) \propto r\rho_1(r)$, from which we report the headline mode r_{mode}, mean $\bar{r}$, standard deviation σ_r, and FWHM (all in Bohr). A compact one-body width-to-scale marker is

$$\gamma \;=\; \sigma_r / r_{\text{mode}}, \tag{5.2}$$

which tracks the width of P_1 relative to its modal radius. It is interpreted alongside shell separation and pair/angular diagnostics, rather than as a universal monotone Wigner marker [7, 12].

5.1.2 Automatic shell detection and topologies

For each frame we sort the single-particle radii $s_1 \le \cdots \le s_N$ (trap units), compute gaps $g_k = s_{k+1} - s_k$, and flag the frame as *two-shell* if the largest gap exceeds a robust threshold:

$$\max_k g_k \;\ge\; \tau\,\text{median}(g_k), \qquad \tau \in [2.5, 3.5]\ (\text{default } 3.0). \tag{5.3}$$

The cut radius is the midpoint of the maximal gap; the inner multiplicity n_{in} defines the topology $(n_{\text{in}}, N - n_{\text{in}})$.

More than two shells. At N=20 and weak confinement a single cut is not enough, and the detector is applied recursively. Having placed one cut, we re-apply the same robust-gap test *within* each resulting group, accepting a further cut only if that group contains at least three particles and its own maximal gap again exceeds $\tau\,\mathrm{median}(g_k)$ computed over the whole frame—so the threshold is global and a shell is never split by a gap that is small on the scale of the configuration. Recursion stops when no group qualifies, which for the systems here never yields more than three shells. A frame is labelled by the resulting occupancy vector $(n_1, \ldots, n_S)$ ordered outward, so that the two-shell case above is the S=2 member of the same family; the N=20 deep-Wigner topology $(1, 7, 12)$ is an S=3 assignment with cuts after the first and eighth particle. Topology fractions and their robustness under τ are reported exactly as in the two-shell case. Shelling and discrete ring occupancies are well known in parabolic Coulomb clusters and quantum dots [8–10]; our detector provides a reproducible, data-driven assignment in that spirit. We report: (i) the fraction of two-shell frames, (ii) the histogram over n_{in} among two-shell frames, and (iii) for selected topologies (e.g. $(1, 5)$ or $(3, 9)$) the fraction among *all* frames.

Shell-resolved reconstruction of $g(r)$. On two-shell frames we decompose pairs into II, IO, and OO and accumulate histograms $g_{\mathrm{II}}, g_{\mathrm{IO}}, g_{\mathrm{OO}}$. With empirical weights $w_{\mathrm{II}}, w_{\mathrm{IO}}, w_{\mathrm{OO}}$ (combinatorial counts) we reconstruct $\tilde{g}(r) = w_{\mathrm{II}}g_{\mathrm{II}} + w_{\mathrm{IO}}g_{\mathrm{IO}} + w_{\mathrm{OO}}g_{\mathrm{OO}}$ and report $\cos\angle(g, \tilde{g})$. Values ≈ 1 indicate that the observed $g(r)$ shape is explained by shell geometry rather than sampling artifacts, in line with shell-based descriptions in [8–10].

5.1.3 Orientational order and “ring stiffness”

On each detected ring (inner or outer) we take the particle angles $\{\phi_k\}_{k=1}^{n}$ in the lab frame. No rotation is removed at this stage: the modulus $|\Phi_m|$ below is already invariant under a global rotation, and the registration of Eq. (5.7) is applied only where the *phase* is needed, which is the cross-shell comparison. Bond-orientational order is

$$\Phi_m \;=\; \frac{1}{n}\sum_{k=1}^{n} e^{\,\mathrm{i}m\phi_k}, \qquad m = n, \tag{5.4}$$

so that $|\Phi_m| \in [0, 1]$ measures n-fold orientational order (e.g. m=5 for $(1, 5)$, m=9 for the outer ring of $(3, 9)$), analogous to standard bond–orientational order parameters used in 2D Coulomb/Yukawa crystals [11]. As a complementary, dimensionless *angular Lindemann* number we use

$$\lambda_\phi \;=\; \frac{\mathrm{std}(\Delta\phi)}{\mathbb{E}[\Delta\phi]}, \qquad \Delta\phi \;=\; \text{nearest-neighbour angular spacing on the ring.} \tag{5.5}$$

Smaller λ_ϕ indicates a stiffer (more Wigner-like) ring. This is the quantity abbreviated “Lind” in Chapter 6; where a shell index is attached, as in λ_{in} and λ_{out}, it is the same estimator evaluated on that shell alone.

Registered and lab-frame order. Two further scalars are used for the multi-shell cases, and both are needed because $|\Phi_{n_s}|$ is defined shell by shell while the question at N=20 is whether the *molecule* is ordered. For a frame with shells $s = 1, \ldots, S$ and occupancies n_s we define a configuration-level order parameter in two frames. In the *lab* frame,

$$C_{\mathrm{global}} \;=\; \Big|\, \Big\langle \tfrac{1}{S'} \textstyle\sum_{s:\, n_s \geq 3} \Phi_{n_s} \Big\rangle_{\mathrm{frames}} \Big|, \tag{5.6}$$

the modulus of the *frame-averaged* complex order parameter, averaged over the S' shells carrying at least three particles. Because a freely rotating molecule visits all orientations, the complex phases cancel and $C_{\mathrm{global}} \to 0$ however ordered the molecule is internally: it measures *pinning* to the lab frame, not order. In the *co-rotating* frame each configuration is first registered by an estimated physical rotation. Because a rigid rotation by α sends $\Phi_{n_s} \mapsto e^{in_s\alpha}\Phi_{n_s}$, the phase of a shell's order parameter advances at n_s times the rotation angle, and the angle itself must be recovered from the outermost shell as $\hat{\alpha} = \arg\Phi_{n_S}/n_S$ (modulo $2\pi/n_S$) before it can be applied to the others:

$$\tilde{C}_{\mathrm{global}} \;=\; \Big|\, \Big\langle \tfrac{1}{S'} \textstyle\sum_{s:\, n_s \geq 3} \Phi_{n_s}\, e^{-in_s\hat{\alpha}} \Big\rangle_{\mathrm{frames}} \Big|, \qquad \hat{\alpha} = \frac{\arg\Phi_{n_S}}{n_S}. \tag{5.7}$$

Dividing the outer phase by n_S is essential: removing $\arg\Phi_{n_S}$ from every shell alike would only correspond to a common spatial rotation when all occupancies coincide, and would otherwise mix shells that rotate together into an apparent misalignment. This retains the *relative* angular arrangement of the shells while discarding the global orientation, so a rigid molecule tumbling freely gives $\tilde{C}_{\text{global}}$ near its single-frame value and C_{global} near zero. The pair $(\tilde{C}_{\text{global}}, C_{\text{global}})$ is therefore the direct test of the rotating Wigner-molecule picture, and it is reported that way in §6.2.5. Finally, $L_{n,\text{reg}}$ denotes the angular Lindemann number λ_ϕ evaluated on the shell of occupancy n after the same registration.

Two of these quantities are insensitive to the registration convention and one is not, and the distinction bounds what may be concluded from each. The shellwise modulus $|\Phi_{n_s}|$ and the nearest-neighbour spacing statistics behind λ_ϕ are invariant under any global rotation, so they carry no dependence on how $\hat{\alpha}$ is estimated. The *cross-shell* quantity $\tilde{C}_{\text{global}}$ does depend on it, since its whole content is the relative alignment between shells of different occupancy. The N=20 values reported in §6.2.5 were produced by an earlier analysis pass whose code has not been retained, so we cannot confirm which convention it used; the per-shell numbers there are unaffected, but the cross-shell $\tilde{C}_{\text{global}}$ should be recomputed under Eq. (5.7) before it is relied on quantitatively.

The random-angle null. None of these is interpretable without a baseline. For n particles at independent uniform angles, Φ_n is a two-dimensional random walk of n unit steps, so $\langle|\Phi_n|\rangle = \sqrt{\pi}/(2\sqrt{n}) \approx 0.886/\sqrt{n}$ and $\langle|\Phi_n|^2\rangle = 1/n$. A five-particle ring with no angular order whatever returns $|\Phi_5| \approx 0.40$. We therefore report $|\Phi_n|$ together with its ratio to this null wherever the ordering claim depends on it (Appendix F.2). This is a Lindemann–type orientational metric tailored to finite rings, motivated by the use of Lindemann criteria to diagnose intershell rotational (angular) melting in mesoscopic Coulomb clusters and dots [9, 12].

5.1.4 Scaling check for $N = 2$

To connect with the classical strong-coupling limit we fit

$$r_{\text{mode}}(\omega) \;=\; C\,\omega^{\alpha} \tag{5.8}$$

by linear regression in $\big(\log\omega, \log r_{\text{mode}}\big)$ and compare the slope with the classical reference $\alpha_{\text{cl}} = -2/3$ (two charges in a harmonic trap at strong coupling; see, e.g., the reviews in [10]). Approach toward α_{cl} as $\omega\downarrow$ is one of our Wigner crossover indicators, consistent with the crossover phenomenology reported in [7, 12].

5.1.5 Reporting and stability checks

For all scalars and histograms we report standard errors calculated over thinned blocks or frames. We verify: (i) acceptance in the target window, (ii) Jensen–Shannon divergence and ℓ_2 distance between halves of production (stationarity), (iii) robustness of topology fractions and $|\Phi_m|$ under $\tau \in [2.5, 3.5]$ and mild pre-smoothing of radii, and (iv) reproducibility of $g(r)$ and $P(r)$ under independent sampler restarts. All diagnostics operate directly on saved bundles $\{\mathbf{X}_{\text{Bohr}},\, r,\, g(r)\}$.

5.2 Representation analysis

We probe what the networks learn by evaluating fixed, trained models on fresh $|\Psi_\theta|^2$ samples. Unless stated otherwise, features are computed in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to match the correlator's internal scaling. All estimates use thinned MCMC frames, split into two equal blocks A and B; PCA fits on A and reports on B (and vice versa) to avoid circularity. We use `float64` throughout.

Feature taps. Let $z_\theta \in \mathbb{R}^D$ denote the concatenated correlator representation

$$z_\theta = \big[\overline{\phi} \mid \overline{\psi} \mid \mathbf{g}\big],$$

where $\overline{\phi}$ are per-particle summaries, $\overline{\psi}$ are pairwise summaries, and $\mathbf{g}$ are global/extras. The readout ρ_θ is a two-layer MLP, not a linear map, so

$$f_{\text{base}}(z) = \rho_\theta(z), \qquad f = f_{\text{base}} + f_{\text{cusp}}, \tag{5.9}$$

and the ablations below re-evaluate ρ_θ on modified inputs rather than manipulating a weight vector. Backflow returns a displacement field $\Delta x \in \mathbb{R}^{N\times d}$ that enters only the Slater determinant; the correlator and cusp are evaluated on the unshifted $\tilde{R}$ (§3.5). The coupling diagnostics below additionally probe the correlator on the shifted coordinates as a counterfactual sensitivity.

5.2.1 Preprocessing and PCA

Given a sample matrix $Z \in \mathbb{R}^{K\times D}$ (rows are frames), we *mean-centre* the columns, $Z_c = Z - \mu$, and take the SVD of Z_c to obtain orthonormal loadings $U = [u_1, \dots, u_D]$; scores are $S = Z_c U$. Centring but not rescaling is deliberate: the readout ρ_θ was trained on the raw features, so every ablation below is performed in raw coordinates and the mean is restored before ρ_θ is applied. Where a per-column standardisation is used—for the branch-power decomposition of §5.2.4, which compares loadings across branches carrying different natural scales—it is stated explicitly. PCA is fitted on block A and reported on the held-out block B unless noted.

5.2.2 Entropy effective rank

Let $\{\lambda_i\}_{i=1}^{D}$ be the eigenvalues of $\mathrm{Cov}(Z_c)$ (on the held-out block). With $p_i = \lambda_i / \sum_j \lambda_j$, the *entropy effective rank* is

$$r_{\text{eff}}(Z) \;=\; \exp\Big(-\sum_{i=1}^{D} p_i \log p_i\Big),$$

which equals 1 for perfectly one-dimensional representations and increases with spectral spread. Two spectral counts appear in this thesis and they are not the same number. $r_{\text{eff}} = \exp(H_1)$ above is the exponential of the Shannon entropy of the spectrum; the *participation ratio* $d_{\text{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ of §2.2.5 is the exponential of the Rényi-2 entropy. Since Rényi entropies decrease in their order, $d_{\text{eff}} \le r_{\text{eff}}$ always, with equality only for a flat spectrum. Appendix E reports r_{eff} for feature covariances and Chapter 7 reports d_{eff} for the tangent metric and for the displacement field, so the same object can carry two slightly different counts in the two chapters; each table names which is used.

5.2.3 PC ablations of the head

To test how many latent axes the readout truly uses, we project the centred features onto the top-k PCs and restore the mean before re-evaluating the trained readout *without refitting*,

$$\Pi_k = \sum_{i=1}^{k} u_i u_i^\top, \qquad z^{(k)} = \mu + \Pi_k\,(z-\mu), \qquad \hat{f}^{(k)} = \rho_\theta\big(z^{(k)}\big). \tag{5.10}$$

Restoring μ matters: ρ_θ is a nonlinear map trained on uncentred inputs, so feeding it a centred or standardised vector would evaluate it off its training domain and the resulting error would measure that displacement rather than the loss of PC content. As a null, the same projection is repeated onto k *random* orthonormal directions and the two error curves are compared; a top-k curve that is not below the random-k curve would mean the leading PCs carry no privileged information for the readout. We report a *normalised* error on block B,

$$\text{rel-MAE}(k) \;=\; \frac{\mathbb{E}_B\,|\,\hat{f}^{(k)} - f_{\text{base}}\,|}{\mathrm{std}_B(f_{\text{base}})},$$

together with the learning curve in k.

5.2.4 PC1 block power decomposition

To attribute the leading axis to branches, we partition u_1 conformably with $z = [\overline{\phi}\,|\,\overline{\psi}\,|\,\mathbf{g}]$ and report

$$\mathrm{power}(\overline{\phi}) = \frac{\|u_{1,\overline{\phi}}\|_2^2}{\|u_1\|_2^2}, \quad \mathrm{power}(\overline{\psi}) = \frac{\|u_{1,\overline{\psi}}\|_2^2}{\|u_1\|_2^2}, \quad \mathrm{power}(\mathbf{g}) = \frac{\|u_{1,\mathbf{g}}\|_2^2}{\|u_1\|_2^2}.$$

These are computed on standardised features so that branches carrying different natural scales are comparable. One confound remains and should be kept in mind when reading them: a branch holding d_b

of the D coordinates would already carry about d_b/D of the squared loading mass under an isotropic random direction, so a large share partly reflects a wide branch. The shares below are therefore read as *relative* shifts across ω at fixed architecture—where the branch widths are constant and the confound cancels—and not as absolute statements that one branch dominates.

5.2.5 Linear probes from PCs to coarse physics

We regress interpretable summaries per configuration on the first k PC scores $S^{(k)}$ using ridge regression (tiny penalty $\alpha = 10^{-6}$):

$$y \approx S^{(k)}\beta, \qquad y \in \{\bar{r},\ \mathrm{Var}(r),\ \Pr(r < r_0),\ \text{shell contrast}\}.$$

We report R^2 on the held-out block; r_0 is a fixed near-contact cutoff in trap units.

5.2.6 Near-field gradient concentration (residual head)

For each frame we compute the minimum pair distance $r_{\min}$ (trap units) and the gradient of the residual head with respect to particle coordinates, $\nabla f_{\text{base}}(\mathbf{y})$. The quantity reported is an *enrichment factor*: the mean gradient power among the closest-contact frames divided by the mean over all frames,

$$\mathcal{E}(q) \;=\; \frac{\mathbb{E}\big[\,\|\nabla f_{\text{base}}\|_2^2 \,\big|\, r_{\min} \le Q_q\,\big]}{\mathbb{E}\big[\|\nabla f_{\text{base}}\|_2^2\big]}, \tag{5.11}$$

with Q_q the q-quantile of $r_{\min}$. This is a ratio of a *conditional* mean to an unconditional one and is therefore unbounded above, unlike an indicator-weighted mass fraction, which could not exceed unity. $\mathcal{E} \approx 1$ means gradient power is spread in proportion to frame count, so the analytic cusp is handling most short-range structure; $\mathcal{E} \gg 1$ means residual stiffness is concentrated near contact.

5.2.7 Backflow geometry and energy-locality shares

Field spectrum. We vectorize the backflow displacement per frame, $\delta = \mathrm{vec}(\Delta x) \in \mathbb{R}^{Nd}$, standardize across frames, and compute the entropy effective rank $r_{\text{eff}}(\Delta x)$ of Sec. 5.2.2. Chapter 7 reports the participation ratio of the same spectrum instead, which is the smaller of the two counts; the ceiling $2N - d$ set by the zero-mean projection (3.19) is common to both.

Where backflow acts. On the same thinned frames we evaluate local energies with and without backflow, $E_{\text{loc}}^{\text{BF}}$ and $E_{\text{loc}}^{\text{noBF}}$, using the *same* coordinates. We form the signed correction $\Delta E_{\text{loc}} = E_{\text{loc}}^{\text{BF}} - E_{\text{loc}}^{\text{noBF}}$ and report near-field shares at quantile q of $r_{\min}$:

$$\text{NF-}\Delta E(q) \;=\; \frac{\mathbb{E}\big[|\Delta E_{\text{loc}}|\,\mathbf{1}\{r_{\min} \le Q_q\}\big]}{\mathbb{E}\big[|\Delta E_{\text{loc}}|\big]}.$$

(Using absolute values isolates magnitude; signed averages are also available.)

5.2.8 Alignment of correlator manifolds (noBF vs BF)

We collect two standardized feature matrices $\tilde{Z}^{\text{noBF}}$ and $\tilde{Z}^{\text{BF}}$ on identical frames (each standardized by its own μ, σ), compute their PC1 loadings u_1^{noBF} and u_1^{BF}, and report the cosine similarity

$$\cos\big(u_1^{\text{noBF}}, u_1^{\text{BF}}\big) \;=\; \frac{\langle u_1^{\text{noBF}},\, u_1^{\text{BF}}\rangle}{\|u_1^{\text{noBF}}\|_2\,\|u_1^{\text{BF}}\|_2}.$$

Cosines near 1 indicate that backflow acts as a structured reweighting along the same dominant axis, rather than rotating the correlator's manifold.

Centred kernel alignment. The PC1 cosine compares one direction. To compare the two feature *geometries* as a whole we use linear centred kernel alignment [51]. With $\tilde{Z}^{\mathrm{noBF}}, \tilde{Z}^{\mathrm{BF}} \in \mathbb{R}^{K\times D}$ column-centred on the same K frames,

$$\mathrm{CKA}\big(\tilde{Z}^{\mathrm{noBF}}, \tilde{Z}^{\mathrm{BF}}\big) \;=\; \frac{\big\|\,(\tilde{Z}^{\mathrm{noBF}})^{\top}\tilde{Z}^{\mathrm{BF}}\,\big\|_F^2}{\big\|(\tilde{Z}^{\mathrm{noBF}})^{\top}\tilde{Z}^{\mathrm{noBF}}\big\|_F\,\big\|(\tilde{Z}^{\mathrm{BF}})^{\top}\tilde{Z}^{\mathrm{BF}}\big\|_F} \;\in[0,1], \tag{5.12}$$

which is invariant to orthogonal transformations and to isotropic scaling of either representation, and equals 1 only when the two Gram matrices are proportional. It is the quantity reported in Table E.2. Two cautions attach to it. CKA compares the *geometry of the feature cloud*, so a high value says the correlator's representation is unchanged by the presence of backflow—not that the backflow occupies the same subspace, and not that the two components are functionally independent. And CKA is known to saturate near 1 whenever both representations are dominated by a few shared high-variance directions, which is exactly the regime here (§5.2.2); values above 0.99 should therefore be read as "no detectable reorganisation" rather than as a quantitative distance.

5.2.9 Uncertainty and stability

For scalars we report standard errors over blocks. For PCA-based quantities and cosines we use a block bootstrap (resampling blocks with replacement) to obtain 68%/95% CIs. Stability checks include: (i) swapping train/held-out blocks for PCA, (ii) varying the near-field quantiles $q \in \{1,5,10\}\%$, (iii) re-running analyses on independent sampler restarts.

Reproducibility package. All diagnostics consume the saved bundles $\{\mathbf{X}_{\mathrm{Bohr}}, r, g(r)\}$ and the model snapshot, and emit JSON/CSV tables with r_{eff}, rel-MAE(k) curves, block-power shares, R^2 probes, near-field shares, $r_{\mathrm{eff}}(\Delta x)$, and noBF/BF cosines. As in Sec. 5.1, we log JSD/ℓ_2 splits to confirm stationarity.

5.3 Tangent-kernel diagnostics

The representation diagnostics above describe the ansatz's *features*; a second family, developed with the geometry of §2.2.5, describes its *tangent space*, and it is these that carry the architecture, optimiser, and paradigm comparisons of Chapter 7. From a batch we assemble the centred score matrix $O_{ki} = \partial_{\theta_i}\log|\Psi_\theta(\mathbf{x}_k)|$ and form the quantum geometric tensor $S = O^{\top}O/B$ from the *centred* O (or, when $B < P$, the tangent kernel $K = OO^{\top}$, whose nonzero spectrum equals that of S up to the overall factor B). From the spectrum $\{\lambda_a\}$ we report the effective dimension $d_{\mathrm{eff}} = (\sum_a \lambda_a)^2/\sum_a \lambda_a^2$ and the condition number $\kappa(S) = \lambda_{\max}/\lambda_{\min}$. Cross-architecture comparisons of d_{eff} are made on a *common probe set*—all models scored on the same configurations, not on each model's own $|\Psi|^2$—so that the comparison reflects the ansatz and not its sampling measure, and we report only participation ratios that have converged in the batch size.

A physical-operator dictionary. Counting tangent directions says how many; naming them says which. We therefore project the leading tangent modes onto a small, fixed dictionary of one- and two-body operators chosen before the fits were run, each written as a scalar function of a configuration whose parameter-space image is obtained by the same score construction as O:

$$\begin{aligned}
&\mathcal{D}_1 = \textstyle\sum_i \tilde{r}_i^2 && \text{breathing / monopole, the generator of } \partial_\omega;\\
&\mathcal{D}_2 = \textstyle\sum_i \tilde{r}_i^2\cos 2\theta_i && \text{quadrupole, the lowest shape distortion;}\\
&\mathcal{D}_3 = \textstyle\sum_{i<j} e^{-\tilde{r}_{ij}^2/2\sigma_h^2} && \text{correlation hole, with } \sigma_h \text{ the fitted hole width;}\\
&\mathcal{D}_4 = \textstyle\sum_{i<j} \tilde{r}_{ij} && \text{pair-linear, the leading long-range pair response.}
\end{aligned} \tag{5.13}$$

Writing $D \in \mathbb{R}^{B\times 4}$ for the centred matrix of these four functions on the probe batch and $v_a \in \mathbb{R}^B$ for the ath tangent-kernel eigenvector, the *naming* score of a mode is the coefficient of determination of its least-squares regression on the dictionary,

$$R_a^2 \;=\; 1 - \frac{\|(I-P_D)\,v_a\|^2}{\|v_a\|^2}, \qquad P_D = D(D^{\top}D)^{-1}D^{\top}, \tag{5.14}$$

and the *dictionary-orthogonal tangent mass* is the eigenvalue-weighted share of the leading A modes lying outside that span,

$$M_\perp \;=\; \frac{\sum_{a\le A} \lambda_a \,\|(I-P_D)v_a\|^2}{\sum_{a\le A}\lambda_a}, \qquad A=10. \tag{5.15}$$

Finally, *correlation-hole capture* is R^2 of Eq. (5.14) computed against the single operator $\mathcal{D}_3$, i.e. the share of the leading mode explained by the hole channel alone.

The dictionary is a *chosen, non-exhaustive* probe, and nothing here establishes that it spans the physically meaningful directions. A large $M_\perp$ means the tangent space is poorly accounted for by these four operators; it does not mean the remaining directions are unphysical. What the comparison licenses is a relative statement between two architectures scored against the same dictionary on the same probe batch, and that is how Chapter 7 uses it.

Reporting the condition number. $\kappa(S)=\lambda_{\max}/\lambda_{\min}$ is quoted with two caveats that its definition forces. Our batches have $B<P$, so S is rank-deficient by construction and $\lambda_{\min}$ is the smallest *sample-supported* eigenvalue rather than an intrinsic one; the value of κ therefore depends on B and is a property of the empirical metric the optimiser actually sees, not of the variational manifold. That is the relevant object for the question we ask of it—whether preconditioning by S^{-1} helps—but it means κ may not be compared across batch sizes, and we hold B fixed wherever we compare it.

To ask whether two trained models are the *same* state rather than merely equal in energy, we use a symmetric, importance-sampled squared overlap,

$$\mathcal{S}^2(A,B)=\big\langle \Psi_B/\Psi_A\big\rangle_{|\Psi_A|^2}\,\big\langle \Psi_A/\Psi_B\big\rangle_{|\Psi_B|^2},$$

each factor evaluated by a log-sum-exp mean over samples of the corresponding state. It equals 1 for identical states, and on the analytic two-electron ground state the trained ansatz reproduces $\mathcal{S}^2=1.000000$, which anchors the tooling.

One convention must be stated, because the ratio Ψ_B/Ψ_A is signed for a fermionic state. We evaluate the ratios from $\log|\Psi|$, so the estimator is strictly a *modulus* overlap, $\langle|\Psi_B/\Psi_A|\rangle\langle|\Psi_A/\Psi_B|\rangle$. For the comparisons made here this is an accurate proxy for the signed overlap: both states share the same Slater reference, and the correlator and cusp enter as a strictly positive exponential, so the sign of Ψ is fixed by the determinant and the two states agree in sign except on the measure-suppressed set where their backflows carry a configuration across a node differently. The direction of the bound matters, and it is worth being explicit about which claim it protects. The modulus overlap is an *upper* bound on the signed one, so a reported $\mathcal{S}^2\to 0$ is conservative—the states are at least that different—while a reported $\mathcal{S}^2\approx 1$ is *not* by itself evidence that they agree in sign. For the $\mathcal{S}^2\approx 1$ readings the weight is carried by the structural argument above (shared Slater reference, strictly positive correlator), not by the bound. These quantities are properties of the state and so are unaffected by how correlation is divided between the Jastrow and the backflow. The energetic price of ablating a component is *not* in that class: two parameterisations that divide correlation differently will report different ablation costs, so it is a reproducible statement about a fixed model and training protocol rather than a state invariant (§7.5). The architectural conclusions of Chapter 7 rest on the state-level quantities first and on ablation as a controlled intervention second, and on the gauge-dependent component ranks of the previous section not at all.

Part IV

Results

The argument of Part IV

The two results chapters run a loop between solver and physics.

We first establish that the trained states reproduce every external energy benchmark that exists for this system. That licenses using them as instruments rather than defending them as answers. Read for the physics they contain, they show a crossover from compact dot to finite Wigner molecule that is *staged*: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them.

We then turn from the state to the solver. Message passing changes the geometry of the variational representation even where the total energy cannot tell two networks apart—the correlator concentrates its tangent weight on fewer and more interpretable directions, and the message-passing backflow avoids a rank collapse that overtakes the conventional one. Architecture, optimiser and sampling measure then meet in a single object, the geometry of the wavefunction's log-derivatives, and natural-gradient preconditioning earns its cost only while that geometry can still be estimated from a batch.

The loop closes when a generic collocation proposal loses the Wigner state. The structural measurements of Chapter 6 supply the remedy: a proposal built from the observed shell geometry carries Markov-chain-free training to $\omega = 0.0035$, where a transfer instability in the Slater–Jastrow balance takes over.

Three differently sized sets of models appear below—the production states that carry the physics, the matched ladder of the architecture study, and the smaller collocation networks—and numbers from one are never compared with numbers from another. Table 3.1 gives all three. The four research questions of §I are answered here:

RQ4 — Physics	§6.2
RQ1 — Architecture	§7.1
RQ2 — Optimiser	§7.2
RQ3 — Markov-chain-free paradigm	§7.3–§7.4

The least initial deviation from the truth is multiplied later a thousandfold.

ARISTOTLE
On the Heavens

Chapter 6

Energies and Wigner-molecule structure

This chapter is about the *state*: whether it is accurate enough to trust, and what physics it turns out to contain. Why the solver behaves as it does is the subject of Chapter 7.

The quantum dot is an unforgiving test. The Coulomb interaction is singular wherever two electrons meet, and the kinetic energy carries second derivatives, which magnify any error in the wavefunction's curvature. The confinement ω then sweeps the physics across three decades of correlation strength: from a compact, kinetic-energy-dominated droplet at $\omega = 1$ to a dilute Wigner molecule at $\omega = 10^{-3}$, where the electrons settle several oscillator lengths apart and their mutual repulsion rather than the trap determines the structure of the state. We work at $N \in \{2, 6, 12, 20\}$ electrons and $\omega \in \{1.0,\ 0.5,\ 0.1,\ 0.01,\ 0.001\}$.

Section 6.1 benchmarks the energies against every published reference that exists: fixed-node diffusion Monte Carlo for $N \in \{6, 12, 20\}$ [52, 53] and the analytic two-electron compiled reference values for $N{=}2$ [52, 54]. Those references reach down to $\omega = 0.1$ for every N, and to $\omega = 0.01$ for $N{=}2$; below that, for $N \geq 6$, no published value exists, and the boundary is marked wherever it occurs. Section 6.2 then uses the trained wavefunction as an instrument and follows the Wigner crossover from two electrons to twenty. Its result is that the crossover is staged: the cloud expands and its shells stiffen radially well before the electrons acquire angular order inside them. How the network internally represents these states is a question about the solver rather than about the physics, and is deferred to Chapter 7 and Appendix E.

6.1 Training protocol and ground-state energies

Unless otherwise stated, we first apply residual-based pretraining to obtain a well-conditioned initialiser, and then run a Stochastic Reconfiguration (SR) tail to refine the last digits of the energy. In the residual phase, the network initially chooses its own target energy as a stabilizer, before we switch to the DMC reference energy (and optionally add a variance penalty). The SR phase then uses on the order of 10^3–3×10^3 iterations with batches of $\sim 3{\times}10^3$ samples per iteration. Throughout this section—and wherever else the thesis says "SR tail" without further qualification—SR is applied to *variational Monte Carlo with* $|\Psi|^2$ *sampling*. This matters because the same optimiser reappears in Chapter 7 on the collocation objective, where the sampling measure differs and SR behaves quite differently: here it is a refinement that buys the last digits; there it is what makes the optimisation viable at all.

We compare two backflow architectures throughout. Both condition each particle's displacement on the interparticle coordinates, but they do so differently: the *conventional* backflow applies a single round of pairwise messages aggregated per particle, whereas the *message-passing* backflow maintains explicit node and edge features and transports information between them through learned maps (a single copresheaf round) before the displacement is read out; the contrast between them is the subject of the backflow analysis in Chapter 7. The architecture size is identical for all N. The correlator is the same in both and is the *separable* network of §3.4.1 (code: `PINN`); the message-passing correlator belongs to the architecture study of §7.1 and does not appear in this chapter. It carries ≈ 54k parameters, the conventional backflow adds ≈ 51k (total ≈ 105k), and the message-passing backflow adds ≈ 182k (total ≈ 236k). The two are not capacity-matched, and at equal width they cannot be: the copresheaf network's persistent edge state is the heavier object by construction (§3.6). The two production ansätze share the same correlator and differ only in this component, so they are named for it throughout: *conv. BF* and *MP BF*. These are not the widths of the controlled architecture comparison, which uses its own matched ladder (Table 3.1); the two sets of counts are kept apart throughout and never compared. All components are fully trainable; the parameter count is independent of N because particle interactions enter through permutation-equivariant features rather than per-particle weights.

These are the *production* networks; they are distinct from both the capacity-matched ladder of §7.1.1 and the smaller collocation models of §7.3, and numbers are never compared across the three (see the programme map on p. 57).

6.1.1 Energies

Across every externally benchmarked system, the trained states reproduce the reference energy to between 10^{-5} and 5×10^{-4} in relative terms, with no systematic deterioration as particles are added or the trap is opened (Tables 6.1 and 6.2, Fig. 6.1). That is the result of this section, and everything after it depends on it: it is what licenses treating the state as an instrument in §6.2 rather than as a claim still under defence.

Two features of the tables carry more than the energies themselves. The first is *where the agreement holds*. A method that matches the reference at $\omega = 1$ and drifts at $\omega = 0.1$ differs from one that holds across the whole benchmarked range, since the physics changes character between those points while the architecture does not. The second is *where the tables stop*: below $\omega = 0.1$ no reference of any kind exists for $N \geq 6$, so the values reported there are variational predictions rather than validated ones, and that boundary is marked wherever it occurs rather than letting a uniformly formatted table imply a uniform level of evidence.

One qualification attaches to the benchmark itself. Where a reference value exists it is not only what we score against: it is also the target that residual pretraining anneals toward (§4.1). The agreement is therefore between a reference and an *independently evaluated variational upper bound on the same ansatz*, not between two blind calculations—weaker than independence, and stronger than circularity. A variational expectation under $|\Psi|^2$ is bounded below by E_0 whatever it was aimed at, so aiming at a reference cannot pull the reported energy below what the ansatz can actually reach, and aiming at one above that minimum would make the number worse rather than better. §8.1 takes the argument up properly.

Table 6.1 begins with the easiest case, and its purpose is diagnostic rather than competitive: two electrons are few enough that the architecture can be tested nearly in isolation. It reports two-electron energies from *residual-only* training, with and without the backflow. For N=2, backflow is *not* required to reach reference accuracy; it slightly reduces the variance and smooths convergence, and nothing more. The closed-shell two-electron singlet has no nontrivial same-spin fermionic node for backflow to repair. This matters because it establishes a baseline against which the backflow's later contribution, at larger N and weaker traps, can be read as a genuine effect rather than as an artefact of having added parameters.

Table 6.1: Ground-state energies (Hartree) for two electrons in a harmonic trap using *residual-only* training—that is, *without* the SR tail. To be read against Table 6.2, which reports the same systems after SR refinement: the comparison isolates what pretraining alone achieves, and the residual bias at $\omega = 0.01$ visible here is the one the SR tail removes. The reference column is the same as in Table 6.2; see its notes for provenance. Parentheses denote 1σ on the last digits.

ω	correlator + BF	correlator only	Reference
1.00	2.99998(3)	2.999940(5)	3.00000(1)
0.50	1.65975(2)	1.65979(3)	1.65977(1)
0.10	0.440795(3)	0.440833(1)	0.44079(1)
0.01	0.0740724(6)	0.0741679(6)	0.073839(2)

Table 6.2 compares diffusion Monte Carlo (DMC) references with our *final* conv.-BF energies after residual pretraining and an SR tail. From the ultra–shallow two-electron case (ω=0.01) to the many-electron systems, absolute deviations are small; for N=2 they lie within DMC uncertainties, and for $N \in \{6, 12\}$ the relative deviations span 0.008–0.05% (see also Fig. 6.1).

Table 6.2: Residual-based pretraining + SR tail. Reference energies and our conv.-BF and MP-BF ground-state energies (Hartree) for selected (N, ω). For $N \in \{6, 12, 20\}$ the references are the fixed-node diffusion Monte Carlo energies of Høgberget [52], as tabulated in Ref. [53]; the N=2 entries are drawn from the same compilation, whose reference column is assembled from Høgberget's DMC and from Taut's analytic two-electron solutions [54], which exist only at a discrete set of frequencies. We do not attribute individual N=2 rows to one source or the other, except at $\omega = 1$, where the tabulated 3.00000 is the known exact result. The fixed-node reading of §8.1 therefore applies wherever the reference is DMC, which is every $N \geq 6$ entry. Parentheses denote 1σ on the last digit(s). The "rel. dev." columns give $100\,(E - E_{\mathrm{ref}})/E_{\mathrm{ref}}$; because the $N \geq 6$ references are fixed-node upper bounds, a negative entry is not necessarily an error, and the columns are labelled deviation rather than error for that reason (§8.1). **These are production energies and the two architectures are not capacity matched** ($\approx$ 105k against $\approx$ 236k parameters); the controlled architecture comparison is §7.1, and this table must not be read as an architecture ablation. A dash in the reference column marks (N, ω) for which no published DMC value exists (all $\omega = 0.001$, and $\omega = 0.01$ for $N \geq 6$); the relative-error columns are correspondingly blank there, as no external benchmark is available to score against.

N	ω	Reference	conv. BF	rel. dev.	MP BF	rel. dev.
2	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	+0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	+0.0014
	0.01	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0011
	0.001	—	0.0137948(8)	—	**0.013778(1)**	—
6	1.00	20.15932(8)$^{\mathrm{H}}$	20.1610(6)	+0.0083	**20.1585(3)**	−0.0041
	0.50	11.78484(6)$^{\mathrm{H}}$	11.7895(4)	+0.0395	**11.7847(2)**	−0.0012
	0.10	3.55385(5)$^{\mathrm{H}}$	3.5549(1)	+0.0295	**3.55388(5)**	+0.0008
	0.01	—	0.69125(2)	—	**0.69036(1)**	—
	0.001	—	0.140913(1)	—	**0.140832(1)**	—
12	1.00	65.7001(1)$^{\mathrm{H}}$	65.717(1)	+0.0257	**65.69556(5)**	−0.0069
	0.50	39.1596(1)$^{\mathrm{H}}$	39.1786(8)	+0.0485	**39.1604(3)**	+0.0020
	0.10	12.26984(8)$^{\mathrm{H}}$	12.2731(2)	+0.0266	**12.2743(2)**	+0.0363
	0.01	—	2.48620(5)	—	**2.47363(4)**	—
	0.001	—	0.515823(3)	—	**0.515365(4)**	—
20	1.00	155.8822(1)$^{\mathrm{H}}$	—	—	**155.8738(7)**	−0.0054
	0.50	93.8752(1)$^{\mathrm{H}}$	—	—	**93.8789(5)**	+0.0039
	0.10	29.9779(1)$^{\mathrm{H}}$	—	—	29.9888(2)	+0.0364
	0.01	—	—	—	**6.14645(5)**	—
	0.001	—	—	—	**1.293033(6)**	—

Reference-energy provenance: values marked $^{\mathrm{H}}$ are fixed-node diffusion Monte Carlo energies computed by Høgberget [52] and tabulated in Ref. [53]. The N=2 values come from the reference column of the same compilation, which draws on both Høgberget's DMC and Taut's analytic solutions [54]; since Taut's construction is quasi-exactly solvable only at particular frequencies, we do not assign individual N=2 rows between the two, beyond noting that 3.00000 at $\omega = 1$ is the exact analytic value. The foundational DMC study of these dots is Pederiva *et al.* [13] (spanning $N \leq 13$; see also its erratum [55]), cited here for the method rather than for these particular numbers. We took the values from the compilation in the master's thesis of Haas [56], which reproduces the same benchmarks; the attribution above is to their origin. Dashes mark (N, ω) with no published value.

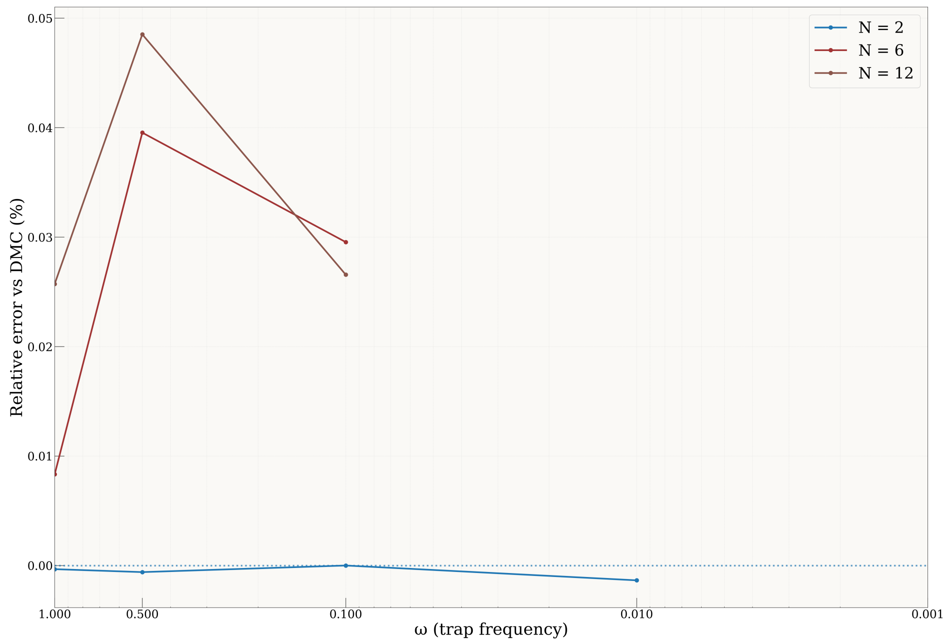


Figure 6.1: Relative deviation of the conv.-BF ground-state energies from the external reference energies, as a function of trap frequency ω, for the cases where a reference exists. The references are of two kinds: the N=6 and N=12 values are fixed-node diffusion Monte Carlo [52], while the N=2 values come from a reference column drawing on both that source and Taut's analytic solutions [54] (Table 6.2). For N=2 the deviations are at the 10^{-3} % level, comparable to the quoted precision of the reference itself; for N=6 and N=12 they span 0.008–0.05 % across ω, with no sign of deterioration in the shallow- or tight-trap limits.

The same benchmark, without a Markov chain. Everything above was trained with a Markov chain: residual pretraining, then an SR tail that draws its samples from $|\Psi|^2$ by Metropolis. A second route removes the chain from the gradient entirely, replacing it with an analytic proposal and importance weights. It reaches comparable energies over part of this range. Its performance, its failure boundary and the mechanism behind both are the subject of §7.3, and the energies are tabulated there (Table 7.5) rather than here, because they are not interpretable until that mechanism is in hand.

Where the tables stop. One subtlety matters here, because the MP-BF ansatz occasionally sits *below* the quoted DMC reference (for example N=12, ω=1: 65.6956 vs. 65.7001 Ha). This does not violate the variational principle, which bounds the energy only from below by the *exact* ground state, $E_{\rm VMC} \geq E_0$. The DMC benchmarks are *fixed-node* calculations and therefore carry their own nodal (systematic) bias; a neural ansatz with a better nodal surface can legitimately lie below a fixed-node DMC value obtained with poorer nodes. We therefore do not treat sitting above the published DMC number as a requirement, and where the CTNN falls below it we read this as a plausibly improved nodal structure rather than an error — though establishing that rigorously would require the DMC trial nodes and a matched uncertainty analysis, which we have not undertaken.

Wherever an external check exists, the pattern is the same. For N=2 the ansatz reaches reference accuracy at every ω for which a reference exists, and for $N \in \{6, 12\}$ the relative deviation stays within 0.008–0.05% across the benchmarked range ($\omega \geq 0.1$; Fig. 6.1). Below $\omega = 0.1$ there is no DMC reference for $N \geq 6$, and we make no claim of absolute accuracy there; what we can say is that the energies join smoothly onto the benchmarked regime and, in the deep-Wigner limit, follow a power law whose fitted exponent, 0.689 for N=6 (§7.4), sits close to the classical Wigner value 2/3. The efficiency is as notable as the accuracy: the SR tail converges in 10^3–3×10^3 iterations of $\sim 3{\times}10^3$ samples each, a modest optimisation budget for a fully many-body wavefunction. We attribute this stability to a small set of conditioning choices— trap-unit scaling, soft-core pair features with $ds/dr \to 0$ at coalescence, and an explicit analytic cusp—each of which keeps the gradients and the Laplacian numerically tame from the shallowest trap to the tightest.

6.2 Wigner–molecule crossover

Up to this point the wavefunction has been the thing under examination. Here it becomes the instrument.

The result of this section is that the crossover is staged. As the trap weakens the cloud first expands and its shells stiffen *radially*—the width-to-radius ratio falls, discrete occupancies emerge, the energy partition approaches classical force balance—while the electrons remain very nearly disordered in angle within those shells. Angular structure then develops in two steps of its own: a small, gradually growing correlation that is measurable well before the Wigner regime, and a large bond-orientational ordering confined to the final decade of confinement. Against the random-angle baseline the shell bond order runs at 1.00–1.02 of its null value from $\omega = 1$ down to $\omega = 0.1$, reaches 1.10 at $\omega = 0.01$ for N=6 (and remains within 1.03 at N=12 and N=20), and only at $\omega = 10^{-3}$ jumps to 1.19–1.65 on every ring at every particle number. Radial localisation and angular crystallisation are not one transition, and the diagnostics below are arranged to separate them.

The subsections take that in order. §6.2.2 shows the cloud expanding and acquiring visible radial structure; §6.2.3 shows interactions taking over and the trap–interaction partition approaching its classical form; §6.2.4 shows the shells stiffening radially; and §6.2.5 shows angular order finally appearing, tested against a null in which the angles are random. §6.2.6 collects what the diagnostics agree on and what they do not establish.

What we are looking for cannot be seen in the observable one would naturally plot. In a rotationally symmetric trap, the electrons of a Wigner molecule localise in *relative* coordinates—concentric rings, polygon-like angular correlations—while the whole pattern is free to rotate. Average over that rotation, as any lab-frame density does, and the rings smear into something close to uniform. The order is there; the density hides it. Every diagnostic below exists to look past that averaging.

We combine one-body densities, label-invariant shell topology, rotation-registered angular order, shell radii and gaps, pair statistics, and Lindemann-type disorder, across $N = 2, 6, 12, 20$ and $\omega \in \{1, 0.5, 0.1, 0.01, 0.001\}$ [7, 10]. No one of them is decisive on its own. They fail in different ways, which is why they are reported together.

6.2.1 Observables and diagnostics

For each (N, ω) we draw $\mathbf{X} \sim |\Psi|^2$ and report (i) the pair distribution $g(r)$, (ii) the single-particle radial density $P(r) \propto r\,\rho(r)$, and summary statistics: r_{mode}, $\langle r \rangle$, σ_r, FWHM, quantiles q_{10}, q_{50}, q_{90}, and a localization ratio $\gamma = \sigma_r / r_{\text{mode}}$ (smaller $\gamma \Rightarrow$ stiffer localization). For multi-electron systems we additionally monitor shell-resolved bond-orientational order $|\Phi_m|$ and a dimensionless Lindemann-type ratio for positional fluctuations on each ring [11].

Density parameter r_s. Following Egger *et al.* [7], we estimate the Wigner–Seitz parameter from the first maximum r^* of the spin-summed pair correlation, $r_s \equiv r^*/a_B^*$. The Fermi–liquid $\to$ Wigner–molecule crossover occurs near $r_s \simeq 4$ in parabolic dots; our weak-trap cases lie well beyond this threshold [7, 12].

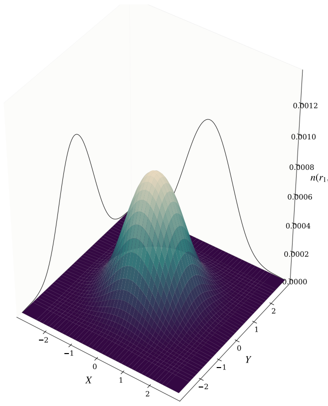
6.2.2 One-body densities: visual crossover

Figures 6.2 and 6.3 compare one-body densities at strong confinement ($\omega = 1$) and weak confinement ($\omega = 10^{-3}$) for $N = 2, 6, 12, 20$. At $\omega = 1$ the density remains compact and the dominant change with N is progressive shell filling under the harmonic trap. At $\omega = 10^{-3}$ it broadens by two orders of magnitude and develops pronounced *radial* structure at every N.

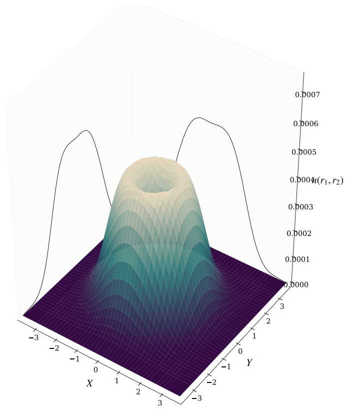
What that structure can and cannot establish follows from the symmetry argument above. The lab-frame density is an average over all orientations of the molecule, so it retains radial information and destroys angular information by construction. These figures are therefore evidence of *radial shell localisation* and are silent on whether the electrons are angularly ordered within those shells—a ring of six electrons at fixed 60° spacing and six electrons sliding freely around the same ring produce the same picture. The angular question is taken up in §6.2.5, where it needs a co-rotating frame and a null model to answer.

6.2.3 Energy-based crossover diagnostics

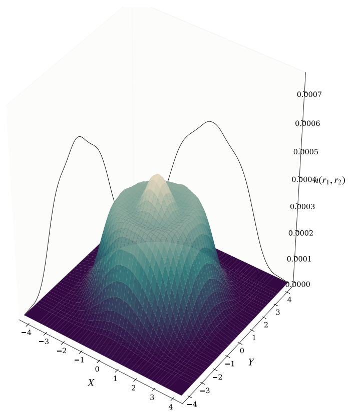
The energy partition carries two messages, and only two.



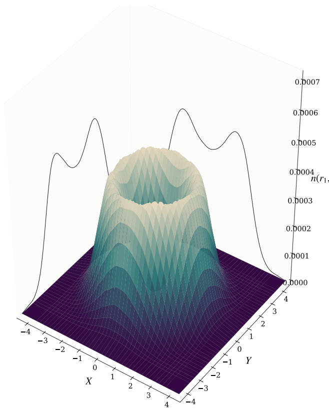
(a) N=2, $\omega = 1.0$



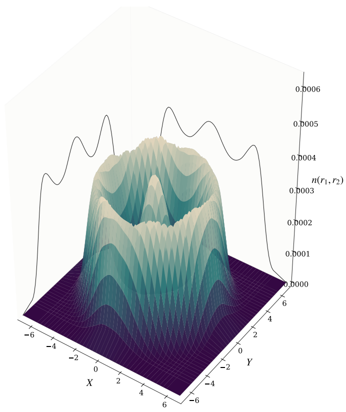
(b) N=6, $\omega = 1.0$



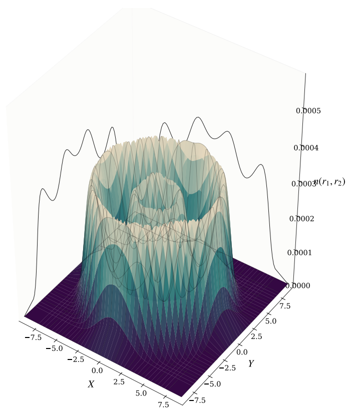
(c) N=12, $\omega = 1.0$



(d) N=2, $\omega = 0.001$

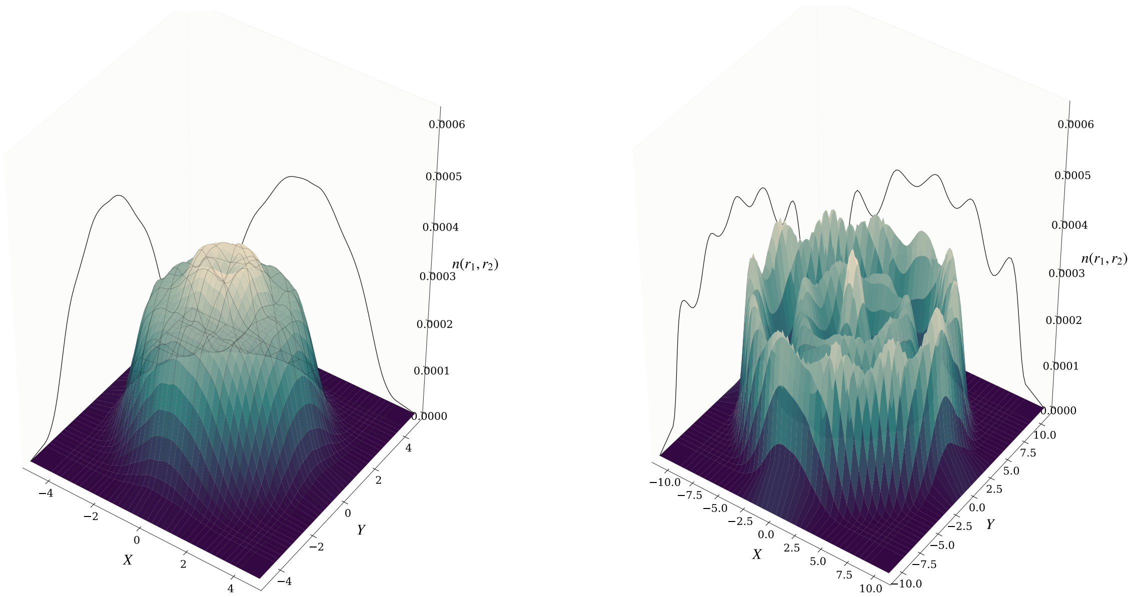


(e) N=6, $\omega = 0.001$



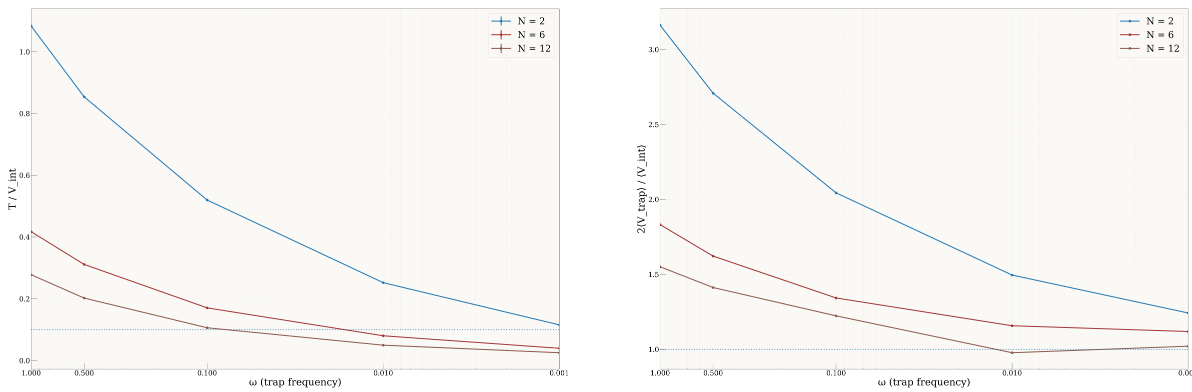
(f) N=12, $\omega = 0.001$

Figure 6.2: One-body densities for $N = 2, 6, 12$ at $\omega = 1.0$ (top) and $\omega = 0.001$ (bottom). The vertical axis is the one-body density $n(x, y)$ over the two Cartesian components of a single particle's position; the axis label carried in the figure files reads $n(r_1, r_2)$ for the same quantity, which collides with the convention used elsewhere in this thesis, where $\mathbf{r}_i$ is the position of particle i. As confinement weakens the densities broaden and develop pronounced *radial* shell structure. Being lab-frame averages over all orientations of the molecule, they cannot show angular order: that requires the registered diagnostics of §6.2.5.



(a) N=20, $\omega = 1.0$ (b) N=20, $\omega = 0.001$

Figure 6.3: One-body densities for N=20 at strong and weak confinement; the vertical axis is $n(x, y)$ as in Fig. 6.2. The weak-trap case shows strong radial broadening and clear multi-shell organisation. As above, this is radial evidence only.



(a) T/V_{int} vs. ω for N=2, 6, 12. The horizontal line marks T/V_{int}=0.1.

(b) $2V_{\text{trap}}/V_{\text{int}}$ vs. ω for N=2, 6, 12. The horizontal line marks the classical virial ratio $2V_{\text{trap}} = V_{\text{int}}$.

Figure 6.4: Energy-based Wigner diagnostics as functions of trap frequency. As ω decreases the dots become interaction dominated ($T/V_{\text{int}} \ll 1$) and the trap–interaction partition approaches the classical virial form, most clearly for N=12.

Interactions take over. The interaction-to-kinetic ratio $\Gamma = \langle V_{\rm int}\rangle/\langle T\rangle$ rises across $\omega \in [1, 10^{-3}]$ from 0.9 to 8.7 at N=2, from 2.4 to 25.4 at N=6 and from 3.6 to 39.9 at N=12 (Fig. 6.4a). The dot stops being a kinetic-energy-dominated droplet and becomes a Coulomb-dominated one.

Classical force balance emerges. At $\omega = 10^{-3}$ the trap–interaction partition $2\langle V_{\rm trap}\rangle/\langle V_{\rm int}\rangle$ reaches 1.25, 1.12 and 1.02 at $N = 2, 6, 12$, so the twelve-electron dot satisfies the classical condition $V_{\rm int} \approx 2V_{\rm trap}$ to within 2% (Fig. 6.4b). The approach is monotone and improves with N, as it should: the classical limit is reached faster when there are more particles to arrange.

Both are consistent with real-space expansion. The rms radius extracted from $V_{\rm trap} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2\rangle$ grows from 1.1–2.0 a_B^* at $\omega = 1$ to 70–169 a_B^* at $\omega = 10^{-3}$, faster than the non-interacting $\omega^{-1/2}$ law, which is interaction-driven swelling. The fitted exponents for the individual energy components and for the radius, which support these statements without adding to them, are collected in Appendix F.1.

6.2.4 Radial localization and pair correlations

Over the weakest three confinements, the typical one-body radius follows $r_{\rm mode} \sim \omega^\alpha$ with $\alpha \simeq -0.626$ at N=2—close to the classical two-body value $-2/3$, which nothing in the ansatz knows about. And the width-to-scale ratio $\gamma = \sigma_r/r_{\rm mode}$ falls, which is the shell stiffening rather than merely dilating (Table 6.3).

Table 6.3: Radial localisation across the confinement range. $r_{\rm mode}$ is the mode of the single-particle radial density in Bohr; $\gamma = \sigma_r/r_{\rm mode}$ is the width-to-scale ratio, which falls as the shell stiffens. Full diagnostics—mean, width, FWHM, quantiles, ring-angle spread and pair Lindemann ratio—are in Appendix F.

	N=2		N=6	
ω	$r_{\rm mode}$	γ	$r_{\rm mode}$	γ
1.00	1.44	0.491	2.09	0.442
0.50	2.30	0.437	3.07	0.445
0.10	3.60	0.480	8.17	0.429
0.01	14.19	0.396	33.15	0.420
0.001	64.29	0.287	113.55	0.421

The exponent is not the sharpest form of this check. Because $r_{\rm mode}$ is the mode of the *single-particle* radial density, it mixes the central electron of a $(1, 5)$ topology with the ring and is not the ring radius. The shell-resolved outer radius is, and comparing it with the classical Wigner ring $1.334\,\omega^{-2/3}$ tests a prefactor as well as a power. That ratio falls monotonically—1.247, 1.188, 1.098, 1.032, 1.017 from $\omega = 1$ to 10^{-3}—so at the weakest trap the quantum ring sits 1.7% outside the classical one, and the fitted exponent on that quantity is -0.637 over the full grid and -0.650 over the weakest three confinements (Appendix F.4). Nothing in the ansatz, the objective or the sampler knows the number 1.334.

The two sizes stiffen differently, and the difference is physical. At $\omega = 10^{-3}$ the N=2 pair sits at separation $r_{12} \approx 122$ with Lindemann ratio $\gamma_{r_{12}} \approx 0.20$ and a relative angle sharply peaked at π: a dilute, anti-aligned dimer. At N=6 the ring's angular fluctuations and pair Lindemann ratio fall monotonically while γ plateaus near 0.42. The shell is stiffening internally even as its radial profile broadens, because the equilibrium radius is growing faster than the width.

6.2.5 Shell topology, angular order, and the rotating Wigner molecule

The densities plotted above look like smooth rings, yet the electrons are localised. Both are true, and the reason is symmetry. Because the Hamiltonian is rotationally invariant, so is its ground state, and any lab-frame average necessarily smears an angularly ordered configuration into a featureless ring. The order is real; it simply lives in relative coordinates, and looking for it in the lab frame is looking in the one place it is guaranteed not to be. Everything below follows from taking that seriously.

Concretely, a Wigner molecule's internal angular order can only be observed after registering frames to a co-rotating reference. For a shell s with n_s particles we compute a rotation-registered bond-orientational parameter $|\Phi_{n_s}|$ and a corresponding Lindemann-type ratio quantifying angular stiffness. Lab-frame densities remain ring-like (the "rotating Wigner molecule" scenario), while the co-rotating frame reveals crystal-like polygon order [7, 10].

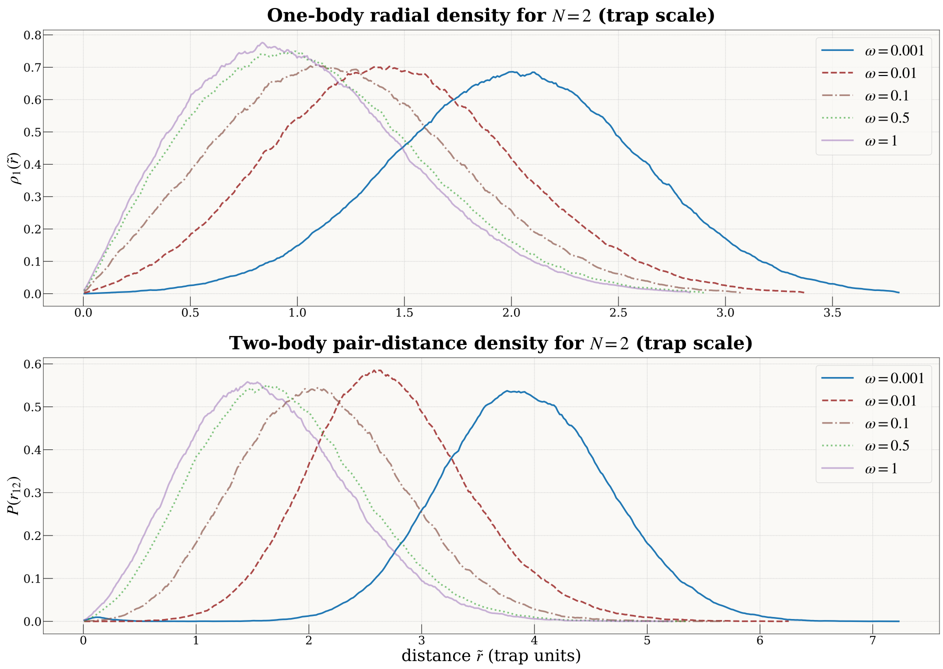


Figure 6.5: Two-electron radial densities $P(r)$ across ω.

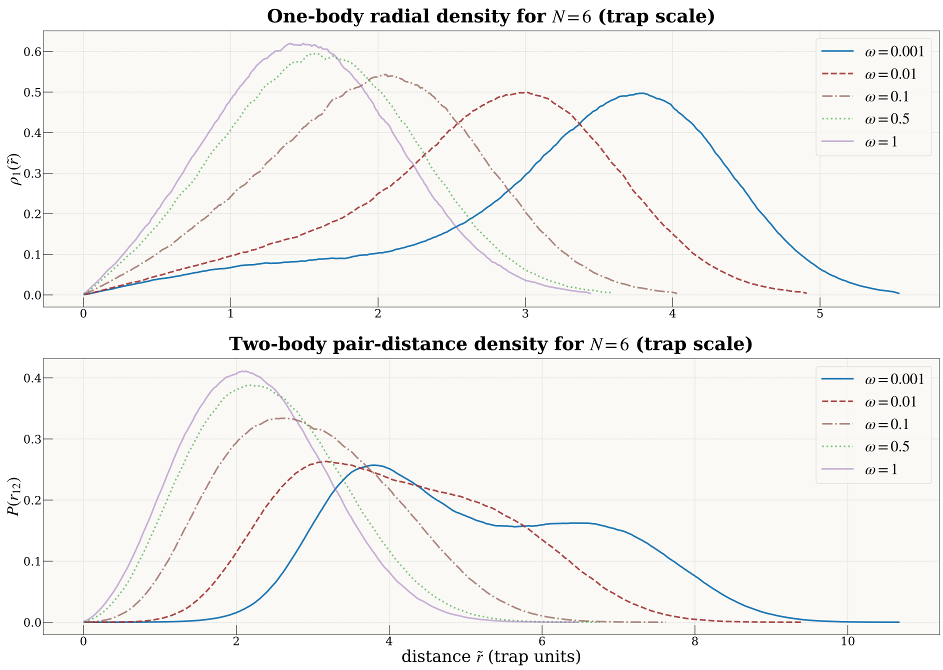


Figure 6.6: Six-electron radial densities $P(r)$ across ω.

N=6**: single ring with sector mixing.** At $\omega = 0.01$ the system fluctuates between a $(1,5)$ sector (fraction ≈ 0.69) and a $(0,6)$ single-ring sector (fraction ≈ 0.31), with bond order $|\Phi_5| = 0.453 \pm 0.214$ and $|\Phi_6| = 0.466 \pm 0.206$ respectively. Two conventions apply here and below. The quoted $\pm$ is the frame-to-frame standard deviation of $|\Phi_m|$, not a standard error on its mean, which with 3×10^4 frames is smaller by two orders of magnitude. And these are *sector-conditioned* values—computed on the frames assigned to that sector—which is why they differ slightly from the modal-topology values tabulated in Appendix F.2. Both must be read against the random-angle null $\langle|\Phi_n|\rangle = 0.886/\sqrt{n}$, which is 0.396 for n=5 and 0.362 for n=6: the ratios here are 1.14 and 1.29, so the measured bond order at $\omega = 0.01$ exceeds the random-angle value by a modest margin rather than sitting on it. At $\omega = 10^{-3}$ the ring stiffens further: $(1,5)$ fraction ≈ 0.80, $(0,6) \approx 0.20$, with $|\Phi_5| = 0.597 \pm 0.224$ (Lind$\,\approx$0.225) and $|\Phi_6| = 0.663 \pm 0.183$ (Lind$\,\approx$0.205)—now far above the random-angle null, by factors of 1.51 and 1.83. This is the confinement at which n-fold angular order becomes *large* rather than merely resolvable (Appendix F.2). The distinction matters because with 10^4–10^6 frames a 2% excess over the null is already resolved at tens of standard errors; detectability is therefore not what separates the regimes, magnitude is. Classically $(1,5)$ is the ground-state topology and $(0,6)$ a proximate competitor; the observed quantum mixture mirrors this hierarchy [8, 9]. In both weak-trap cases the global $g(r)$ is reproduced to numerical precision by the appropriate inner–inner / inner–outer / outer–outer (II/IO/OO) mixtures.

N=12**: two shells and commensurability.** At $\omega = 0.01$ two-shell formation is already frequent (fraction 0.761 at radial-gap threshold τ=3.0), distributed across sectors $(1,11) = 32.6\%$, $(2,10) = 25.7\%$, $(3,9) = 13.7\%$. At $\omega = 10^{-3}$ shelling becomes ubiquitous ($f_{\text{2shell}} \approx 0.99$–0.92 for $\tau = 2.0$–3.0) and the *commensurate* $(3,9)$ split emerges as modal (43.3% of quality-cut frames), with strong bond order on both rings: $|\Phi_3| = 0.687 \pm 0.229$ ($\text{Lind}_{\text{in}} = 0.223$), $|\Phi_9| = 0.526 \pm 0.194$ ($\text{Lind}_{\text{out}} = 0.224$). Whether the shell picture actually explains the pair structure needs a test that could fail, and the obvious one cannot. Recombining the inner–inner, inner–outer and outer–outer histograms with the combinatorial weights of the ideal $(3,9)$ topology reproduces the global $g(r)$ essentially exactly—but the sector histograms and the total are accumulated from the same frames, so that recombination is close to exact by construction. It establishes that the sector bookkeeping is self-consistent, nothing more, and it is recorded in Appendix F.3 for that reason.

The test that can fail is generative and held out (Table 6.4). Shell radii and widths are fitted on half the frames; each shell is modelled as a regular n_s-gon carrying a per-frame global rotation and a fitted angular jitter; and the model is scored on the other half. It reproduces the pair distribution to cosine similarity 0.987–0.999 across the whole grid, against 0.936–0.996 for an angle-randomised null built from the same fitted radial distributions. The difference between them, $\Delta\cos$, is what *angular* order buys over shell radii alone, and it grows monotonically as the trap weakens at every particle number—at N=12 from 0.006 at $\omega = 1$ to 0.026 at $\omega = 10^{-3}$, at N=6 from 0.012 to 0.058, at N=20 from 0.003 to 0.010. Shell radii alone account for most of the pair structure at strong confinement; angular structure accounts for a growing remainder as the molecule forms.

The shape of that growth is worth noting, because it is not the same shape the bond-order ratios show. $\Delta\cos$ rises by a factor of 1.1–2.1 per decade at every particle number, monotonically and without a threshold, whereas the bond order stays flat at its null and then jumps. The two are not in conflict: $\Delta\cos$ responds to *any* angular correlation that improves the pair-distance density, while $|\Phi_n|$ responds specifically to n-fold order on a registered ring. Weak angular correlations therefore build gradually from strong confinement onward, and crystalline n-fold order emerges from them only in the final decade. That is a three-stage reading, and it is better supported than a single threshold would be.

N=20**: three-shell molecule and rapid topology collapse.** At $\omega = 10^{-3}$ the dominant shell topology is $(1,7,12)$—a clear three-shell Wigner molecule with widely separated rings at radii $r_0 \approx 38$, $r_1 \approx 132$, $r_2 \approx 247$ (gaps ≈ 94 and ≈ 115). The lab-frame pinning metric remains small ($C_{\text{global}} \approx 0.06$), as expected for a freely rotating molecule. The cross-shell quantity $\tilde{C}_{\text{global}}$ is omitted: it depends on a registration convention whose original analysis code was not retained (§5.1.3). The per-shell moduli and angular-spacing diagnostics below are rotation-invariant and carry no such dependence. Lindemann disorder on the two occupied outer shells is low: $L_{7,\text{reg}} \approx 0.35$ and $L_{12,\text{reg}} \approx 0.27$.

Already at $\omega = 10^{-2}$ the radial gap detector returns a $(1,19)$ partition rather than a three-shell one: the shell radii contract by an order of magnitude ($r_0 \approx 10$, $r_1 \approx 45$), the outer-shell angular disorder rises to ≈ 0.57. The wording matters here. At $\omega = 10^{-3}$ the $(1,7,12)$ label describes a Wigner *topology*, because it is accompanied by separated shells, registered angular order and low Lindemann disorder. At $\omega = 10^{-2}$ and above, with angular order at its random-angle null and disorder high, $(1,19)$ is what the

Table 6.4: The structural test that can fail. Shell radii and widths are fitted on half the frames; each detected shell is then modelled as a regular n_s-gon carrying a per-frame global rotation and a fitted angular jitter; and the resulting pair-distance density is scored on the *held-out* half. $\cos_{\text{poly}}$ is the polygon model's agreement with the empirical density and $\cos_{\text{uni}}$ that of an angle-randomised null with the same fitted radial distributions, so $\Delta \cos$ isolates what polygonal order contributes over shell radii alone. It grows monotonically as the trap weakens at every particle number. Full table including residuals in Appendix F.3.

N	ω	occ.	$\cos_{\text{poly}}$	$\cos_{\text{uni}}$	$\Delta \cos$
6	1.0	$1, 5$	0.9988	0.9866	**0.0122**
6	0.5	$1, 5$	0.9986	0.9836	**0.0150**
6	0.1	$1, 5$	0.9974	0.9741	**0.0234**
6	0.01	$1, 5$	0.9922	0.9536	**0.0386**
6	0.001	$1, 5$	0.9942	0.9358	**0.0585**
12	1.0	$1, 11$	0.9993	0.9937	**0.0056**
12	0.5	$1, 11$	0.9990	0.9927	**0.0063**
12	0.1	$1, 11$	0.9980	0.9896	**0.0085**
12	0.01	$1, 11$	0.9929	0.9805	**0.0124**
12	0.001	$3, 9$	0.9868	0.9604	**0.0264**
20	1.0	$1, 19$	0.9992	0.9964	**0.0028**
20	0.5	$1, 19$	0.9990	0.9958	**0.0032**
20	0.1	$1, 19$	0.9982	0.9942	**0.0040**
20	0.01	$1, 19$	0.9947	0.9900	**0.0047**
20	0.001	$1, 7, 12$	0.9915	0.9820	**0.0095**

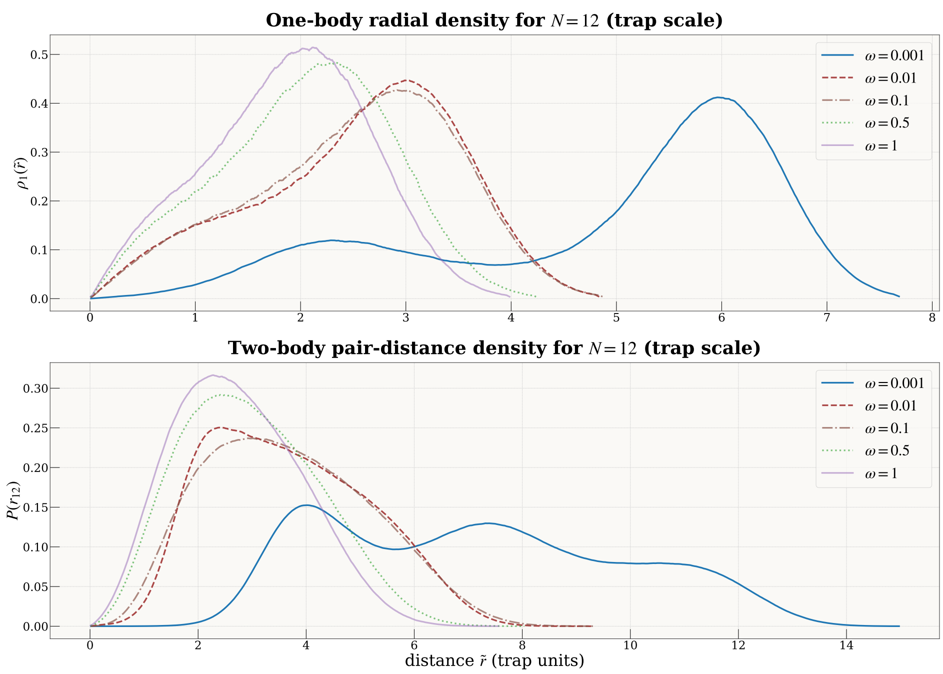


Figure 6.7: Twelve-electron radial densities $P(r)$ across ω.

radial classifier returns on a state that is not angularly ordered at all—a decomposition of the radial density, not a crystalline shell structure. We use the occupancy notation in both cases because the detector does, but only the weakest-trap case earns the word topology. At $\omega = 1$ the classifier returns the same $(1, 19)$ partition, with compact radii ($r_0 \approx 0.55$, $r_1 \approx 2.3$) and high angular disorder ($L_{\text{out}} \approx 0.69$). The sharpness of this transition—a single decade in ω separates a stiff three-shell crystal from a disordered single ring about a central electron—is the most dramatic structural collapse observed in our system-size series.

A caveat on total spin. Our reference states are the closed-shell singlets of the non-interacting problem, with $N_\uparrow = N_\downarrow$ held fixed throughout. This is safe at weak coupling, but in the deep-Wigner regime the ground-state *spin* can change: for parabolic dots the interaction can drive spin transitions as it strengthens (for instance an $S = 0 \to S = 1$ change for $N = 6$ at strong coupling) [7]. Fixing $N_\uparrow = N_\downarrow$ does not by itself pin the total spin, since an $S = 1$ state also has an $S_z = 0$ component. We do not scan total spin here, so the weak-trap energies and structures below should be read as the lowest states we find *within the fixed spin-projection sector*, not as proven global-spin ground states. Determining the true spin ordering across the crossover, by comparing sectors of different S, is left to future work.

6.2.6 The crossover across diagnostics

Six diagnostics, applied at four particle numbers and five confinements, agree. As ω falls from 1 to 10^{-3} the densities expand and develop rings, and the shell topology resolves into the discrete finite-N patterns that classical Coulomb clusters predict—$(1, 5)$ at N=6, $(3, 9)$ at N=12, $(1, 7, 12)$ at N=20 [8, 9]. The shells separate and stiffen. The Lindemann disorder falls monotonically throughout, while rotation-registered bond order stays within 2% of its random-angle null down to $\omega = 0.1$ and reaches 1.19–1.65 times it only at $\omega = 10^{-3}$ (Appendix F.2)—radial stiffening precedes n-fold angular ordering rather than accompanying it. Lab-frame order stays near zero throughout, as expected for a freely rotating molecule. The energy partitioning approaches the classical virial form, with Γ reaching $\sim$40 for N=12 at the weakest trap. None of these was built into the ansatz. The commensurate $(3, 9)$ sector at N=12 carries real weight in $(1, 11)$ and $(2, 10)$ besides, which is the quantum rotational mixing the rotating-Wigner-molecule picture predicts [10, 12]. At N=20 the three-shell structure and its collapse to a single ring about a central electron within one decade in ω is the sharpest transition in the series. The $(1, 7, 12)$ configuration itself is not new—it is the known classical shelling and has been reported in quantum calculations of comparable dots—and no novelty is claimed for it. What we have not found reported elsewhere for $N \geq 12$ parabolic dots is the combination of *sector-conditioned* bond-order and Lindemann statistics, read against their random-angle null, with a held-out polygon reconstruction of $g(r)$ scored against an angle-randomised baseline.

All of it is bounded by two qualifications. The states are the lowest we find within a fixed spin-projection sector, and we have not scanned total spin, so these are not proven global-spin ground states. And the combinatorial reconstruction of $g(r)$ establishes decomposability rather than sampling correctness; the held-out polygon fit of Appendix F.3 is the version that could have failed, and sampling correctness itself rests on the independent restart and chain diagnostics.

What §6.2 establishes. The crossover is staged, and the two angular diagnostics stage it slightly differently, which is itself informative. Radial localisation—expansion, shell separation, falling width-to-radius ratio, classical force balance—develops progressively from $\omega = 1$ down. Bond-orientational order is flat at its null through $\omega = 0.1$, departs slightly at $\omega = 0.01$ and only at N=6, and becomes large at 10^{-3} everywhere. The held-out polygon model, which is sensitive to any angular correlation rather than to n-fold order specifically, instead grows monotonically throughout, roughly doubling per decade. Read together these say that weak angular correlations build gradually across the whole range while *crystalline* n-fold order appears only in the last decade—a three-stage picture rather than a two-stage one. What this does *not* establish is the total spin of these states, which we did not scan, nor that the occupancy labels denote crystalline topologies outside the weakest-trap cases.

These structural facts also enter the method. The shell radius approaching its classical $\omega^{-2/3}$ scaling—to within 1.7% of the classical prefactor at $\omega = 10^{-3}$, Appendix F.4—and the angular order sharpening in the last decade are the quantities a sampler must respect in order to reach the Wigner regime without MCMC. Chapter 7 uses them directly: an origin-centred proposal aims where the wavefunction no longer has any mass, whereas a proposal built from the measured shell radius and angular order tracks it, and carries MCMC-free training far into the correlated regime (§7.4). What the trained model reveals about the physics is what later lets us train it more cheaply.

That is the state seen from outside. What the network had to hold internally in order to produce it is a question about the solver, and Chapter 7 takes it up.

Chapter 7

Architecture, optimisation, and sampling geometry

The previous chapter showed that the ansatz works: it reaches reference quality wherever a reference exists, and the states it produces carry the staged Wigner-molecule structure the diagnostics were built to detect. Why the solver can represent those states, and what governs whether they can be trained at all, are separate and harder questions.

Three observations from that chapter remain unexplained. Two networks of the same size, trained the same way and landing on the same energy, need not be the same object. A natural-gradient step that is decisive in one training regime does nothing in another. And an importance-sampled objective reproduces a Markov chain's answer in one corner of the phase diagram and departs from it in the next.

All three can be examined through the per-sample, column-centred log-derivative matrix

$$O_{ki} \;=\; \frac{\partial \log |\Psi_{\boldsymbol{\theta}}(\boldsymbol{x}_k)|}{\partial \theta_i}, \qquad k = 1, \ldots, B, \;\; i = 1, \ldots, P, \tag{7.1}$$

and on its two Gram matrices, the quantum geometric tensor $S = O^\top O/B$ and the neural tangent kernel $K = OO^\top$ [27, 28]. That gives the chapter its structure, and it is worth stating in one line each:

Architecture determines the shape of the tangent space.
The *optimiser* acts on it through its metric.
The *sampling measure* determines how well that metric can be estimated.

The three questions of this chapter are the same three objects, and the chapter takes them in that order: architecture in §7.1, the optimiser in §7.2, the sampling measure and the training paradigm it makes possible in §7.3–§7.4.

The instrument throughout is the *effective dimension* of the tangent space—the participation ratio of the QGT spectrum $\{\lambda_a\}$,

$$d_{\text{eff}}(S) \;=\; \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2}, \tag{7.2}$$

which counts how many directions actually carry Fisher weight, whatever the parameter count says. Beside it sit the local-energy variance $\mathrm{Var}(E_L)$, zero only for an exact eigenstate; the effective sample size; and the condition number $\kappa(S)$, which is reported in §7.2 and turns out to be large everywhere and therefore diagnostic of nothing by itself.

Architectures are compared on a *common probe set*: the same configurations for every model, never each model's own $|\Psi|^2$. Backflow comparisons are seeded and held at reference quality. Below $\omega = 0.1$ every relative error is measured against our own converged variational energies, and is therefore a statement about internal consistency rather than about accuracy. The tooling is anchored on the analytic two-electron ground state, where the ansatz returns $E = 3.000127$ Ha (+0.004%) at squared overlap 1.000000.

Component-level quantities such as the backflow's rank are read at the evidential level set out on p. 57; §7.5 gives the evidence for that ordering.

7.1 Message passing and the variational geometry

What does a message-passing network learn that a separable one cannot? Both learnable objects of the ansatz can be built in either form, so the question is asked twice, changing one object at a time. In the first comparison only the *correlator* is swapped—a separable DeepSets network against the message-passing CTNN—while the backflow, the Slater reference, the analytic cusp, the training protocol and the capacity tier all stay as they are. In the second only the *backflow* is swapped—single-round pairwise messages against the copresheaf CTNN—with the correlator now among the things held fixed. Neither comparison varies the two together.

7.1.1 The correlator: tangent-space compression

This subsection establishes four separate things, and it is easier to hold them apart if they are named first.

1. **The energy cannot distinguish the two architectures.** Both reach the reference, across a capacity ladder, to four figures.
2. **The tangent geometry can.** At strong confinement the message-passing correlator reaches the same energy along about half as many effective tangent directions.
3. **The local-energy variance separates them at every confinement**, including those where the tangent dimension no longer does.
4. **So does the physical content of the tangent space.** Far more of the message-passing network's tangent variance falls inside a compact dictionary of the Hamiltonian's collective modes.

The first two are strong-confinement statements; the last two hold throughout. Taken together they are why the architectural claim is made on $\mathrm{Var}(E_L)$ and the dictionary, and not on the direction count that first suggested it.

Energy cannot separate them. At $N = 6$, $\omega = 1$ the two Jastrow families are *energetically indistinguishable*: trained to DMC quality across a 20k–164k parameter ladder, every model lands within $\pm 0.1\%$ of the reference, and the best raw energy belongs to a 164k DeepSet—twice the CTNN parameter count. Total energy is a blunt discriminator; both have saturated the variational floor.

The tangent geometry can. On a common probe set the message-passing correlator reaches the ground state in $d_{\text{eff}} \approx 1.2$–$1.7$ (three seeds), while every DeepSet needs $d_{\text{eff}} \approx 2.9$–$3.6$—a factor of ~ 2. The gap is robust to the probe measure (it moves by ≤ 0.26 whether scored on CTNN, DeepSet, or pooled points) and is not closed by capacity: scaling the DeepSet eightfold, from 20k to 164k parameters, moves its effective dimension only from 3.6 to 2.9, and the 89k DeepSet that matches the CTNN in size sits at 3.6. The compression is architectural, not a matter of size or sampling mass. It is also a well-posed comparison despite d_{eff} not being a reparameterization-invariant absolute dimension (§2.2.5): both families are compared under matched normalisation conventions and a common probe measure, so the factor-of-two gap reflects how each architecture spends its tangent directions, not a change of coordinates.

The gauge caveat of §7.5 does not apply to this comparison. It is made between Jastrow families with *no backflow present*, so there is no correlation to divide between two substitutable channels. What remains is a comparison of how two architectures spend their tangent directions on the same ground state, and it is stable across seeds and across the probe measure—a property of the correlator, not an accounting artefact.

On the common pooled probe at $\omega = 1$ the three CTNN runs give $d_{\text{eff}} = 1.20, 1.50, 1.66$ and the four DeepSets, spanning 20k to 164k parameters, give 3.63, 3.37, 3.65, 2.94. The two sets do not overlap, and the closest approach between them—the largest DeepSet against the largest CTNN—is still a factor of 1.8. With three and four runs respectively we report the values rather than a sigma-separation, which at these sample sizes would borrow more authority than the design supports. The gap is a *strong-confinement* effect: it closes toward the Wigner crystal, where both families reach ~ 3.7 but occupy *different* tangent subspaces. A single-seed dimension "inversion" at the crystal, reported in an earlier draft, did *not* survive seeding and is retracted here as a worked cautionary result (see §7.5).

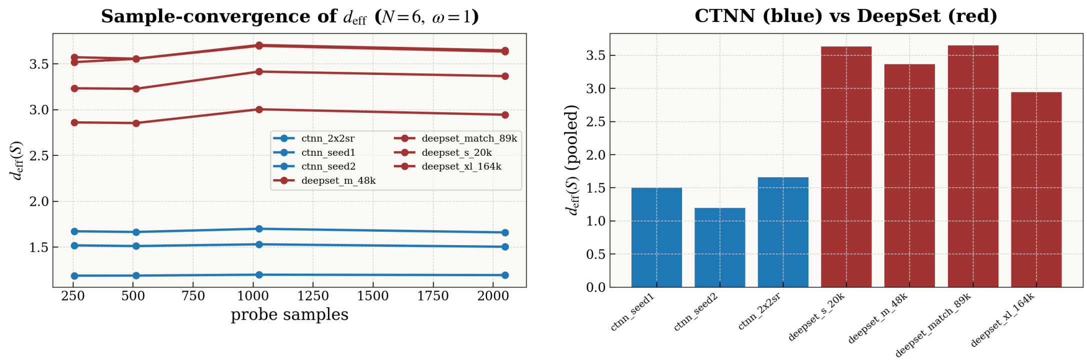


Figure 7.1: Message passing compresses the correlator (N=6, ω=1). *Left:* the effective rank of the quantum geometric tensor on a common pooled probe, converged in the number of probe samples. *Right:* the same pooled measure per model. The CTNN (blue) reaches the ground state in $d_{\rm eff} \approx 1.2$–1.7, while every DeepSet (red) needs ≈ 2.9–3.6—and the gap does not close as the DeepSet is scaled from 20k to 164k parameters. The compression is architectural, not a matter of capacity.

Local-energy variance. $d_{\rm eff}$ stops separating the families at the crystal, and rank will turn out to be gauge-like (§7.5). The local-energy variance does neither: it is defined on the state, vanishes only for an exact eigenstate, and is directly comparable between any two ansätze evaluated on their own $|\Psi|^2$. Measured at matched training protocol, $\mathrm{Var}(E_L)$ is lower for the message-passing correlator at every confinement (Table 7.1), by factors between 1.2 and 3.4, while the two are indistinguishable in energy.

Two qualifications fix the size of that effect, and they pull in opposite directions. At the crystal a seeded replication is available, and we report it in preference to the single cascade run: three CTNN runs give (1.95, 1.63, 1.96) $\times 10^{-5}$ against three DeepSet runs at (3.40, 3.23, 3.41) $\times 10^{-5}$. The two sets do not overlap, which is the finding; but the ratio of their means is 1.8, not the 7.4 obtained by pairing the single lowest-variance CTNN cascade run against its DeepSet counterpart. The seeded ratio is the one we quote. Capacity works the other way. The first four rows compare 79.8k CTNN against 47.7k DeepSet, a 1.7× advantage to the message-passing network that a reader is entitled to discount. Repeating $\omega = 1$ against the capacity-matched 89.4k DeepSet—averaging the Adam and SR arms of the optimiser study on both sides, so that nothing but architecture differs—gives 2.75×10^{-2} against 9.38×10^{-2}, a ratio of 3.4. The separation is therefore not an artefact of the CTNN being the larger network; at matched size it is wider than the production comparison shows. Intermediate confinements ($\omega = 0.28, 0.05, 0.03$) reproduce the same ordering at ratios 1.2–1.8. The one place the ordering is not resolved is the smallest capacity tier used for the depth analysis (12k CTNN vs 9k DeepSet), where the two sit within 8% of each other at $\omega = 1$ and 0.5—so this is a statement about trained models at production width, not about the architectures at any size.

Table 7.1: Local-energy variance at N=6, matched training protocol, both architectures at reference-quality energy ($|\mathrm{err}| \le 0.13\%$). Lower is better; $\mathrm{Var}(E_L) \to 0$ only for an exact eigenstate. The evidential basis differs by row and should be read accordingly. The $\omega = 0.01$ row is a mean over three seeds per architecture; the rows at $\omega = 1$, 0.5 and 0.1 are single runs of the production cascade, for which no seeded replication was run. Since the single cascade run at $\omega = 0.01$ turned out to overstate the ratio fourfold against its own seeded mean, the unseeded rows should be read as establishing the *ordering*—which never reverses, here or at the three intermediate confinements—and not the precise magnitude of the factor. Capacities are 79.8k (CTNN) against 47.7k (DeepSet) except in the last row. The last row repeats $\omega = 1$ against the *capacity-matched* 89.4k DeepSet, where the ordering is if anything stronger.

ω	CTNN	DeepSet	ratio	basis
1.0	1.09×10^{-2}	2.51×10^{-2}	2.3	single run, 79.8k vs 47.7k
0.5	4.62×10^{-3}	1.01×10^{-2}	2.2	single run, 79.8k vs 47.7k
0.1	5.63×10^{-4}	1.14×10^{-3}	2.0	single run, 79.8k vs 47.7k
0.01	1.85×10^{-5}	3.35×10^{-5}	1.8	3-seed mean, 79.8k vs 47.7k
1.0	2.75×10^{-2}	9.38×10^{-2}	3.4	matched, 79.8k vs 89.4k

Naming the leading tangent directions, at every size. The leading tangent directions are not abstract. At the exact $N = 2$ anchor an operator decomposition identifies them, to $R^2 = 1.00$, as the physical collective modes of H—a breathing mode (∂_ω), a correlation-hole mode, and the first relative excitation. This is not special to two bodies. Projecting the leading tangent mode of trained correlators onto a dictionary of physical operators (breathing/monopole, quadrupole, correlation-hole, pair-linear), seed-averaged over well-trained checkpoints (energy within 1.5% of reference), the CTNN reaches $R^2 \gtrsim 0.93$ at $N = 6$ and $R^2 \approx 1.00$ at $N = 12$ across the whole confinement range $\omega \in [0.01, 1]$, and both architectures reproduce the $N = 2$ naming at strong confinement (Fig. 7.2). The message-passing correlator finds the ground state by moving along the handful of coordinates the Hamiltonian actually couples.

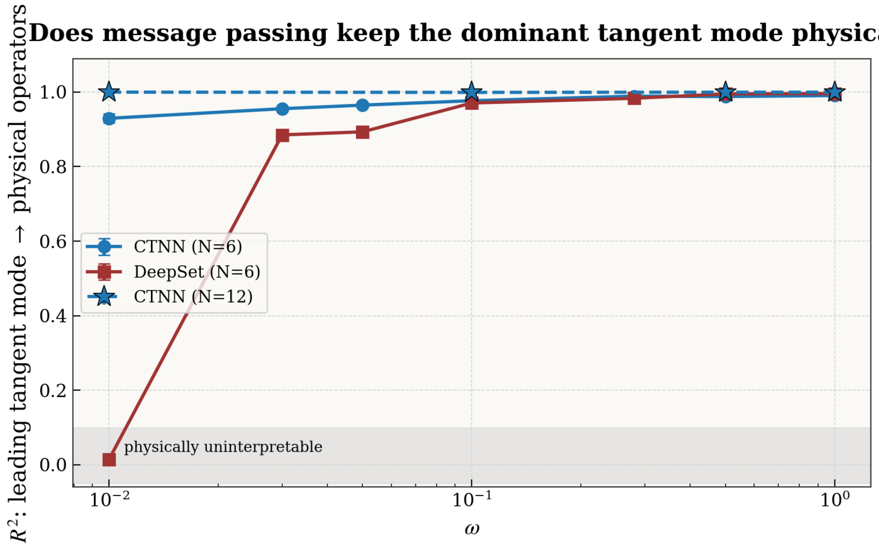


Figure 7.2: The leading tangent mode is a nameable physical mode. R^2 of the correlator's leading tangent-kernel eigenmode onto the physical-operator dictionary, seed-averaged over well-trained checkpoints. The CTNN stays physical at every ω and both sizes; the DeepSet matches it down to $\omega \approx 0.1$ and then loses the leading mode entirely in the deep-Wigner regime. Three seeds at $\omega = 0.01$.

A physical leading mode does not imply a physical tangent space. The two figures that follow can look contradictory and are not, so the distinction is worth making before the second one. Figure 7.2 scores only the *dominant* eigenmode against the dictionary; Figure 7.3 asks where the eigenvalue-weighted mass of the leading modes *as a whole* lies. A network can place its single largest mode squarely inside the dictionary and still spend most of its remaining tangent budget outside it, and that is exactly what the separable network does.

The leading-mode score is accordingly generous to it: at $\omega = 1$ the DeepSet is nominally the more physical of the two ($R^2 \approx 0.996$ vs. 0.991), with energies identical to four figures. A correlation-specific probe exposes the deficit that score hides. Write the eigenvalue-weighted fraction of the top tangent modes lying *outside* the physical-operator span as the *dictionary-orthogonal tangent mass*. The DeepSet carries a large excess at *every* confinement—0.44 against the CTNN's 0.06 at $\omega = 1$, a gap of ~ 0.3–0.5 that persists from the strongly-confined limit down to the crystal (Fig. 7.3). The reading is that a separable network reaches the same energy—correlation being a small fraction of the total at strong confinement, so the energy looks converged—while leaving nearly half its tangent budget along directions a compact physical dictionary does not resolve. A direction outside the dictionary need not be unphysical; the robust statement is that the CTNN's tangent space is far more completely accounted for by a small set of physical operators than the DeepSet's. The gap is present at every confinement rather than appearing at weak coupling; weak confinement merely removes the disguise, because there correlation *is* the energy. At $N = 12$ the contrast is starker still: the CTNN's dictionary-orthogonal mass stays $\lesssim 0.02$ across ω, while a matched DeepSet (trained to +0.2%) sits at ≈ 0.77 at $\omega = 1$—the gap *widens* with system size rather than closing.

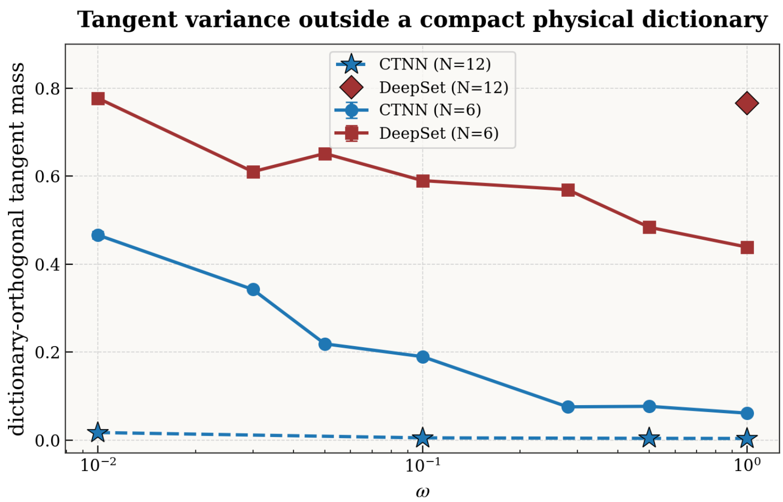


Figure 7.3: A limitation visible where the energy cannot see it. Dictionary-orthogonal tangent mass—the eigenvalue-weighted tangent variance outside the span of a compact physical-operator dictionary—versus ω. At $\omega = 1$ the two correlators are indistinguishable in energy (-0.001%) and in leading-mode physicality, yet the DeepSet already spends 44% of its tangent variance on directions not resolved by the chosen physical-operator dictionary, against the CTNN's 6%. The excess is roughly constant in ω: the gap is present throughout, disguised at high ω only because correlation is a small slice of the energy there.

7.1.2 The backflow: rank collapse and its absence

The same relational channel is decisive, in a sharper form, in the backflow. Here we fix the correlator, the data, the protocol and the optimiser—the matrix-free CG solve of §4.4, using the Woodbury identity to apply $(S + \lambda I)^{-1}$ through the $B \times B$ kernel rather than the $P \times P$ metric, which is what makes it affordable at $B \ll P$—and vary only the backflow architecture (conventional single-round messages vs. the copresheaf network), training both to convergence.

Unlike the correlator comparison, this one is *not* capacity-matched, and the reader should hold that in mind for the next two tables. At the shared production width the copresheaf network carries 182k parameters against the conventional network's 51k, because a persistent edge state is intrinsically heavier than a per-particle aggregate (§3.6); no matched pair was trained. A raw energy gap between the two is therefore not by itself evidence about architecture. What does separate the two explanations is set out after the rank measurement, and it is internal to the conventional network rather than a comparison between networks. At $\omega = 1$ both reach reference quality, but they diverge as the trap weakens (Table 7.2): the CTNN backflow stays within 0.01–0.12% across all N, ω, while the conventional backflow degrades to +0.58% at $N = 6, \omega = 0.01$ and $\sim 6\times$ the CTNN error at $N = 20, \omega = 0.01$ (+0.69% vs. +0.12%).

Table 7.2: Relative energy error (%) of the two backflow architectures, seed-averaged.

	$N = 6$		$N = 12$		$N = 20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	−0.03	+0.06	−0.01	+0.04	+0.06	+0.07
0.1	−0.05	+0.16	+0.01	+0.16	+0.02	+0.18
0.01	−0.04	+0.58	−0.01	+0.54	+0.12	+0.69

The energy gap has a sharp structural signature. Let the *backflow rank* be the participation ratio of the displacement field $\Delta\mathbf{x}$ (how many independent directions the backflow moves electrons along). The field lives in $2N$ dimensions—12, 24 and 40 at $N = 6, 12, 20$—and the zero-mean projection of Eq. (3.19) removes two of them.

One detail of the implementation makes the two columns of Table 7.3 sit under *different* ceilings, and it has to be stated because otherwise the table appears to contain impossible numbers. The diagnostic is evaluated on the displacement field as each network returns it. The copresheaf network applies the zero-mean projection inside its forward pass, so its output is already COM-free and its participation ratio is bounded by $2N-2$ (10, 22, 38). The conventional network does not: the projection is applied by the caller when $\tilde{R}'$ is formed, so the field this diagnostic sees still carries the COM component and is bounded by $2N$ (12, 24, 40). That is why the conventional column reaches 10.4 at $N=6$ and 22.2 at $N=12$—legal against its own ceiling of 12 and 24, and not against $2N-2$. The consequence for what follows is that the two columns may not be compared *in level*: roughly one unit of the conventional network's rank is the centre-of-mass direction that the copresheaf network has already removed. It does not affect the finding, which is a collapse to rank 1—two orders below either ceiling.

The obvious control is to apply the same zero-mean projection to both fields before computing the participation ratio, which would put them under a common $2N-2$ ceiling and make the levels directly comparable. That was not possible here: the diagnostic was computed during training and the trained checkpoints for this sweep were not retained, so it cannot be recomputed from the saved artefacts. We therefore make no comparison of level between the two columns anywhere, and the reader should not either. Toward Wigner the *conventional* raw field undergoes a spectral collapse to rank 1, while the CTNN backflow stays near the ceiling (Table 7.3). The endpoint is the same at every size—rank 1, from ceilings of 12, 24 and 40 at $N=6$, 12 and 20—so what grows with N is the height of the fall, not the sharpness of it. At $N=6$ and $N=12$ both seeds return exactly 1.00; the $N=20$ entry comes from a separate re-run of that cell and is single-sourced (see the table note). The tangent dimension tells the same story from the other side: toward Wigner d_{eff} grows for CTNN ($\sim 4 \to 10$) and collapses for the conventional backflow ($\sim 6 \to 1$). The collapse to *exactly* rank 1 is seed-robust where two seeds exist ($N=6$ and $N=12$). Rank is not the discriminator above the crystal: at $\omega \geq 0.1$ the conventional field is nominally the higher-rank of the two (Table 7.3), though that comparison is confounded by the differing ceilings noted above, and the two architectures are then also within 0.2% of each other in energy. It is the collapse, not the level, that carries the finding.

Why a simple global capacity shortage is unlikely to explain the collapse. The conventional network is a third the size of the copresheaf one, so the obvious alternative reading is that it simply runs out of parameters at weak confinement. The conventional network's own ω-dependence makes a simple global shortage unlikely. Capacity does not vary with ω: the same 51k network, trained on the same protocol, holds a displacement rank of 10.4–10.8 at $N=6$, 21.7–22.3 at $N=12$ and 33.7–38.7 at $N=20$ for every $\omega \geq 0.1$—within a unit or two of its own ceiling of 12, 24 and 40—and then falls to exactly 1.00 below $\omega=0.1$. This abrupt change disfavors the reading that the network is simply too small throughout the grid. It does not exclude a regime-dependent capacity limitation: without a widened conventional-backflow control, the change in target complexity at weak confinement remains a confound.

The same argument does *not* rescue Table 7.2. The energy gap there is a between-network comparison at unequal size, and nothing in our data separates the architecture from the extra 131k parameters. We report it as what the two production ansätze achieve—which is a real and useful statement, since these are the models that produced the energies of Chapter 6—and not as an architecture ablation. The missing experiment is a widened conventional backflow at matched capacity; it was not run.

Table 7.3: Backflow displacement rank (participation ratio), seed-averaged, evaluated on the field as each network returns it. The ceilings differ by construction: $2N-2$ (10/22/38) for the copresheaf network, which projects out the centre of mass inside its forward pass, and $2N$ (12/24/40) for the conventional one, which does not, so the columns are comparable in their collapse and not in their level. The two networks are also not capacity-matched (51k against 182k; §3.6), which is why the collapse is argued from the conventional network's own ω-dependence rather than from the comparison. Provenance: all entries are seed means over two seeds from the scaling sweep, except the $N=20$, $\omega=0.01$ column, whose cells exhausted GPU memory in that sweep and were re-run separately; the per-run rank artefacts of that re-run were not retained, so unlike every other entry it cannot be regenerated from the saved outputs.

	$N=6$		$N=12$		$N=20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	8.8	10.4	20.8	22.2	36.9	35.4
0.1	7.7	10.4	18.2	21.7	33.8	37.6
0.01	9.8	**1.0**	20.8	**1.0**	35.2	**1.0**

Pricing each component by deletion. We price each component by ablation on a *common probe*: configurations are drawn once from the full model's $|\Psi|^2$ and every arm is re-evaluated on them, so the trap and Coulomb terms are held fixed by construction and each figure below is the change in the kinetic term at fixed measure—the instantaneous cost of removing a component, not the energy of a re-relaxed state. Two seed-stable facts emerge. (i) The **relational work migrates from the nodes to the amplitude** as the trap weakens. Deleting the backflow costs +5.6% at $\omega = 1$ and +1.4% at $\omega = 0.1$, but only +0.013–0.017% at $\omega = 0.01$ (three seeds), while deleting the Jastrow messages runs the other way: +3.9% at $\omega = 1$ rising to +9–15% at $\omega = 0.01$. At the crystal the backflow is very nearly inert—with the Jastrow messages already removed, deleting the displacement as well changes the energy by under 0.01%. "Message passing buys kinetic energy through the nodes" is a high-ω statement only. (ii) **Within the backflow, the transport maps *are* the displacement**: zeroing the inter-particle transport maps $(\rho_{v\to e}, \rho_{e\to v})$ perturbs $\Delta\mathbf{x}$ by as much as deleting it outright—the mean $|\Delta\mathbf{x}_{\text{abl}} - \Delta\mathbf{x}|$ is comparable to the mean $|\Delta\mathbf{x}|$ at every (N, ω)—so what displacement the network produces comes from the messages rather than from the per-particle path. Both architectures condition each particle on the relative coordinates $\mathbf{r}_{ij}$, so neither is blind to inter-particle structure; the difference is that the conventional backflow uses a single round of pairwise messages aggregated per particle, whereas the CTNN maintains and transports explicit edge state across the graph. Empirically the single-round backflow does not *sustain* a full-rank relative displacement at the crystal—it collapses to a single collective displacement mode (rank 1)—while the CTNN's copresheaf transport holds the full-rank response, and the ablation shows that this transport carries essentially the entire effect.

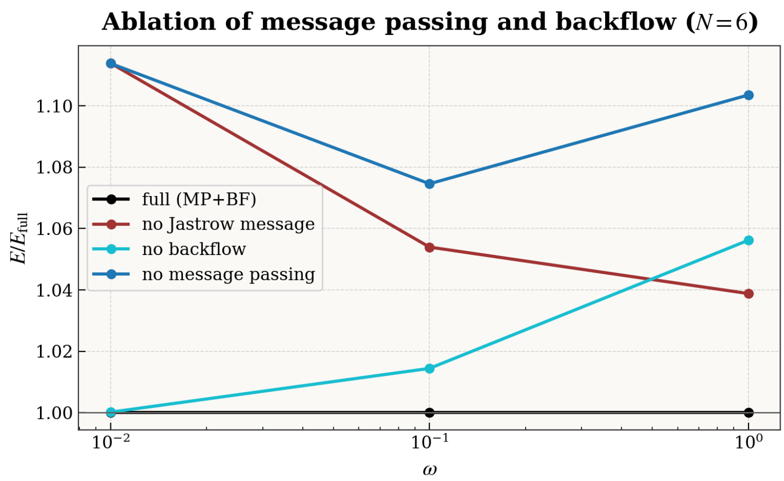


Figure 7.4: The *fixed-measure instantaneous ablation cost* of each component (trained networks, N=6). Configurations are drawn once from the full model's $|\Psi|^2$ and every arm is re-evaluated on them, so the trap and Coulomb terms are identical across arms by construction and the plotted ratio $\Delta E_{\text{probe}}/E_{\text{full}}$ is the instantaneous change in the kinetic term when a component is switched off. It is *not* the energy of a network retrained without that component, which would be smaller, since nothing is allowed to re-relax. Removing the backflow (cyan) costs $\sim$6% at ω=1 but under 0.02% at ω=0.01, while removing the Jastrow messages (dark red) grows more expensive as the trap weakens, from 3.9% to 11.4%; the two curves cross near $\omega \approx 0.4$. Removing all message passing (blue) tracks the Jastrow-message curve and meets it at ω=0.01: once the correlator's messages are gone, deleting the displacement as well costs nothing measurable. The relational work migrates from the nodes to the amplitude: at the crystal it is the correlator's messages, not the backflow, that carry it. This is a statement about how *this* trained model divided the correlation between two substitutable components, and so is conditional on the staged training protocol rather than gauge-invariant (§7.5); the ablation costs themselves are reproducible at fixed protocol.

7.1.3 A common relational bottleneck

The two effects have a candidate common cause in how relational information is *routed*, and the correlator-swap experiment below is what raises that from a resemblance to a supported reading. The baselines are not blind to inter-particle structure—the DeepSet pools independently-encoded pairs, and the conventional backflow aggregates a single round of pairwise messages—but neither maintains a persistent, transported edge representation: each pairwise term is computed and discarded. The copresheaf CTNN does maintain one, and moves it back and forth between nodes and edges through learned transport maps ($\rho_{v\to e}, \rho_{e\to v}$); the message-passing correlator iterates this exchange over a V-cycle, while the backflow applies a single such copresheaf round. Where a compact representation exists this *compresses* the correlation onto fewer tangent directions (Jastrow: smaller d_{eff}); where the physics demands a genuinely collective, full-rank response the single-round baseline instead collapses to a single collective displacement mode (backflow: rank 1), which the copresheaf transport avoids. Both effects are visible only in the tangent-kernel geometry, not in the total energy, until weak confinement makes the geometric deficit an energetic one.

The two channels fail together, and an intervention shows it. Two components failing at similar confinements is a resemblance. What makes it evidence is that moving one of them moves the other. The experiment has four steps.

Observation 1. The correlator's grip on the physical correlation modes gives way at some confinement, and where depends on its architecture (§7.1.1).

Observation 2. The conventional backflow's displacement rank collapses at some confinement (§7.1.2).

Intervention. Hold the backflow *architecture* fixed—the conventional network in both arms—and swap only the correlator, retraining each arm to convergence. Then track both quantities down the ω-sweep (Fig. 7.5).

Result. The backflow's failure boundary moves with the correlator. With a separable correlator both give way together near $\omega \approx 0.05$—hole capture from 0.68 to 0.07, displacement rank from $\sim$10 to 1 across the same step. With a message-passing correlator both hold half a decade further, to $\omega \approx 0.01$.

The inference is that a single relational capability underlies both, since intervening on one component relocates the other's failure. Note what the intervention *is*: each arm is a separately trained network, not one network reconfigured, so "held fixed" means the same backflow architecture and the same recipe, never the same weights. The correlator's grip on the physical correlation modes and the backflow's ability to sustain a full-rank displacement are the same capability seen from two sides: when the amplitude model can no longer resolve the correlation hole, the nodal deformation can no longer stay high-rank, and a better correlator postpones *both*. The evidence—coincident transitions that shift together under a correlator swap—supports a common relational bottleneck beneath both rather than proving them literally identical.

7.1.4 What the trained representation looks like

One descriptive result belongs beside these controlled comparisons, provided it is read at its own evidential level. Examining a single high-quality trained solution (Appendix E) shows the correlator occupying a feature space of one to three effective dimensions across the entire grid, and never expanding: at $\omega = 10^{-3}$, where the real-space density has developed rings and discrete occupancies, the latent rank is no larger than at $\omega = 1$. For N=6 and N=12, what changes across the crossover is which branch the leading axis draws on—pair features at moderate coupling, single-particle and global features at strong coupling—not how many axes there are. The N=20 component diagnostic is an explicit exception (Appendix E). The displacement field is the opposite kind of object, spanning nearly all the spatial directions available to it, and it leaves the correlator's feature geometry almost unperturbed (centred kernel alignment above 0.98 everywhere).

These are component-level measurements on one training run, so they are gauge-like in the sense of §7.5 and none of the architectural conclusions above depends on them. They are worth recording because a low effective covariance rank in the tapped representation is not what one would have predicted for a state whose real-space structure is this elaborate, and because the reorganisation happens near $\omega \approx 0.1$—the same confinement at which the energy partition crosses over (§6.2.3) and at which the ablation costs

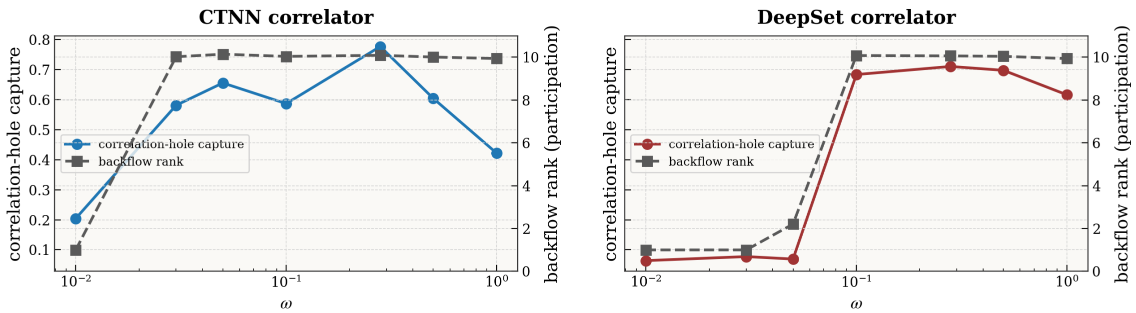


Figure 7.5: Correlator physicality and backflow rank collapse together ($N = 6$; the backflow architecture is held fixed, only the correlator changes). For each correlator, the correlation-hole capture (left axis) and the backflow displacement rank (right axis) give way at the same ω: $\omega \approx 0.05$ for the DeepSet, $\omega \approx 0.01$ for the CTNN. A correlator that keeps its physical structure lets the backflow keep its rank.

of Fig. 7.4 exchange order. Whether that coincidence reflects one mechanism or three we cannot say from a single run.

What §7.1 establishes. Message passing supplies a relational channel whose effect is geometric before it is energetic. In the correlator it shows as a tangent space that is smaller at strong confinement and better accounted for by a compact, physically motivated operator dictionary, with a local-energy variance lower at every confinement by factors of 1.2 to 3.4. In the backflow it shows as a collapse averted, costing half a percent where it is not. The correlator-swap intervention ties the two together. What this does *not* establish is that the two architectures reach different states: they agree in energy to four figures, and we did not measure their overlap in this comparison.

7.2 Conditioning and the sampling measure

Two design choices remain, and the tangent geometry settles both at once. The optimiser question—what does a natural-gradient step buy over a diagonal one—and the paradigm question—what changes when the loss is defined by an analytic proposal instead of by $|\Psi|^2$—turn out to be the same question asked twice. Stochastic reconfiguration inverts the metric S. The sampling measure is what builds S in the first place. Neither can be understood without the other, so we take them together.

7.2.1 Natural-gradient preconditioning

Stochastic reconfiguration preconditions the gradient by the inverse QGT, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, S^{-1}\nabla_{\boldsymbol{\theta}} E$; geometrically it *whitens* the tangent kernel. At $N = 2$ this is exact: the SR step aligns with the imaginary-time direction to numerical precision. The question is where whitening changes the outcome.

Fixing the ansatz and varying only the optimiser from a common stage-3 base: **for $|\Psi|^2$-VMC, SR and Adam are indistinguishable**—exact at $N = 2$ (var $\sim 10^{-9}$), and at $N = 6$ the energy gap $E_{\mathrm{Adam}} - E_{\mathrm{SR}}$ is $\lesssim 0.002$ across ω and both seeds; in the matched pair at $\omega = 1$ the two agree to four decimals in relative error. **For weak-form collocation from the same common base, SR is decisive**: there $E_{\mathrm{Adam}} - E_{\mathrm{SR}}$ is large and positive (e.g. $+2.9$ at $N = 6, \omega = 1$), and plain Adam does not converge the objective.

That statement is about a controlled comparison, and it needs reconciling with §7.3.1, where Adam is the production default and the entire 30-run reliability campaign uses it successfully. Both are true, and the difference between them is the rest of the pipeline. Stripped to a bare objective and a fixed proposal, the collocation gradient is badly enough behaved that a diagonal step stalls and preconditioning rescues it. The production recipe never runs in that condition: adaptive proposal refitting, $16\times$ oversampling, reward normalisation, instability rollback and cascade warm-starting between confinements each attack the gradient's conditioning at its source, and once they are in place a diagonal step suffices. The two findings are therefore complementary—SR substitutes for sampling control, and sampling control substitutes for SR—with one asymmetry that decides the production choice: below $\omega \simeq 0.01$ the sampling that builds S collapses, so the substitution stops working in one direction only (§7.3.3). Adam is the default because it degrades gracefully where SR does not.

A large condition number alone does not explain the difference. Table 7.4 reports $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measured on the trained states under $|\Psi|^2$ sampling, and it is enormous—10^7 to 10^{10}—for *both* architectures and at every capacity in the ladder, including precisely the models on which Adam and SR agree to four decimals. Whatever makes SR unnecessary under $|\Psi|^2$ sampling, it is not that the metric is well conditioned.

Table 7.4: Condition number of the empirical quantum geometric tensor at $N{=}6$, $\omega = 1$, measured on each trained state's own $|\Psi|^2$ samples at fixed batch $B = 768$. Because $B < P$, $\lambda_{\min}$ is the smallest sample-supported eigenvalue and κ is a property of the metric the optimiser actually sees (§5.3); the values are comparable to one another but not across batch sizes. Adam and SR reach the same energy on every row of this table.

architecture	params	$\kappa(S)$	d_{eff}
CTNN (seed 1)	79.8k	6.9×10^7	1.46
CTNN (seed 2)	79.8k	5.2×10^7	1.39
DeepSet (small)	20.2k	3.2×10^9	3.56
DeepSet (medium)	47.7k	3.4×10^7	4.35
DeepSet (matched)	89.4k	3.2×10^{10}	3.61
DeepSet (large)	164k	1.4×10^7	4.76

SR matters when the tangent metric is ill-conditioned and still estimable, and it is the second half of that sentence that does the work. Anisotropy is certainly present under $|\Psi|^2$ sampling, where we measured it; we did not measure κ under the collocation measure at matched batch size, so what the table licenses is the negative half of the argument—that conditioning does not distinguish the regime where SR is unnecessary—rather than a matched comparison between them. What separates the two regimes, on the evidence we do have, is whether a batch can resolve the metric. Under $|\Psi|^2$ the samples are drawn from the very measure that defines S, so the directions carrying Fisher weight are also the directions the batch populates, and Adam's diagonal step already moves along them. Under a fixed covering proposal the estimator of S is built from reweighted points that increasingly miss the target, and preconditioning supplies structure the diagonal cannot. We have not measured κ under the collocation measure at matched batch size, so this is an argument from the estimator, not a second condition-number comparison; what the table establishes is the negative half—that raw conditioning does not distinguish the two regimes.

"More ill-conditioned $\Rightarrow$ more useful SR" cannot hold without bound, since inverting S requires S itself to be *reliably estimated.* SR helps when the tangent metric is anisotropic but its empirical estimate is trustworthy; once the sampling that builds S collapses—as the importance weights do deep in the Wigner regime—the empirical S becomes noisy and rank-deficient, and preconditioning by its inverse amplifies that noise rather than the signal. This is exactly the behaviour seen at ultra-weak confinement, where CG-SR on the collocation objective becomes unstable (§7.3.3). SR pays off in the window where the metric is both ill-conditioned and estimable, and fails once estimation itself breaks down.

The VMC equivalence of Adam and SR is established here at $N = 2$ and $N = 6$, and we have not verified that it survives to $N = 12$ and $N = 20$. The mechanism we propose—that $|\Psi|^2$ sampling makes the metric *estimable* on the batch, whatever its conditioning, and so lets Adam's diagonal preconditioning suffice—predicts that it should, since the argument turns on the sampling measure and not on dimension. But it is a prediction there, not a measurement, and testing it is the obvious extension. The same applies to the reverse reading of Table 7.4: κ is measured at one N and one ω, and we do not know how it moves with either.

7.2.2 Consequences of the sampling measure

Between $|\Psi|^2$-sampled VMC and importance-sampled collocation the wavefunction family is identical; only the measure that defines the loss changes. That change has three consequences, and underneath them one cause.

Target sampling aligns the estimator with the metric; a covering proposal does not. Sampling from $|\Psi|^2$ makes the estimators of S and of the gradient share the measure that defines them, so the directions carrying Fisher weight are also the directions the batch populates. That is an alignment between estimator and objective, not preconditioning in the sense of applying S^{-1}. A fixed covering proposal instead drives a monotone *ESS collapse* as N and $1/\omega$ grow: at $N = 6, \omega = 0.01$ the effective sample size falls to 0.11% of the batch: about 5 of the 4096 *retained* points. Two normalisations are in use in this chapter and they differ by the oversampling factor—ESS as a fraction of the $n_{\text{coll}} = 4096$ resampled points, as here, and ESS as a fraction of the $m\, n_{\text{coll}}$ raw proposal draws, as in §7.3.4. Each figure states which. When ESS collapses the gradient estimator is degenerate—a hard limit, not undertraining.

§7.3.4 below nonetheless reports usable collocation energies at the same (N, ω)—+0.24–0.29% across three seeds, Table 7.7—and the two situations differ in the sampler and in the quantity being described. The sampler first: the collapse measured above is for the *static* origin-centred mixture, while the campaign's robust recipe refits its proposal every 30 epochs and oversamples 16×. The campaign's *baseline* rows, which keep the static proposal, are the ones that blow up (+3.9 to +7.2% at $\omega = 0.001$). Second, and more interesting, the two statements are not about the same quantity. The energy stays within a third of a percent while the state overlap ceases to be resolvable—which is the sharpest illustration in this thesis of why energy alone does not certify a state. We stop short of saying the overlap *is* zero: $\mathcal{S}^2$ is a ratio estimator built from the same degenerate weights, so at this ESS it reports *unresolved* rather than orthogonal.

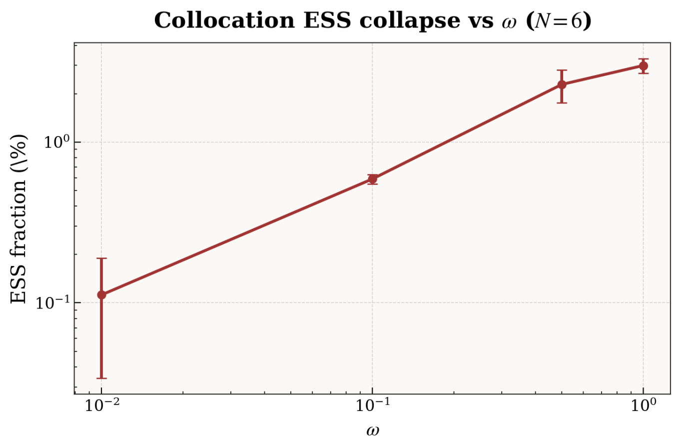


Figure 7.6: The naive collocation proposal collapses in the Wigner regime (N=6). The effective sample size (as a percentage of the batch) falls by more than an order of magnitude from ω=1 to ω=0.01, where it reaches ∼0.1%—about five usable points in a batch of 4096. This is a property of the sampling measure, not of optimisation, and it is what the physics-informed ring proposal (§7.4) is designed to repair.

State agreement degrades with proposal overlap. We test whether the two paradigms reach the same *state* (not merely the same energy) via the symmetric overlap $\mathcal{S}^2 = \langle \Psi_B/\Psi_A \rangle_{|\Psi_A|^2} \langle \Psi_A/\Psi_B \rangle_{|\Psi_B|^2}$. At $N = 2$ they are identical ($\mathcal{S}^2 > 0.997$ for every ω, both optimisers, both seeds): the paradigm is efficiency-only, collocation paying ∼5× the variance for the same wavefunction. At $N = 6$ they *diverge*: under Adam $\mathcal{S}^2$ falls from 0.85 at $\omega = 1$ to 0.77 at $\omega = 0.1$ (both seeds), tracking the ESS collapse, and at $\omega = 0.01$ the estimator returns values near zero but is no longer resolved—at an ESS of about five points it is a ratio of two degenerate importance-weighted averages, and we record it as *unresolved* rather than as a measured orthogonality. SR partially rescues the $\omega = 0.1$ value ($\mathcal{S}^2 \approx 0.95$) but does not close the gap. Weak-form collocation thus has a **domain of validity**—equivalent to VMC when the importance sampling is healthy, degrading as correlation strengthens. The two halves of this section meet here: the paradigm that needs SR is the one that breaks down at scale. An adaptive proposal that tracks $|\Psi|^2$ is the natural remedy, pursued in §7.4.

Why the collocation energy readout is biased. For an exact eigenstate the local energy $E_L = \hat{H}\Psi/\Psi$ is constant over configuration space, so $\mathrm{Var}(E_L) \to 0$ as the ansatz approaches the ground state—

the zero-variance principle, and the reason $\mathrm{Var}(E_L)$ is a sharp quality measure. That cancellation is a property of the *full* local energy, in which the kinetic (Laplacian) and potential terms combine to a constant; it is not a property of the Laplacian term alone. The collocation objective does *not* discard the Laplacian: it is a score-function (REINFORCE) estimator that *forms* the full E_L as a detached reward and differentiates only $\log|\Psi|$ (§7.3.1). What it gives up is the *measure*, and it is worth separating the three distinct reasons the importance-sampled readout is not a trustworthy convergence criterion, because they are not equally fundamental. First, the estimator is *self-normalised*: we divide by $\sum_i w_i$ because $|\Psi_\theta|^2$ is unnormalised, and a ratio of two Monte Carlo estimates carries an $\mathcal{O}(1/n)$ finite-sample bias even when both are unbiased. This vanishes with sample size and is the mildest of the three. Second, the tail controls *deliberately* change the target: tempering $w \to w^\alpha$ with $\alpha < 1$ and clipping the log-weights at an upper quantile both trade bias for variance on purpose, so the quantity estimated is no longer $\langle E_L \rangle_{|\Psi|^2}$ at all. Third, and dominant in the regimes that matter here, proposal–target mismatch inflates the variance so far that the realised estimate is unreliable at any sample size we can afford (§7.3.4). Ordinary importance sampling is not biased merely for being importance sampling; it is the last two that make the readout unusable as a stopping rule. The reliable criteria are $\mathrm{Var}(E_L)$ and the state overlap, both evaluated on $|\Psi|^2$ samples.

What §7.2 establishes. The value of a natural-gradient step is not predicted by the conditioning of the metric, which is large under $|\Psi|^2$ sampling on exactly the models where SR buys nothing. What predicts it is whether the batch can resolve the metric it inverts. This is a claim about the estimator, and its negative half is directly measured while its positive half—that collocation occupies an anisotropic but still estimable middle regime—is inferred, since we did not measure κ under the collocation measure at matched batch size.

7.3 Training without a Markov chain

The conditioning story has a practical edge. If the sampling measure is what makes the metric estimable, then choosing that measure is a design decision, not a fact of nature—and the obvious thing to try is a measure with no Markov chain in it at all.

Two walls. What follows is easier to hold if the destination is given first. The Markov-chain-free route runs into two limits of entirely different kinds, and almost every result in §7.3–§7.4 is located against one or the other.

The first wall is statistical. A fixed analytic proposal loses overlap with $|\Psi_\theta|^2$ as the trap opens; the importance weights grow heavy-tailed; the gradient estimator comes to be dominated by a handful of draws; the empirical metric becomes unreliable and preconditioning by it does harm. Everything on that chain is a sampling problem, and a better proposal moves the wall—from $\omega \approx 0.01$ with an origin-centred mixture to $\omega = 0.0035$ with one built out of the measured Wigner geometry.

The second wall is not. Below $\omega = 0.0035$ a converged state fails to survive transfer to a weaker trap even when the sampler is initialised directly on the correct shell. The Wigner peak exists only through a near-cancellation between an exponentially suppressed determinant and a Jastrow that amplifies it back, and a 23% change in ω tips that balance. No change of proposal repairs it, because the proposal is not what fails.

§7.3 establishes the first wall and the mechanism behind it; §7.4 moves it and then runs into the second.

MCMC is what makes neural variational Monte Carlo sequential. Chains must equilibrate, successive samples are correlated, and mixing times grow with particle number and with correlation strength—which is to say, they grow fastest exactly where the physics gets interesting. Draw the configurations instead from a fixed analytic proposal and reweight them, and the whole gradient step becomes embarrassingly parallel. The question is what that costs.

Two ways of posing the problem behave very differently, and the difference decides everything downstream. The *strong form* asks the wavefunction to satisfy the local Schrödinger equation pointwise, minimising the variance of E_L and differentiating through the Laplacian to do it. The *weak form* uses a score-function (REINFORCE) gradient in which the Laplacian enters only as a reward and is never differentiated. Appendix C shows that the strong form meets a structural obstruction the moment it is asked to train a backflow: the learning signal lives at the nodal surface, where E_L diverges for any imperfect node, and no combination of sampling, loss or preconditioning breaks an accuracy floor near

0.3%—a floor set by the Jastrow alone, with the backflow contributing almost nothing. That is a property of the objective, not of the idea. Everything below uses the weak form.

On the same Slater–Jastrow–backflow ansatz, and with no Markov chain in any gradient, the weak form recovers the variational energy to statistical consistency with DMC at strong confinement and to a few tenths of a percent of our gold standard well into the correlated regime for $N \in \{6, 12\}$ (Table 7.5). The energies are collected here, where the method that produced them is explained, rather than beside the SR–VMC energies of §6.1: without the failure mechanism of §7.3.4 in hand, the pattern of where they hold and where they give way is not interpretable.

Table 7.5: Ground-state energies from Markov-chain-free collocation training, all rows using the backflow+Jastrow ansatz. Each run selects its best checkpoint using a VMC probe (≥20k samples) and reports a final IS evaluation; the latter is the table's scoring estimator. For N=6 the headline number is the *reliability of that evaluated protocol*, not its luckiest run: the campaign of §7.3.4 fixes a 1200-epoch budget and the hyperparameters and varies only the seed, and both the three-seed mean ± standard deviation and the best of the three are given. The N=2 rows are single collocation runs, zero-variance extrapolated. The N=12 rows are marked †: no campaign was run at that size, so they are the best of the independent runs and are *not* protocol-controlled—read them as what the method reached, not as what it reaches reliably. The reference column is the external value where one exists and, where none does (N=6 at $\omega \leq 0.01$ and N=2 at $\omega = 10^{-3}$; see Table 6.2), our own converged SR–VMC energy, in which case the deviation measures internal consistency rather than absolute accuracy. Deviations are $100\,(E - E_{\text{ref}})/E_{\text{ref}}$.

				relative deviation (%)	
N	ω	Reference	Best energy	best	mean ± s.d.
2	1.0	3.00000	3.00035	+0.012	—
	0.5	1.65977(1)	1.66014	+0.022	—
	0.1	0.44079(1)	0.44077	−0.005	—
	0.01	0.073839(2)	0.073840	+0.001	—
	0.001	0.013778	0.013799	+0.152	—
6	1.0	20.15932(8)	20.1620(56)	+0.013	$+0.050 \pm 0.049$
	0.5	11.78484(6)	11.7850(48)	+0.001	$+0.054 \pm 0.055$
	0.1	3.55385(5)	3.5632(7)	+0.263	$+0.286 \pm 0.034$
	0.01	0.69036	0.69200(1)	+0.238	$+0.269 \pm 0.028$
	0.001	0.140832	0.14118(1)	+0.247	$+0.281 \pm 0.044$
12†	1.0	65.7001(1)	65.706(4)	+0.009	—
	0.5	39.1596(1)	39.169(4)	+0.024	—
	0.1	12.26984(8)	12.2824(6)	+0.102	—

7.3.1 The weak-form objective

The collocation training used here is exactly the MCMC-free pipeline specified in the Methods: a weak-form (REINFORCE [46]) objective in which the local energy enters as a detached reward (§4.5, Eq. (4.8)), an analytic Gaussian-mixture proposal reweighted to $|\Psi_\theta|^2$ with ESS / Pareto-$\hat{k}$ monitoring [47] and tail controls (§4.5.1), and—in the Wigner regime—the physics-informed ring proposal with cascade warm-starting (§4.5.2). Two methodological findings shape what follows. First, among the optimisers (§4.4), **Adam** was the robust default across all regimes, whereas **CG-SR** converged faster at $\omega \geq 0.1$ but diverged for $\omega \leq 0.01$, where the empirical metric becomes statistically unresolved. Second, **cascade warm-starting**—initialising each confinement from the converged checkpoint of a stronger one—was decisive for reaching the deep-Wigner regime, exploiting the transfer of correlation structure across ω.

Three ansatz families were tried within this framework, all built on the same Slater–Jastrow core and differing only in what dresses it; their definitions and sizes are in §3.6. The **backflow+Jastrow** family produced the best results at N=6 and N=12 and carries every number below. The **Jastrow-only** family was the fastest and most robust to train, served as the initialiser for the backflow runs, and becomes the better choice at N=20 for the memory reason of §7.3.5. The **neural Pfaffian** [48], which replaces the determinant with a more general antisymmetric form and in principle offers richer nodal topology, reached only +0.43–0.50% at N=6, ω=1 against +0.013% for backflow+Jastrow: its extra expressivity was not exploitable under this sampling regime, because the larger parameter space amplifies importance-sampling gradient noise faster than the richer nodes repay.

Why the weak form is stable where the strong form is not. The catch-22 analysis (Appendix C) identified a fundamental obstacle for direct PDE-residual training of the many-body Schrödinger equation: the backflow learning signal is concentrated near nodes and coalescence points where the local energy E_L diverges, but these are precisely the configurations where second-derivative pathways are ill-conditioned. Backpropagating through the Laplacian in such regions produces pathological gradients that destabilise training.

The REINFORCE formulation resolves this by construction. The Laplacian enters only through the *reward* (the local energy in Eq. (4.2)), and gradients act exclusively on the score function $\nabla_\theta \log|\Psi|$. The stability is subtler than the score simply being bounded. For a *parameter-dependent* (backflow) node the score itself can diverge— $\nabla_\theta \log|\Psi| \supset (\nabla_\theta D)/D$ grows like $1/d_\theta$ as the learned determinant $D \to 0$ near its node. What matters is that this divergence is *integrable* under the $|\Psi|^2$ sampling measure, which weights it by $|\Psi|^2 \sim d_\theta^2$ (and the near-node points are further suppressed by the importance screening). REINFORCE avoids the *stronger*, non-integrable divergence of the strong-form residual, in which the Laplacian term $\nabla^2 D/D$ enters the gradient directly (Appendix C). Keeping the Laplacian in the *reward* but out of the *gradient flow* is what lets the collocation pipeline train backflow and Jastrow parameters stably in regimes where direct residual minimisation fails.

The superiority of pure REINFORCE (β=0) over the hybrid objective (β>0) has a related explanation. The direct weak-form term $\tilde{E} = \frac{1}{2}\|\nabla \log \Psi\|^2 + V$ is cheaper to evaluate (no Laplacian needed). Averaged under the current $|\Psi|^2$ measure, it equals the variational energy by integration by parts. The implemented direct path, however, differentiates $\tilde{E}(R)$ at retained samples without the score contribution from the changing sampling measure; $V(R)$ then has no explicit parameter derivative and the update behaves as a kinetic-only pathwise gradient. At nonzero β, the resulting gradient conflict between the two objectives manifests as oscillatory behaviour in the energy, with the REINFORCE term pulling toward the variational minimum and the direct term pulling toward a kinetic-energy minimum that is generally distinct.

7.3.2 Collocation performance

N=6**: near-reference accuracy across all** ω**.** For six electrons, collocation training reaches DMC-level accuracy where DMC exists: at $\omega \geq 0.5$ the errors are $\lesssim 0.01\%$, which is statistically consistent with the reference once the collocation uncertainty is included—though it is several times the DMC error bar taken alone, so "consistent" here is a statement about the combined uncertainty and not about matching DMC's own precision. At the most extreme confinements, $\omega \in \{0.01, 0.001\}$, where the electrons are separated by $\sim$30–120 a_0^* (a few oscillator lengths) and no DMC reference is available, the energies stay within 0.1–0.25% of our converged variational gold standard; this is agreement between two independent training paradigms on the same ansatz, not a certificate of absolute accuracy. The best campaign result at ω=1.0 is E=20.1620 ± 0.0056 (+0.013%), which lies within 0.5σ of the reference on the combined uncertainty—dominated by ours, since the deviation is some tens of times the DMC error bar of 8×10^{-5}.

These results are achieved without any Markov chain in the training gradients. MCMC enters only outside the gradient loop: a small Metropolis probe periodically selects the best checkpoint, and the final heavy evaluation certifies the energy. The training itself uses 4096 importance-resampled collocation points per epoch with an oversampling factor of 8, requiring $\sim$32k proposal evaluations per gradient step. That is ten times the $\sim 3\times10^3$ samples per step used by the SR route, but each is a single independent amplitude evaluation rather than a step of a correlated chain that must first equilibrate. The collocation batch is also embarrassingly parallel where the chain is not, though we did not benchmark the two pipelines against a common wall-clock budget.

Staged continuation is what makes those budgets pay, and an exploratory run at ω=1.0 shows the effect cleanly: an initial BF+Jastrow joint REINFORCE stage (900 epochs, lr=5×10^{-4}), a low-LR continuation (750 epochs, lr=3×10^{-4}) and a hard-focus stage (500 epochs, lr=2×10^{-4}), each inheriting the previous checkpoint, improve monotonically through +0.08%, +0.05% and +0.010%. That run was hand-tuned and is not among the protocol-controlled numbers of Table 7.5; we report it because it is the observation that put cascade warm-starting into the campaign's robust recipe (Table 7.6).

N=12**: sub-**0.03% **at strong confinement.** For twelve electrons, the BF+Jastrow collocation results track DMC closely at $\omega \geq 0.5$, with best-run errors of +0.009% and +0.024% at ω=1.0 and 0.5 respectively

(Table 7.5). No reliability campaign was run at this size, so these are the best of the independent runs and not protocol-controlled in the way the N=6 rows are. At ω=0.1 the error increases to +0.102%, reflecting the growing difficulty of importance sampling in 24 dimensions at weak confinement. The ESS at ω=0.1 is typically 5–15 per epoch (out of 4096 kept points), indicating that the Gaussian mixture proposal still maintains usable overlap with $|\Psi|^2$.

Attempts to extend to $\omega \leq 0.01$ at N=12 were unsuccessful: both direct starts and cascade warm-starts from ω=0.1 failed to converge, with the ESS collapsing to 1 and the rollback mechanism rejecting all gradient updates. This marks the boundary of the current collocation approach for N=12.

N=20 **exposes the memory–sampling trade-off.** Twenty electrons in two dimensions push the collocation pipeline against a hardware wall rather than a conceptual one. The backflow's edge-feature tensor scales as $\mathcal{O}(N^2 \times h_{\text{bf}})$, which at N=20 forces the hidden dimension, the collocation budget, or the oversampling factor down far enough that the gradient quality suffers—so much so that a Jastrow-only ansatz, freed of that cost, trains better than the backflow ansatz it should in principle beat. Because the limitation is one of memory and gradient noise rather than of the method, we report the N=20 collocation results, and the reversal that exposes the bottleneck, in Appendix D.2 rather than here.

7.3.3 Ablations and failed variants

Most of the method space we explored did not work. The failures are reported here beside the successes, because a working configuration tells you only that some set of choices is sufficient, whereas a failure chased far enough to name its mechanism tells you which choice was load-bearing—and that transfers to systems other than this one. Each attempt below was, at the time, an intuitively reasonable idea. Each turned out to violate an assumption the method had never been forced to state.

Table 7.6: Collocation variants tried, and what each one settled. Errors are relative to the reference of Table 7.5. The three marked † are taken up in prose below; the rest are recorded for completeness.

Attempt	Outcome	Mechanism
Pure REINFORCE (β=0) over the hybrid objective	adopted	The direct path omits the sampling-measure score contribution; at $\beta > 0$ it contributes a kinetic-only pathwise gradient that conflicts with REINFORCE, and the two gradients oscillate against each other.
Cascade warm-start from higher ω	essential	Cold starts never converged at ω=0.01 for $N \geq 6$. Cascading from ω=0.1 cut the initial error from ∼30% to ∼5% and made +0.15% reachable in ∼1000 epochs.
ESS-adaptive oversampling with instability rollback	essential	Raises the proposal budget when overlap degrades and reverts on energy spikes. Without both, ω=0.001 diverged within a few hundred epochs.
CG-SR below $\omega \simeq 0.01$ †	abandoned	+4–36% against +0.15–0.24% from Adam. Not ill-conditioning but unestimability; see below.
Langevin proposal refinement †	abandoned	Worse in every configuration tested (+153% against +5.7% at N=20, ω=0.1). The weights are computed against a proposal the samples have stopped following; see below.
Finite-difference residual loss	abandoned	+0.22% at N=6, ω=1 against +0.010% from REINFORCE. Sensitive to the step size and to noise in the Laplacian estimate.
Neural Pfaffian ansatz [48]	abandoned	+0.43% at N=6, ω=1, forty times the BF+Jastrow error. The larger parameter space amplifies importance-sampling gradient noise faster than the richer nodal topology repays.
Backflow at N=20 †	abandoned	+18% against +1.32% for Jastrow-only. A memory limit rather than a conceptual one; see §7.3.5.

Why natural gradients stop helping. CG-SR gave the fastest convergence and the best final energies at $\omega \geq 0.1$, and then failed outright below it. The obvious explanation is ill-conditioning: at weak confinement the Fisher matrix $F_{ij} = \mathbb{E}[\partial_i \log |\Psi|^2 \, \partial_j \log |\Psi|^2]$ has eigenvalues spanning many orders of magnitude, because parameters controlling short-range correlation and parameters controlling the orbital envelope act on scales separated by $1/\sqrt{\omega}$.

Ill-conditioning alone does not account for the failure, because ill-conditioning is what makes natural gradients useful in the first place; a well-conditioned metric is one Adam handles unaided. What disqualifies them is the failure to *estimate* the metric. F is built from importance-resampled points, and when the ESS is low a handful of high-weight samples dominate it, leaving it noisy and rank-deficient. Tikhonov damping stabilises the inversion at the cost of washing out the curvature information that motivated it.

Three regimes follow. Where the metric is well conditioned, SR is sound and unnecessary. Where it is anisotropic and still estimable, SR is at its most valuable—which is the collocation case of §7.2. And where importance-sampling collapse makes the empirical metric unreliable, inverting it amplifies the estimation noise and SR does active harm. The failures above are the third regime, not the second, and the practical implication is about the estimator rather than the curvature: any second-order method applied to a weakly confined many-body system must certify that the metric it inverts is statistically resolved at the sample sizes in use.

Why pushing the samples uphill makes things worse. Langevin refinement was motivated by an intuition that is hard to argue with. If the Gaussian proposal overlaps $|\Psi|^2$ poorly, evolve the samples up the score $\nabla_{\mathbf{R}} \log |\Psi|^2$ for K steps before reweighting, and the overlap should improve. It failed in every configuration tested, and the reason is standard in the MCMC literature and easy to miss in an importance-sampling setting.

The weights $w_i = |\Psi(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$ are valid only if the $\mathbf{R}_i$ came from q. After K steps from an out-of-equilibrium start the samples follow some intermediate $\tilde{q}$ that depends on K and ε, while the weights still divide by q. The resulting bias is not noise. It pushes the wavefunction toward the regions the Langevin dynamics favours, which moves $\tilde{q}$ further the same way, and the loop closes on itself—which is what the +153% in the table is. Ten to twenty steps in Nd=40 dimensions do not equilibrate. Fixing it means running to equilibrium, adding a Metropolis–Hastings step, or computing annealed weights along the non-equilibrium path. The first two are MCMC under another name; the third costs enough to forfeit the advantage that made the route attractive.

What the failures have in common. Read together these are not separate disappointments. Four of them—CG-SR, Langevin, the Pfaffian's parameter count, and the backflow at N=20—fail through one channel: a quantity that must be *estimated* from a finite sample degrades faster than the quantity it was meant to improve. Natural gradients need an estimate of the metric. Langevin needs samples that have actually equilibrated. A larger parameter space needs a correspondingly better-resolved gradient. None of these methods is wrong in principle, and each would be sound given samples we cannot afford. What fails is the estimate, and it fails precisely in the regime where the correction was supposed to help. That thread is what §7.3.4 follows deliberately, from the proposal mismatch at one end to the evaluation bottleneck at the other.

7.3.4 Reliability analysis: the gradient-quality chain

The results in Table 7.5 report the *best* energies obtained by collocation training, selected from multiple runs on different seeds. A natural question is whether these results are *typical* or whether they reflect rare fortunate outcomes from a high-variance process. This subsection reports a systematic 30-run reliability campaign designed to answer that question, and exposes a mechanistic chain that accounts for both the successes and failures of the MCMC-free collocation approach.

Experimental design

All runs use the BF+Jastrow ansatz (∼49k parameters) initialised from the same pretrained checkpoint. The grid is N=6 across $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$ with three random seeds per cell (42, 137, 314), giving $5 \times 2 \times 3 = 30$ runs. Two training recipes are compared head-to-head:

- **Config A (baseline):** Adam optimiser with lr=$5{\times}10^{-4}$, $\mathrm{lr_{jas}}$=$5{\times}10^{-5}$, 4096 collocation points, 8× oversampling, local-energy clipping at 5σ, no reward normalisation, no rollback, fixed Gaussian mixture proposal (Eq. (4.9)).
- **Config B (robust):** Identical to Config A plus four stabilisation mechanisms: 16× oversampling (doubling the proposal budget), reward normalisation (standardising the REINFORCE reward to

zero mean and unit variance before the gradient step), instability rollback (reverting to the previous checkpoint when the energy jumps exceed 3σ, with learning-rate decay of 0.95), and an adaptive Gaussian mixture proposal (8 GMM components refitted every 30 epochs to track the evolving $|\Psi|^2$).

All runs train for 1200 epochs. Evaluation uses a 20 000-sample VMC probe every 50 epochs, with the best-VMC checkpoint restored for a final 20 000-sample IS evaluation.

Main results

Table 7.7 presents the complete results of the 30-run campaign. For each (ω, recipe) combination the three seeds are listed individually, together with their mean, standard deviation and worst case.

Table 7.7: Reliability campaign: relative energy deviations for N=6 across five trap frequencies, scored against the reference column of Table 7.5—external at $\omega \geq 0.1$, and our own converged SR–VMC energy at $\omega \leq 0.01$, where the deviation then measures internal consistency rather than absolute accuracy. Each cell is the final IS-evaluated deviation for one $(\omega, \text{recipe}, \text{seed})$ combination. Bold marks the best run at each ω. The last two columns give the mean ± standard deviation and the worst of the three seeds; these quantify the reliability of the final IS evaluator, not trained-state quality independently of that evaluator. The standard deviation is reported in place of a coefficient of variation because the latter is uninformative when the mean deviation approaches zero.

ω	Recipe	Seed 42	Seed 137	Seed 314	mean ± s.d.	worst
1.0	Baseline	+0.073	+0.045	+0.060	0.059 ± 0.014	0.073
	Robust	+0.032	**+0.013**	+0.105	0.050 ± 0.049	0.105
0.5	Baseline	+0.106	+0.060	+0.108	0.091 ± 0.027	0.108
	Robust	**+0.002**	+0.112	+0.047	0.054 ± 0.055	0.112
0.1	Baseline	+0.419	+0.547	+0.427	0.464 ± 0.072	0.547
	Robust	+0.325	**+0.263**	+0.270	0.286 ± 0.034	0.325
0.01	Baseline	+0.500	+0.379	+0.746	0.542 ± 0.187	0.746
	Robust	+0.289	+0.282	**+0.237**	0.269 ± 0.028	0.289
0.001	Baseline	+3.876	+7.187	+4.733	5.265 ± 1.718	7.187
	Robust	**+0.250**	+0.331	+0.263	0.281 ± 0.044	0.331

Several patterns are immediately apparent. First, the robust recipe dramatically improves the low-ω regimes: at ω=0.001 the mean error drops from +5.3% (baseline) to +0.28% (robust), a factor of 19 improvement. Second, the best single result at every ω is a robust run, achieving +0.013% at ω=1.0, +0.002% at ω=0.5, +0.263% at ω=0.1, +0.237% at ω=0.01, and +0.250% at ω=0.001. Third, the robust recipe's IS-evaluated worst case improves at every confinement, and most dramatically where sampling is hardest: at $\omega = 10^{-3}$ the worst robust seed (+0.331%) is better than the *best* baseline seed (+3.876%) by an order of magnitude. At high ω the two recipes are within each other's seed spread, so the stabilisation mechanisms neither help nor hurt there.

The gradient-quality chain

The reliability campaign exposes a mechanistic chain that accounts for the quality of collocation-trained wavefunctions:

$$\boxed{\begin{aligned} &\text{proposal mismatch} \longrightarrow \hat{k}\ \text{inflation} \longrightarrow \text{gradient noise} \\ &\qquad\qquad\qquad\quad \longrightarrow \text{optimisation stochasticity} \longrightarrow \text{run-to-run variance} \end{aligned}} \tag{7.3}$$

Each link can be checked quantitatively against the training logs, and each is supported below. One caveat on how strongly to read the arrows: the robust recipe changes four things at once—oversampling, reward normalisation, rollback and adaptive proposal refitting—so the chain is a mechanism consistent with every measurement we have, not a set of individually isolated causal links. We did not ablate the four stabilisers separately.

Step 1: proposal–target mismatch. As described in §4.5.1, the REINFORCE estimator in Eq. (4.8) requires samples from $|\Psi_\theta|^2$, but collocation draws from the analytic Gaussian mixture proposal $q(\mathbf{R})$ in Eq. (4.9) and reweights via importance sampling. The quality of this reweighting depends on how well q overlaps with $|\Psi_\theta|^2$.

For the harmonic quantum dot, $|\Psi|^2$ has full support but concentrates its mass in a region whose geometry depends on ω: at ω=1.0, electrons occupy a compact cloud of radius of order the oscillator length $\ell = 1/\sqrt{\omega}$, and the three-component Gaussian mixture in Eq. (4.9) provides adequate coverage. At ω=0.001, the electron cloud extends to $\sim 100\, a_0^*$ (several oscillator lengths, $\ell \approx 31.6\, a_0^*$), the pair correlation function develops sharp shell structure (Section 6.2), and $|\Psi|^2$ concentrates in a thin, strongly correlated region that the isotropic Gaussian proposal covers inefficiently.

The mismatch manifests directly in the importance weights. At ω=1.0, weights are relatively uniform: the effective sample size is typically ESS $\sim$ 5000–6000 out of the $\sim$65k *raw proposal draws* (16× oversampling of 4096 points)—the second of the two normalisations noted in §7.2—and the Pareto shape parameter $\hat{k}$ of the weight tail is 0.4–0.5, indicating a well-behaved thin-tailed distribution. At ω=0.001, the ESS collapses to 1–6 out of the same 65k draws, and $\hat{k}$ rises above 1.0, indicating a catastrophically heavy fitted tail and a sample far from the regime in which the reweighted estimator is reliable.

Step 2: $\hat{k}$ inflation and gradient quality. The tail index $\hat{k}$ of the importance weights turns that mismatch into a number (§4.5.1): below 0.7 the reweighted estimator is trustworthy, above 1.0 the fitted tail is heavy enough that a handful of samples dictates the update. Two qualifications keep the reading honest. $\hat{k}$ is fitted to the *weights*, not to the full gradient integrand $w\,(E_L - b)\,\nabla_\theta \log|\Psi_\theta|$, whose tail index we do not measure; and because w is a ratio of proper densities its mean is finite in the limit, so $\hat{k} > 1$ diagnoses a sample far from that limit rather than a gradient that fails to exist. What the campaign establishes is severe proposal–target mismatch and a gradient estimator dominated by a few draws. That is enough to explain everything downstream, and it is all we claim.

Table 7.8 summarises the average $\hat{k}$ across the campaign. The correspondence between $\hat{k}$ and final accuracy is striking.

Table 7.8: Average Pareto $\hat{k}$ of the importance weight distribution during training, by regime and recipe. Lower is better. The fitted tail has finite variance only for $\hat{k} < 1/2$ and a finite mean only for $\hat{k} < 1$; $\hat{k} < 0.7$ is the practical reliability threshold, and $\hat{k} > 1$ indicates a sample so far from the asymptotic regime that the estimator behaves as though no finite mean existed (§4.5.1).

	$\hat{k}$ (avg)		Mean error (%)	
ω	Baseline	Robust	Baseline	Robust
1.0	0.75	0.50	0.059	0.050
0.5	0.83	0.54	0.091	0.054
0.1	1.48	0.70	0.464	0.286
0.01	2.63	1.39	0.542	0.269
0.001	3.13	1.86	5.265	0.281

The robust recipe reduces $\hat{k}$ by a factor of 1.5 to 2.1, with the larger reductions at the confinements where the baseline is worst. At ω=1.0 this takes $\hat{k}$ from 0.75 (borderline) to 0.50 (reliable), resulting in a modest improvement. At ω=0.1 it takes $\hat{k}$ from 1.48 (broken) to 0.70 (borderline), corresponding to the transition from mediocre (+0.46%) to marginal (+0.29%) accuracy. At ω=0.001 even the robust recipe has $\hat{k}$=1.86, but the adaptive proposal and rollback mechanisms prevent the catastrophic divergence seen in the baseline (+5.3%).

Step 3: from gradient noise to optimisation stochasticity. When $\hat{k}$ is large, most of the gradient signal comes from a few samples with extreme importance weights. Which samples are extreme depends on the random proposal draw at each epoch—a function of the random seed. Two seeds that happen to draw different extreme samples will follow different optimisation trajectories, even though they share the same architecture and hyperparameters.

This is not the usual seed dependence of stochastic gradient descent, where the law of large numbers ensures that gradient estimates are unbiased and the trajectory variance decreases with batch size. When $\hat{k} > 1$ the weight distribution is so heavy-tailed that the realised estimator behaves as though its variance were unbounded: increasing the batch size reduces the spread sublinearly, and extreme draws continue to dominate the update at every batch size we could afford. The result is optimisation trajectories that diverge qualitatively across seeds, explaining the large run-to-run variance observed at low ω.

Step 4: the evaluation bottleneck. An additional subtlety is that the final energy evaluation itself uses importance sampling from the same proposal distribution. Even if the model has found an excellent wavefunction, a noisy evaluation can report a misleading energy.

The data expose this directly. Table 7.9 compares the VMC-best energy seen during training (a 20 000-sample Metropolis probe from the model's own $|\Psi|^2$, used for checkpoint selection) with the final IS evaluation that uses the Gaussian proposal. At high ω where IS is reliable, the two estimates agree closely. At low ω, the gap between them grows, and its sign is unpredictable—the IS evaluation can overestimate or underestimate the true energy.

Table 7.9: Gap between VMC-best checkpoint energy (Metropolis sampling from $|\Psi|^2$) and final IS evaluation, for selected runs. Positive gap means IS reports higher (worse) energy than the VMC probe saw.

ω	Recipe	Seed	VMC-best (%)	Final IS (%)	Gap (%)
1.0	Baseline	314	+0.001	+0.060	+0.059
1.0	Robust	137	+0.002	+0.013	+0.011
0.5	Robust	42	+0.037	+0.002	−0.035
0.1	Robust	137	+0.136	+0.263	+0.127
0.01	Baseline	42	+0.320	+0.500	+0.180
0.001	Baseline	314	+4.114	+4.733	+0.619

The first row is particularly instructive: this baseline run with seed 314 achieved a VMC-best energy of +0.001%—essentially exact—during training, but the IS evaluation reports +0.060%. The wavefunction is excellent; the evaluation is noisy. This demonstrates that some nominally "bad" runs may contain excellent wavefunctions measured poorly, and some nominally "good" results may benefit from evaluation noise in the favorable direction.

Summary of the chain. The chain in Eq. (7.3) provides a unified account of the full landscape of collocation results. At high ω, the proposal matches the target well ($\hat{k} \sim 0.5$), gradients are reliable, optimisation is stable, and all seeds converge to near-DMC accuracy. At low ω, the proposal-to-target mismatch grows, $\hat{k}$ inflates, gradients become noisy, optimisation becomes seed-dependent, and the evaluation itself becomes unreliable. The robust recipe acts on multiple links of this chain simultaneously: higher oversampling improves the raw ESS, reward normalisation stabilises the gradient scale, rollback prevents catastrophic divergence, and the adaptive proposal reduces the fundamental mismatch. No single mechanism is sufficient; each addresses a different link in the chain.

7.3.5 The memory wall at N=20

The reversal of the backflow advantage at N=20 highlights an important architectural trade-off. The message-passing backflow (`CTNNBackflowNet`, §3.5.2) stores edge features of size $\mathcal{O}(N^2 \times h_{\text{bf}})$, which at N=20 with h_{bf}=128 requires $\sim$10× the memory of the Jastrow-only network. This forces a choice: reduce h_{bf} (losing expressivity), reduce the collocation budget (losing gradient quality), or reduce the oversampling factor (losing ESS). All three options degrade the result.

The Jastrow-only network, with $\sim$25k parameters and $\mathcal{O}(N^2)$ edge cost (no hidden-dimension multiplier), permits full-size collocation budgets (n_{coll}=4096, oversample=8–16) even at N=20. The resulting gradient is less noisy, and the simpler loss landscape (fewer parameters) makes optimisation more reliable.

This does not mean that backflow is unnecessary at N=20; it means that the current training pipeline cannot supply enough gradient signal to train backflow effectively at this scale. The implication is that scaling collocation to larger systems requires either (i) memory-efficient backflow architectures (e.g. sparse attention, low-rank message passing), or (ii) improved sampling that provides higher ESS at constant computational cost.

7.4 A physics-informed proposal and its limit

The ESS collapse of §7.2 is a limitation of the *proposal*, not of collocation itself, and it can be pushed back a long way. A physics-informed proposal carries MCMC-free collocation deep into the Wigner regime; what limits it, and the wall that survives, follow from the same physics.

7.4.1 The origin-centred proposal misses the Wigner shell

Every proposal in the standard pipeline is a mixture of *origin-centred* Gaussians,

$$q(\mathbf{R}) \;=\; \frac{1}{K}\sum_{k}\mathcal{N}\big(\mathbf{R};\,0,\,\sigma_{f,k}^2/\omega\big)\,. \tag{7.4}$$

But as $\omega \to 0$ the electrons localise into a Wigner molecule on a *shell* far from the origin. The classical $N = 6$ minimum is the $(1, 5)$ configuration: one central electron and five on a ring of radius $1.334\,\omega^{-2/3}$ [8, 9]. In oscillator units that ring sits at 1.96, 2.87 and $4.22\,\ell$ for $\omega = 0.1$, 0.01 and 0.001, moving outward as the trap weakens and past the reach of a width-2ℓ origin Gaussian. The proposal aims where $|\Psi|^2$ has no mass; hence the collapse.

7.4.2 The Wigner-ring proposal

Before the numbers, one distinction has to be exact, because "MCMC-free" is a claim of this thesis. The Wigner geometry enters in three roles and only the third sits inside the training loop. *(i)* A proposal can be *evaluated* against a trusted state obtained by MCMC, which is how Table 7.10 is produced—a measurement, not a training step, and one that uses a chain by design. *(ii)* Its geometry can be *fitted* to $|\Psi_\theta|^2$ samples as a diagnostic. *(iii)* In production it is refitted from the current importance-weighted batch alone, with no chain anywhere; that is the genuinely chain-free adaptation used in the cascade of §7.4. Throughout, a Metropolis probe still runs outside the gradient loop to select checkpoints and certify the final energy, and no sample it produces ever reaches a parameter update. We replace the origin mixture with a *Wigner-ring* proposal: a radial Gaussian at the shell radius times an angular distribution for the ring electrons, plus a central Gaussian. On trusted states this lifts the effective sample size by 15–32× over the origin mixture (Table 7.10), turning a collapsed sampler into a usable one—without any training. Fitting the ring to $|\Psi|^2$ samples (the simplest adaptive form) doubles the ESS again; a $(1, 5) + (0, 6)$ mixture *hurts* at $N = 6$, since the state is $(1, 5)$ and the extra channel dilutes the proposal mass.

Table 7.10: ESS (of 4096) against a trusted $N = 6, \omega = 0.01$ state. The origin mixture is already collapsed; the ring proposal covers the shell.

proposal	ESS
origin-Gaussian mixture (standard)	1.5
Wigner ring $(1, 5)$, hand-tuned	22.5
Wigner ring $(1, 5)$, fitted to $\lvert\Psi\rvert^2$	47.5

7.4.3 Angular crystallisation defeats the uniform-angle ring proposal

Why does the ring work at $\omega = 0.01$ but fail at $\omega = 0.001$? The molecule *crystallises angularly* as the trap weakens. The nearest-neighbour angular-gap standard deviation of the five ring electrons is 40.8°/33.3°/25.5°/6.7° at $\omega = 1/0.1/0.01/0.001$: a "ring liquid" at moderate coupling becomes an angular crystal (electrons pinned near 72° apart) at $\omega = 0.001$. A proposal with independent, uniform angles almost never lands on the crystalline configuration—so the angular distribution must crystallise too. This is captured MCMC-free by modelling the five gaps (which sum to 2π) as Dirichlet(α): an exact tractable density whose single concentration α interpolates liquid (α small) to crystal (α large), fitted to the measured gap variance. This diagnosis *explains* the collocation wall that is otherwise only observed.

7.4.4 An MCMC-free cascade reaches $\omega = 0.0035$

Combining the ring proposal (radial Gaussian × Dirichlet gaps) with importance oversampling, a *continuous* adaptive re-fit of the proposal to the current importance-weighted batch (every few steps, no MCMC), and a gentle ω-cascade seeded from a converged $\omega = 0.01$ state, MCMC-free collocation produces states that lie on the Wigner energy-scaling curve down to $\omega = 0.0035$—four cascade rungs, with heavy-VMC energies within 2% of the scaling estimate, and within 1% on three of the four rungs (Table 7.11), well past where the origin proposal collapses (ESS $\approx$ 1.5 of 4096 at $\omega = 0.01$, Table 7.10).

What certifies the frontier state, and what does not, is worth being exact about. The agreement in Table 7.11 is against a scaling curve fitted to our own $\omega = 0.01$ energy, so it establishes continuity rather than accuracy. Two further checks are structural: the radial density peaks at 3.34ℓ against a classical ring at 3.42ℓ, and initialising the sampler on the shell converges to the same peak rather than to a second basin. What is missing is an *angular* diagnostic at $\omega = 0.0035$—the bond-order and Lindemann machinery of §6.2.5 was run on the production grid and not on the cascade rungs, and those states were not retained. So "a Wigner molecule at $\omega = 0.0035$" rests here on energy continuity and radial geometry, not on demonstrated angular order, and the natural next check is to measure $|\Phi_5|$ against its random-angle null on that state.

One further qualification concerns the state's ancestry rather than its certification. The chain-free claim is a claim about the *gradient*, and it holds exactly: no Metropolis sample reaches a parameter update at any rung. It is not a claim that the frontier state is reference-free. Each rung is warm-started from the converged state of a stronger confinement, that ladder runs back to $\omega = 0.01$ and beyond it to the strong-confinement runs, and the whole cascade begins from a pretrained checkpoint (§7.3.4) whose residual objective used an external energy as its target wherever one exists. So the parameters arriving at $\omega = 0.0035$ inherit structure shaped, several confinements upstream, by reference-guided pretraining—which is the same inheritance the SR–VMC states carry (§8.1), and which no measurement here separates from what collocation itself contributed. Training a rung from scratch at weak confinement is the experiment that would settle it; §7.4 reports that this does not converge, which is itself part of the finding. The essential ingredient is the continuous re-fit: it keeps ESS healthy ($\sim 10^2$) *while* the energy descends, where a static or slowly-updated proposal collapses to ESS ~ 1 as the wavefunction moves away from it.

Table 7.11: MCMC-free Wigner-ring cascade, $N = 6$: heavy-VMC energy vs. the fitted Wigner scaling curve $E(\omega) \approx 0.690\,(\omega/0.01)^{0.689}$, whose exponent is close to the classical 2/3 but whose prefactor is anchored at $\omega = 0.01$, where no external reference exists. The agreement is therefore internal scaling consistency, not validation, and it is retained through $\omega = 0.0035$.

ω	E	scaling est.	ratio
0.0077	0.577	0.577	1.00
0.0059	0.481	0.480	1.00
0.0045	0.400	0.398	1.01
0.0035	0.340	0.335	1.02

7.4.5 The remaining limit: the Slater–Jastrow balance

Below $\omega = 0.0035$ the cascade becomes unstable, and—informatively—the instability is *non-monotonic* across proposal variants: wide-exploratory, tempered-tracking, and outward-anchored proposals each converge some rungs and diverge others. That pattern locates the obstacle not in the proposal but in the warm-start, and four transfer diagnostics (taking the converged $\omega = 0.0035$ rung down to $\omega = 0.0027$, a 23% step) pin down its mechanism precisely.

First, the frontier state is a genuinely Wigner-localised one on the evidence available: at its own $\omega = 0.0035$ its radial density peaks at 3.34ℓ against a classical ring at 3.42ℓ, and it occupies a *single* basin—initialising the sampler on the shell converges to the same 3.34ℓ. We stop short of calling it a Wigner molecule in the sense Chapter 6 defines, because that definition requires registered angular order and no angular diagnostic was run on this state. Transferred to $\omega = 0.0027$, however, the wavefunction collapses to $\sim 1\ell$, and two tests show *where* the collapse lives. It is not the sampler: initialising the Metropolis walkers directly on the classical shell (3.57ℓ) still relaxes inward to 1.1ℓ, so the collapsed density is a property of the ansatz, not of sampler initialisation. And it is not the backflow: its displacement is small ($|\Delta\mathbf{x}| \approx 0.13\ell$, radially inward) and essentially *preserved* across the transfer ($0.15\ell \to 0.13\ell$)—it scales correctly with ℓ but is far too small to bridge the 2.4ℓ gap between the Slater core ($\sim 1\ell$) and the Wigner shell.

The mechanism is a delicate cancellation. The bare harmonic-oscillator determinant suppresses the shell by $\exp\!\big(-\omega r^2/2\big) \sim \exp(-5.6)$ in amplitude ($\sim 10^{-5}$ in density); the Wigner peak exists only because the Jastrow amplifies it back by a comparable factor, with the backflow contributing a small inward nudge. A 23% change in ω perturbs both the Slater confinement and the Jastrow correlation length, the fine-tuned balance tips, and the density falls back to the non-interacting shell—from which the collocation re-fit cannot reliably climb out (hence the non-monotonic proposal behaviour). This rules out the natural fix of an ω-robust warm-start by coordinate rescaling: because the ansatz relaxes inward even when handed shell configurations, no coordinate transform preserves the state across the step. It does not rule out representing the $\omega = 0.0027$ state at all—that would require optimising it from scratch there, which we did not do. Reaching $\omega < 0.0035$ would require per-ω re-optimisation against a shell-*amplitude*-anchored objective, or a genuinely ℓ-covariant parametrisation—a distinct problem left to future work. The contribution here is concrete: a physics-informed proposal that carries MCMC-free collocation from the origin-proposal wall down to a radially localised Wigner-regime state at $\omega = 0.0035$, and a precise account of the angular-crystallisation and Slater–Jastrow-balance effects that bound it.

7.5 Gauge dependence and robust observables

Not every number above can be leaned on equally, and the ones that read most vividly are among the weaker. The three-way hierarchy set out at the start of Part IV (p. 57) is developed here, with the evidence for it.

Quantities defined on a single component—the backflow's rank, its displacement magnitude $|\Delta\mathbf{x}|$, the per-component d_{eff}—are **gauge-like**, because the Jastrow and the backflow are substitutes and the division of correlation between them is a nearly flat, path-dependent direction of the loss. Such quantities record the route the training took rather than the state it reached. This is not a hypothetical worry: every claim that reversed during this project was of this kind, from the d_{eff} inversion at the crystal that did not survive seeding, through a mislabelled backflow, to an artefactual rank collapse produced by training both components jointly from scratch. That last one needs separating from the collapse reported in §7.1.2, since they are the same observable. The artefact appeared only under joint from-scratch training, affected *both* architectures, moved with the seed, and carried no energy penalty; the reported collapse appears under the staged protocol used throughout this chapter, affects only the conventional backflow, is seed-robust at all three particle numbers, and is accompanied by the half a percent of energy in Table 7.2. It is the last of those that makes it a finding rather than a diagnostic curiosity.

The measurements used here fall into three classes, and the conclusions rest on the first two. *State-level* quantities—the energy, the local-energy variance $\mathrm{Var}(E_L)$, and the symmetric overlap between two trained states—are properties of the wavefunction and are unaffected by how the parameters are arranged. $\mathrm{Var}(E_L)$ does most of the work, because it separates the two families where the energy cannot and where d_{eff} no longer does: at matched protocol it is lower for the message-passing ansatz at every confinement, by factors between 1.2 and 3.4 (Table 7.1), with non-overlapping seeds at the crystal and the widest separation at matched capacity. *Controlled interventions*—deleting a component and pricing the loss in energy—are not gauge-invariant in the same sense, since two parameterisations that divide correlation differently will report different ablation costs; they are reproducible statements about a fixed model and training protocol, and are used as such. *Component-level* diagnostics—rank, displacement magnitude, per-branch loadings—are the gauge-dependent class.

Rank belongs to that third class. It appears throughout this chapter as the clearest signature of the mechanism, a field collapsing onto a single mode being easy to see, but what carries the finding is the half a percent of energy that collapses with it.

Part V

Discussion

Convictions are more dangerous enemies of truth than lies.

Friedrich Nietzsche

Human, All Too Human

Chapter 8

Discussion

8.1 Geometry between architecture and optimiser

The three parts of this thesis converge on one object. Architecture fixes the shape of the variational tangent space; the optimiser acts on that space through its metric; and the sampling measure decides how well the metric can be estimated. Read in that order, results that arrived separately become a single chain.

Its first link is the one the energy cannot see. Two correlators of matched or deliberately mismatched capacity reach the same reference-quality state at strong confinement, agreeing to four figures, and arrive there through tangent spaces of different size and different content. Counting directions is the weaker half of that observation—it says how many the correlator uses, and it is measured on one of two substitutable components. Which directions is the stronger half. The message-passing correlator's leading tangent modes lie inside a four-operator physical dictionary to $R^2 \approx 1$ across confinement and up to N=12, where a separable network of equal or greater capacity leaves close to half its tangent budget outside it. A direction outside a chosen dictionary need not be unphysical, so what the measurement supports is that one tangent space is far more completely accounted for by a small set of collective operators than the other—and that a smaller d_{eff} makes such a dictionary easier to cover cannot explain the gap, since the two effective dimensions converge toward 3.7 at weak confinement while the dictionary gap persists.

The second link is where the optimiser meets that space. The result there is narrower than the usual one about natural gradients. Under $|\Psi|^2$ sampling the empirical metric is badly conditioned—$\kappa \sim 10^7$–10^{10} on precisely the models where Adam and SR agree to four decimals—so conditioning cannot be the criterion. What separates the regimes is whether a batch resolves the metric at all. Under $|\Psi|^2$ the samples come from the measure that defines S; under a fixed collocation proposal they do not, and preconditioning becomes what makes the optimisation converge; and once the proposal loses the target, inverting the metric amplifies noise instead of signal. The negative half of that is measured. The positive half—that collocation under a fixed proposal occupies an anisotropic but still estimable middle regime—is inferred from the estimator, since κ was never obtained under the collocation measure at matched batch size.

Whether the chain generalises is a fair question and the honest answer is that we cannot settle it here. Message-passing neural quantum states have been adopted for extended fermion systems on the argument that relational structure scales [5, 6]; what this thesis adds is a measurement of *where* that structure shows up—in the tangent geometry before the energy—and a mechanism for why the effect should grow rather than shrink with correlation strength. The dictionary gap widens with N over the range we can measure. If that continues, the architectures that scale are those maintaining a transported edge representation rather than pooling pairwise features and discarding them. Three sizes is not a scaling study, and the comparison is seeded at N=6.

What agreement with the benchmarks supports. The energies agree with fixed-node DMC over the benchmarked grid, and that agreement is neither exactness nor a meeting of two blind calculations: where a DMC value exists it is also the target of residual pretraining. What protects the comparison is the variational bound. The reported energies are expectations of $\hat{H}$ under $|\Psi_\theta|^2$ taken after a pure-energy SR tail, and such an expectation lies above E_0 whatever it was aimed at, so agreement from above cannot be manufactured by aiming; aiming at a target above the ansatz's reachable minimum would only make the number worse. The reference itself is variational with respect to its own imposed nodes, which is why several of our energies sit below it without contradiction, and why the tables report deviation rather than error. Below $\omega = 0.1$ for $N \geq 6$ no external value exists and those states are predictions.

8.2 Physics returned to the solver

The most useful thing this thesis did was not to train a wavefunction but to let the wavefunction correct the method that produced it.

The order matters. A generic three-component Gaussian proposal covers the strong-confinement droplet adequately and collapses in the Wigner regime, where the effective sample size falls to about 1.5 of 4096 and the gradient estimator dies with it. Nothing in the objective or the architecture diagnoses that. What diagnoses it is the structure measured from the trained states themselves: a single ring at N=6 with a well-determined radius, angular order that appears well after radial stiffening, and shell occupancies that are read rather than assumed. Rebuilding the proposal as a radial Gaussian at the measured shell radius times a Dirichlet angular model lifted the effective sample size by 15–32× with no training at all, and carried Markov-chain-free training from $\omega \approx 0.01$ to $\omega = 0.0035$.

That is a loop, and its two halves fail in different ways, which is what makes it informative. Everything up to $\omega = 0.0035$ lies on one causal chain: proposal–target mismatch inflates the weight tail, the tail destroys the gradient estimator, the destroyed estimator makes the optimisation seed-dependent and the metric unreliable, and inverting an unreliable metric amplifies noise. Langevin refinement joins that chain at the first link by breaking the proposal the weights are computed against; CG-SR joins at the last. Even the N=20 reversal, where a Jastrow-only ansatz beats the backflow that should outperform it, reaches the same chain through memory rather than confinement. All of it is a sampling problem, and a better proposal moves it.

Below 0.0035 the obstacle changes character. Metropolis walkers started on the classical shell relax inward regardless; the backflow displacement transfers correctly and is simply too small to bridge the gap; the collapse is non-monotonic across proposal variants in a way a sampling failure would not be. What fails is the warm start. At these confinements the Wigner peak exists only because the Jastrow amplifies an exponentially suppressed determinant back up by a comparable factor, and a 23% change in ω perturbs both sides of that cancellation until the density falls back to the non-interacting shell. The failure is one of covariance in ℓ under a change of trap—of transfer and parameterisation, not of expressivity. We have not shown the ansatz cannot represent the $\omega = 0.0027$ state; the remedy we suggest, per-ω re-optimisation against a shell-amplitude-anchored objective, presumes it can.

Both halves outlive the quantum dot. An importance-sampled scheme whose proposal increasingly misses a strongly correlated target should meet the first chain in that order, and any second-order method applied to such a system ought to certify that the metric it inverts is statistically resolved at the sample sizes in use.

The staged crossover. The physics that drove the loop is itself worth stating against what was known. That parabolic dots form Wigner molecules with shell structure is established [7, 8, 10, 12]; that the crossover is *staged*—radial expansion and shell stiffening completing well before angular order appears within the shells—is what the topology-resolved diagnostics here make visible, because they condition on the shell partition before measuring angular correlation rather than averaging over it. The N=6 ring approaches its classical radius $1.334\,\omega^{-2/3}$ monotonically to within 1.7% while its bond order is still at the random-angle null, which is the separation stated quantitatively. Whether the same staging survives a total-spin scan we cannot say.

What "Markov-chain-free" claims. The claim is about the gradient: no configuration entering a parameter update comes from a Markov chain. A small Metropolis probe runs outside the gradient loop to select checkpoints and certify the final energy, and no sample it produces reaches an update. It is not a claim of freedom from Monte Carlo, since importance sampling and resampling are themselves Monte Carlo; nor of freedom from the reference, since where a DMC value exists the residual pretraining anneals toward it and the rollback monitors against it. Nor, at the deep-Wigner rungs where no reference exists, is the state reference-free in ancestry: its warm starts descend from benchmark-guided states at stronger confinement. Against the SR route, collocation is competitive for $N \lesssim 12$ and $\omega \gtrsim 0.1$, a few times worse at N=12, ω=0.1, and not competitive at N=20. The two look complementary rather than rival: collocation for fast, parallel, chain-free exploration and pretraining; SR for final accuracy where correlation is strong.

8.3 Scope, and what to do next

Four limits bound everything above. All states are the lowest found within a fixed spin-projection sector, and total spin is not scanned. The architecture comparison is seeded at N=6, rests on fewer models at N=12 and on a single production state at N=20. The backflow variants are not capacity-matched and cannot be at equal width, so the cross-architecture energy gap does not isolate architecture. And one entry in the backflow rank table, at N=20 and $\omega = 0.01$, comes from a re-run whose diagnostics were not retained.

Four things follow that we would do next.

Make the proposal learned rather than designed. Up to the frontier the proposal is the bottleneck, not the objective, which argues for a better sampler rather than a better loss—normalising flows trained against $|\Psi|^2$, or the shell-informed construction of §7.4 generalised beyond ring geometries. Past the frontier the target changes: what is needed there is an ℓ-covariant parameterisation, or an objective anchored on the shell amplitude, so that a converged state survives being carried to a weaker trap.

Make rotation equivariance exact. Permutation equivariance holds by construction and the centre of mass is projected out, but the displacement head reads a d-vector from a network that has seen absolute coordinates, so SO(2) equivariance is only encouraged—by relative-vector edge features and by the random in-plane rotation applied to every batch—and never enforced. In a rotationally invariant trap whose weak-confinement physics is a freely rotating molecule, that is the wrong symmetry to leave approximate. The fix is local to one module: build the displacement as $\Delta\mathbf{r}_i = \sum_{j\neq i} g_\beta(\text{invariants}_{ij})\,\mathbf{r}_{ij}$, a learned scalar gain on each relative vector, so the output rotates exactly with the input and the augmentation can be dropped. We did not run it.

Break the quadratic backflow. What stops the method at twenty electrons is the $\mathcal{O}(N^2 h_{\text{bf}})$ edge tensor, a memory limit rather than a physical one. Sparse or low-rank message passing, a local backflow without all-to-all edge features, or the sample-side reductions of minSR and the kernel formulation [44, 45] would extend the reach without giving up the nodal correction that is the reason for having a backflow at all.

Match the backflow capacities. A widened conventional backflow at 182k parameters is the one control that would turn the rank-collapse observation into an architecture ablation. It is a single training run.

Part VI

Conclusion

All things are implicated with one another, and the bond is holy; and there is hardly anything unconnected with any other thing.

MARCUS AURELIUS
Meditations

Chapter 9

Conclusions

The work formed a feedback cycle between solver and physics. A variational state was constructed and checked against every external benchmark available; the checked state was then used as an instrument, and resolved the shell geometry of the Wigner regime—rings at definite occupancies, n-fold angular order becoming large only in the last decade of confinement, a shell radius approaching its classical $\omega^{-2/3}$ scaling. Those measurements are what a sampler needs in order to place its mass correctly. Incorporating the measured geometry into the proposal extended Markov-chain-free training from the failure of the origin-centred sampler at $\omega \approx 0.01$ down to $\omega = 0.0035$, into states the original method could not reach.

Architecture. Message passing supplies a relational channel whose effect appears in the tangent geometry before it appears in the energy (§7.1). At strong confinement a message-passing correlator reaches a reference-quality solution along half as many effective tangent directions as a separable network of matched or larger capacity, and its leading directions are far more completely accounted for by a compact dictionary of collective operators; the separable network spends close to half its tangent budget outside that dictionary, matches the energy to four figures anyway, and carries a local-energy variance higher by factors of 1.2 to 3.4. In the nodes the same channel shows as a collapse averted: toward the Wigner regime a conventional backflow's displacement field collapses spectrally to rank one and loses half a percent of the energy, where the copresheaf field keeps its rank. Correlator and backflow give way at the same confinement, and a controlled correlator swap moves that boundary—which points to one relational bottleneck rather than two deficits.

Optimiser. Natural-gradient preconditioning earns its cost where the tangent metric is anisotropic and still estimable (§7.2). That is narrower than the usual claim and it has three regimes rather than two. Under $|\Psi|^2$ sampling the empirical metric is badly conditioned, $\kappa \sim 10^7$–10^{10}, on the very models where Adam and SR agree to four decimals, so conditioning alone is not the criterion. Under a fixed collocation proposal SR is what makes the optimisation work at all, until proposal adaptation and gradient stabilisation make Adam sufficient and more robust. And once the sampling that builds the metric collapses, inverting it amplifies noise rather than signal. Estimability, not conditioning, is what the data identify as decisive.

Paradigm. An accurate wavefunction can be trained with no Markov chain in the gradient, inside a regime whose edges we found by crossing them (§7.3). The weak-form objective reaches DMC accuracy at $N \in \{6, 12\}$ under strong confinement and stays within a few tenths of a percent of our own converged energies well into the correlated regime. It fails through one chain of causes—proposal mismatch, weight-tail inflation, gradient noise, an unestimable metric—which a better proposal moves but does not remove, and then through a second and different failure of transfer between confinements.

Physics. The trained state serves as an instrument (§6.2). The crossover it resolves is staged: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them. At $\omega = 10^{-3}$ the N=6 dot shows persistent mixing between the $(1, 5)$ and $(0, 6)$ sectors, the N=12 dot a two-shell molecule with a modal $(3, 9)$ split, and the N=20 dot a three-shell molecule that collapses to a single ring about a central electron within one decade in ω. The N=6 shell radius approaches the classical value $1.334\,\omega^{-2/3}$ monotonically, to within 1.7%.

The measurements then returned to the solver. Incorporating the observed shell geometry into the collocation proposal lifted the effective sample size by 15–32× without any training and carried Markov-chain-free training from $\omega \approx 0.01$ to $\omega = 0.0035$—into states the original sampler could not reach.

9.1 Limitations and further work

All states are the lowest found within a fixed spin-projection sector; total spin is not scanned. The architecture comparison is seeded at N=6 and thinner above it, and the two backflow variants are not capacity-matched, so the cross-architecture energy gap does not isolate architecture. Below $\omega = 0.0035$ the limit is a transfer instability in the Slater–Jastrow cancellation that no change of proposal repairs; whether the ansatz could represent such a state under per-ω re-optimisation is a question we did not test. The interventions most likely to move the picture are a backflow that is not quadratic in particle number, a proposal that adapts on its own, an exactly SO(2)-equivariant displacement, and the capacity-matched backflow control that would settle the last question above. §8.3 sets these out.

9.2 Concluding perspective

At the widths reached for the well-resolved $N \leq 12$ studies, raw representational capacity was rarely the first bottleneck; alignment, conditioning and sampling were. It was not irrelevant everywhere—at N=20 the memory cost of the backflow's edge tensor forces a capacity cut that degrades the collocation gradients—but what determined the outcome across architecture, optimisation and sampling alike was alignment with the low-dimensional collective structure the Hamiltonian already possesses: breathing, correlation-hole and relative modes at strong confinement, resolving at weak confinement into the shell geometry of the Wigner molecule. That the same alignment matters in all three places is probably a fact about the system rather than about neural networks.

Several of these results were reached only after an initial reading of the same data proved wrong: a rank collapse that survived seeding where a dimension inversion did not, a transfer failure first attributed to the sampler and later to the ansatz, and an importance-sampling error whose signature was indistinguishable from an architectural deficiency. Each was identified by a diagnostic that could separate the competing explanations. That is the methodological point these chapters have tried to make concrete: a variational calculation is only as trustworthy as the tests that can tell its failure modes apart.

Part VII

Appendices

Chapter A

Laplacian conditioning

A.1 Residual training for Schrödinger

We consider the stationary many-body Schrödinger equation on configuration space $\mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N) \in \Omega \subset \mathbb{R}^D$ with $D = dN$ (for our quantum dots, $d = 2$):

$$\hat{H}\Psi(\mathbf{R}) = E\Psi(\mathbf{R}), \qquad \hat{H} = \hat{T} + \hat{V}, \qquad \hat{T} = -\frac{1}{2}\sum_{i=1}^{N} \Delta_i, \tag{A.1}$$

where $\hat{V}$ collects external and interaction potentials.

In *strong residual minimization* one introduces

$$L := \hat{H} - E \quad \Rightarrow \quad L\Psi = 0, \tag{A.2}$$

and minimizes a squared residual loss of the schematic form

$$\mathcal{L}_{\text{res}}(\theta) = \frac{1}{2}\|L\Psi_\theta\|^2_{L^2(\Omega;w)} = \frac{1}{2}\int_\Omega |(L\Psi_\theta)(\mathbf{R})|^2 \, w(\mathbf{R})\, d\mathbf{R}, \tag{A.3}$$

for a nonnegative weight w induced by the sampling strategy. A key message of the numerical-analysis viewpoint is that strong residual losses can be *severely ill-conditioned*, and training speed is tightly linked to spectral properties of an operator/matrix induced by L [40].

Which of the thesis's objectives this analyses. Three objectives appear in the main text and Eq. (A.3) is exactly one of them. Stage I pretraining (§4.3) minimises $\frac{1}{B}\sum_b (E_L(\tilde{R}_b) - E_{\text{eff}})^2$ over configurations drawn from $|\Psi_\theta|^2$; since $E_L = \hat{H}\Psi/\Psi$, that sample average estimates $\int |\Psi_\theta|^2 \, (\hat{H}\Psi_\theta/\Psi_\theta - E)^2 = \|(\hat{H} - E)\Psi_\theta\|^2_{L^2}$, which is Eq. (A.3) with $w \equiv 1$. Everything in this appendix therefore applies to stage I directly. It does *not* apply in the same way to the other two. Stage II is variational energy minimisation (§4.4), whose objective is linear rather than quadratic in the residual and whose conditioning is governed by the quantum geometric tensor measured in §7.2, not by the Δ^2 stiffness derived below. The collocation route (§4.5) replaces the strong residual with a weak-form score-function objective under an analytic proposal, for the reason developed in §A.4 and Appendix C: the weak form is what removes the second-derivative amplification, and it is the escape from the mechanism analysed here rather than an instance of it. Where the two production objectives inherit anything from this analysis, it is through the single channel they share with stage I—the Laplacian inside E_L—and not through the squared-residual structure.

A.2 Linearized geometry and the matrix A

To expose the mechanism, follow the standard linearization argument used in conditioning analyses [40]. Fix an expansion point θ_0 and define tangent (feature) functions

$$q_j(\mathbf{R}) := \partial_{\theta_j}\Psi_\theta(\mathbf{R})\big|_{\theta=\theta_0}, \qquad j = 1, \ldots, P. \tag{A.4}$$

For small parameter increments $\delta\theta$, we have the first-order model

$$\Psi_{\theta_0+\delta\theta}(\mathbf{R}) \approx \Psi_0(\mathbf{R}) + \sum_{j=1}^{P} q_j(\mathbf{R})\,\delta\theta_j, \qquad \Psi_0 := \Psi_{\theta_0}. \tag{A.5}$$

Applying the residual operator gives

$$(L\Psi_{\theta_0+\delta\theta})(\mathbf{R}) \approx (L\Psi_0)(\mathbf{R}) + \sum_{j=1}^{P} (Lq_j)(\mathbf{R})\,\delta\theta_j. \tag{A.6}$$

Introduce the weighted inner product (complex-valued wavefunctions allowed)

$$\langle f, g\rangle_w := \int_\Omega f(\mathbf{R})^*\, g(\mathbf{R})\, w(\mathbf{R})\, d\mathbf{R}. \tag{A.7}$$

Let the Jacobian J be the linear map whose jth column is Lq_j. Then the loss (A.3) becomes (to second order in $\delta\theta$)

$$\mathcal{L}_{\text{res}}(\theta_0 + \delta\theta) \approx \frac{1}{2}\|r_0 + J\,\delta\theta\|_w^2, \qquad r_0 := L\Psi_0. \tag{A.8}$$

with normal-equation matrix

$$A := J^* J, \qquad A_{ij} = \langle Lq_i,\, Lq_j\rangle_w. \tag{A.9}$$

Here A is Hermitian positive semidefinite; in practice it can be close to singular, so one often effectively works with a damped variant $A + \lambda I$ (implicit or explicit).

Now consider gradient descent on the quadratic model (A.8) (with step size η). The error in parameter space evolves as

$$\delta\theta^{(t+1)} - \delta\theta^\star = (I - \eta A)\left(\delta\theta^{(t)} - \delta\theta^\star\right), \tag{A.10}$$

so each eigen-direction of A contracts by a factor $(1 - \eta\lambda)$. Hence convergence speed is governed by the eigenvalue spread of A, typically summarized by the (effective) condition number

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \tag{A.11}$$

(over the nonzero/regularized spectrum). Large $\kappa(A)$ implies stiffness: a step size small enough to be stable for $\lambda_{\max}$ yields slow progress along directions associated with $\lambda_{\min}$. This is precisely the conditioning bottleneck emphasized in physics-informed training analyses [40].

A.3 Bi-Laplacian scaling

The defining feature of Schrödinger is that L contains a second-order differential operator. To isolate the kinetic contribution, consider the regime where the residual is dominated by the Laplacian part,

$$L \approx -\frac{1}{2}\Delta_{\mathbf{R}}. \tag{A.12}$$

From L to L^*L. The matrix A in (A.9) is a Gram matrix in the feature space $\{Lq_j\}$. The corresponding operator-level object is the *Hermitian square* L^*L (with adjoint taken in the $L^2(\Omega; w)$ inner product) [40]. For general non-constant w, the adjoint L^* differs from the formal differential adjoint by weight-induced lower-order terms; the cleanest picture is obtained for constant w and standard boundary conditions (periodic/Dirichlet), where the Laplacian is self-adjoint.

In the Laplacian-dominant regime (A.12), one has (up to a constant factor)

$$L \approx -\tfrac{1}{2}\Delta \quad\Rightarrow\quad L^*L \approx \tfrac{1}{4}\,\Delta^2, \tag{A.13}$$

i.e. a bi-Laplacian scaling. The prefactor does not affect conditioning ratios.

Fourier/eigenmode derivation of k^4. On a periodic domain (or, more generally, in a Laplacian eigenbasis) $-\Delta\phi_{\mathbf{k}} = |\mathbf{k}|^2\phi_{\mathbf{k}}$, so $\Delta^2\phi_{\mathbf{k}} = |\mathbf{k}|^4\phi_{\mathbf{k}}$. If an error $\delta\Psi$ expands as $\delta\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}}\phi_{\mathbf{k}}$ and $L \approx -\frac{1}{2}\Delta$, then

$$\|L\,\delta\Psi\|_{L^2}^2 \approx \frac{1}{4}\sum_{\mathbf{k}} |\mathbf{k}|^4 |c_{\mathbf{k}}|^2. \tag{A.14}$$

Thus the squared residual penalizes high-frequency components with a *quartic* weight $|\mathbf{k}|^4$. If the relevant spectrum spans $|\mathbf{k}| \in [k_{\min}, k_{\max}]$, the induced eigenvalue ratio scales like

$$\kappa \sim \left(\frac{k_{\max}}{k_{\min}}\right)^4, \tag{A.15}$$

which is a sharp mathematical expression of "Laplacian-driven stiffness". This quartic growth is consistent with the Laplacian/Poisson toy examples and frequency-dependent conditioning effects emphasized in [40].

Length-scale interpretation. Let ℓ denote the smallest length scale that must be represented. Then $k_{\max} \sim 1/\ell$ and (A.15) implies

$$\kappa \text{ grows rapidly as } \ell \downarrow 0. \tag{A.16}$$

In many-body Schrödinger problems, small ℓ arises from nodal structure, strong localization, and short-range correlation features; in all such regimes, the Laplacian makes the strong residual extremely curvature-sensitive.

A.4 Barron/spectral perspective: the Laplacian

The conditioning story above can be rephrased in approximation terms. In spectral Barron-type viewpoints [37], function classes are controlled by norms that weight the Fourier transform by powers of frequency (or related spectral weights). A key implication is that applying derivatives systematically increases the importance of high-frequency tails [40]. Since Δ multiplies a Fourier mode by $|\mathbf{k}|^2$, and the squared residual effectively involves L^*L (hence Δ^2 in Laplacian-dominant regimes), the residual can require substantially more "spectral budget" than the wavefunction itself. In practical terms: even if Ψ is approximable in a given network class, *approximating its strong residual* can be much harder, because the loss implicitly demands accurate high-frequency curvature (cf. (A.14)). This is one of the motivations, in numerical analysis of PINNs, for considering weak/variational formulations that reduce derivative order when the strong form is too stiff [40].

A.5 Implications for Stochastic Reconfiguration

Stochastic reconfiguration (SR) updates parameters using a natural-gradient-like preconditioner [24, 26, 57]. Writing $O_i = \partial_{\theta_i}\log\Psi_\theta$, one common SR update is

$$S(\theta)\,\delta\theta = -g(\theta), \qquad S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = 2\,\mathrm{Cov}_{p_\theta}[E_L, O_i], \tag{A.17}$$

where $p_\theta \propto |\Psi_\theta|^2$ and $E_L = (\hat{H}\Psi_\theta)/\Psi_\theta$. SR improves conditioning in parameter space by using the (quantum) information geometry of the variational family [24, 57]. However, the Laplacian-induced stiffness described above is an *operator-level* phenomenon: strong residuals magnify curvature errors via Δ and Δ^2 (cf. (A.14)), which remains a fundamental challenge even when parameter updates are geometrically preconditioned.

Chapter B

Coulomb conditioning

B.1 Coulomb structure and singularity

For our two-dimensional electronic systems ($\mathbf{r}_i \in \mathbb{R}^2$), the interaction contains Coulomb terms of the form

$$\hat{V}_{\mathrm{C}}(\mathbf{R}) = \sum_{1\le i<j\le N} \frac{1}{r_{ij}}, \qquad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|. \tag{B.1}$$

This is a *singular multiplication operator*: it diverges at particle coalescence $r_{ij} \to 0$. The exact eigenfunction remains finite (or vanishes by antisymmetry) but develops non-smooth short-distance structure (a cusp) so that divergences in kinetic and potential contributions cancel in physically meaningful quantities such as the local energy [26].

> Coulomb does not merely add "another term" to the residual; it introduces *singular short-range physics* that forces delicate cancellations with the Laplacian. Strong residual objectives square these effects and become extremely sensitive to tiny cusp errors.

B.2 2D cusp conditions: antiparallel vs parallel spins

We present the 2D cusp logic in a way that separates the two spin channels. The cleanest statement (especially for practical ansätze) is in terms of the short-distance behaviour of a *Slater–Jastrow* factorization

$$\Psi(\mathbf{R}) = D(\mathbf{R})\, J(\mathbf{R}), \qquad J(\mathbf{R}) = \exp\Big(\sum_{i<j} u(r_{ij})\Big), \tag{B.2}$$

because (i) the Slater part D enforces antisymmetry and determines whether Ψ vanishes at coalescence, and (ii) the Jastrow slope $u'(0)$ is what cancels the $1/r$ Coulomb singularity in the local energy [26].

Fix a pair (i, j) and introduce relative/centre-of-mass coordinates in 2D:

$$\mathbf{R}_{ij} := \frac{\mathbf{r}_i + \mathbf{r}_j}{2}, \qquad \mathbf{r} := \mathbf{r}_i - \mathbf{r}_j, \qquad r := |\mathbf{r}|. \tag{B.3}$$

A direct computation gives, in any dimension,

$$\Delta_i + \Delta_j = \frac{1}{2}\Delta_{\mathbf{R}_{ij}} + 2\,\Delta_{\mathbf{r}}, \tag{B.4}$$

hence the singular short-distance kinetic behaviour is controlled by the relative term $-\Delta_{\mathbf{r}}$.

A key 2D identity is

$$\Delta_{\mathbf{r}} r = \frac{1}{r} \qquad (r > 0,\ \mathbf{r} \in \mathbb{R}^2). \tag{B.5}$$

Antiparallel spins (distinguishable in the Slater part). If the two electrons have opposite spin, they enter different determinants in D and generically D does *not* vanish at $\mathbf{r} \to 0$. Then $\Psi(0) \neq 0$ and a direct cusp expansion in 2D is legitimate:

$$\Psi(\mathbf{r}, \ldots) = \Psi(0, \ldots)\Big(1 + a\, r + \mathcal{O}(r^2)\Big), \qquad r \to 0. \tag{B.6}$$

Using (B.5), the singular part of the relative Laplacian is

$$\Delta_{\mathbf{r}}\Psi \approx \Psi(0)\, a\, \Delta_{\mathbf{r}} r = \Psi(0)\, \frac{a}{r}, \qquad \frac{\Delta_{\mathbf{r}}\Psi}{\Psi} \approx \frac{a}{r}. \tag{B.7}$$

Since the pair kinetic contribution contains $-\Delta_{\mathbf{r}}$, the singular part of the local energy is

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi}{\Psi} \supset \left(-\frac{a}{r} + \frac{1}{r}\right) = \frac{1-a}{r}. \tag{B.8}$$

Thus finiteness of E_L at coalescence requires

$$a = 1, \qquad \text{i.e.} \qquad \partial_r \log \Psi|_{r=0} = 1 \quad \text{(2D, antiparallel spins)}. \tag{B.9}$$

Equivalently, in a Slater–Jastrow form with $D(0) \neq 0$, this is the same as $\partial_r \log J|_0 = 1$ for the ij-pair.

Parallel spins (identical fermions, node at coalescence). For same-spin electrons, antisymmetry forces D (and hence Ψ) to vanish as $\mathbf{r} \to 0$. Generically one has a *linear* zero:

$$D(\mathbf{r}, \ldots) \;=\; \boldsymbol{\eta}(\ldots) \cdot \mathbf{r} \;+\; \mathcal{O}(r^2), \qquad r \to 0, \tag{B.10}$$

with some nonzero vector $\boldsymbol{\eta}$ depending on the other coordinates. In this case, the cancellation of the Coulomb singularity is controlled by the Jastrow factor through cross terms in $-\Delta_{\mathbf{r}}(DJ)/(DJ)$. A standard short-distance analysis (carried out explicitly for 2D quantum dots) shows that the divergent kinetic contribution behaves like

$$-\frac{\Delta_{\mathbf{r}}(DJ)}{DJ} \supset -\frac{(2k+d-1)}{r}\, \partial_r \log J \qquad \text{with} \qquad d = 2, \;\; k = 1 \text{ (identical particles)}, \tag{B.11}$$

while the Coulomb potential contributes $+1/r$ to E_L as before. Hence finiteness of the local energy requires

$$\partial_r \log J|_{r=0} = \frac{1}{2k+d-1} = \frac{1}{3} \quad \text{(2D, parallel spins)}. \tag{B.12}$$

This 1 (antiparallel) versus 1/3 (parallel) 2D cusp statement is the standard quantum-dot/Jastrow–Slater cusp condition [58].

What to remember. In 2D:

$$\text{antiparallel spins:} \quad \partial_r \log \Psi|_0 = 1 \quad \Longleftrightarrow \quad \partial_r \log J|_0 = 1, \tag{B.13}$$

$$\text{parallel spins:} \quad \Psi(0) = 0 \text{ (linear node)}, \quad \partial_r \log J|_0 = \frac{1}{3}. \tag{B.14}$$

B.3 Why Coulomb is ill-conditioned (2D emphasis)

The cusp conditions above show *how* Coulomb creates stiffness: it forces kinetic and potential contributions to be individually large and singular, while their *sum* is finite only under a short-distance constraint. Strong residual minimization is therefore a *cancellation problem*.

Sensitivity amplification in the local energy. If the learned antiparallel-spin cusp slope is $a_\theta = 1 + \delta a$, then (B.8) implies

$$E_L(\mathbf{R}) \supset -\frac{\delta a}{r}, \tag{B.15}$$

so an arbitrarily small cusp mismatch produces an arbitrarily large local-energy spike as $r \to 0$.

For parallel spins, if the learned Jastrow slope is $\partial_r \log J|_0 = \frac{1}{3} + \delta b$, then the same short-distance analysis leading to (B.12) implies

$$E_L(\mathbf{R}) \supset -\frac{3\,\delta b}{r}. \tag{B.16}$$

Thus both channels generate $1/r$ spikes unless the cusp is satisfied.

Why squaring is especially dangerous in 2D. A $1/r$ defect in E_L (or in $(\hat{H}-E)\Psi$) is concentrated near coalescence. But in 2D the squared singularity is not integrable—it diverges logarithmically:

$$\int_{r<\varepsilon}\left(\frac{1}{r}\right)^2 d^2\mathbf{r} = \int_0^{\varepsilon}\frac{1}{r^2}\,(2\pi r\,dr) = 2\pi\int_0^{\varepsilon}\frac{dr}{r} = +\infty.$$

Therefore, without effectively resolving the cusp, *strong* squared-residual objectives can be dominated (or even made ill-defined in the continuum limit) by the coalescence region. In Monte Carlo practice, this manifests as heavy-tailed fluctuations and extreme sensitivity to rare close-pair samples.

Residual expansion and cancellation structure. Writing $L = \hat{T} + (\hat{V} - E)$ and expanding the squared residual norm:

$$\|L\Psi\|_w^2 = \|\hat{T}\Psi\|_w^2 + \|(\hat{V}-E)\Psi\|_w^2 + 2\,\mathrm{Re}\langle\hat{T}\Psi,\ (\hat{V}-E)\Psi\rangle_w, \tag{B.17}$$

we see that near coalescence both $\hat{T}\Psi$ and $\hat{V}_{\mathrm{C}}\Psi$ are large. The exact solution relies on cancellation between these contributions (cf. (B.8) and (B.12)), which means (B.17) demands high *relative* accuracy in the cusp region: if either term is off by a small fraction, the net residual can be large in absolute magnitude. This is a classic recipe for ill-conditioning: the loss landscape contains directions where small parameter changes cause huge residual changes in a small region.

Operator-level picture (L^*L). From the same viewpoint as Appendix A, strong residual training induces matrices/operators tied to L^*L [40]. For $L = \hat{T} + (\hat{V} - E)$, one has (formally)

$$L^*L = \hat{T}^2 + (\hat{V}-E)^2 + \hat{T}(\hat{V}-E) + (\hat{V}-E)\hat{T}. \tag{B.18}$$

Coulomb enters here in two destabilizing ways:

1. the positive term $(\hat{V} - E)^2$ involves a $1/r^2$-type singular weight;
2. the mixed terms contain derivatives of V through product rules, e.g. $\hat{T}(V\Psi)$ generates $\nabla V\cdot\nabla\Psi$ and ΔV contributions, which are even more singular/distributional at coalescence.

This formal expansion explains why the Coulomb singularity can severely inflate the effective spectral spread of the training operator, aligning with the operator-conditioning viewpoint [40].

B.4 Barron/spectral perspective: the Coulomb singularity

Coulomb forces the *true* 2D solution to have cusp-like, non-smooth short-range structure, so approximating the *strong residual* requires representing high-frequency content that is not well captured by globally smooth/low-complexity function classes.

In spectral/Barron-style approximation theories [37], approximation quality is governed by spectral decay and norms that weight the Fourier tail [40]. A cusp (or a linear node times a cusp in J) implies slower spectral decay than a globally smooth function. Strong residual training then *further* amplifies the high-frequency tail because it applies Δ (and effectively L^*L), which multiplies Fourier modes by powers of $|\mathbf{k}|$ (Appendix A). Hence, even if Ψ can be approximated reasonably in function value, the strong residual can remain large unless the cusp structure is represented accurately. This is exactly the type of regularity mismatch that motivates weak/variational formulations in numerical analysis of PINN-type methods, since weak forms reduce the derivative order demanded of the approximation [40].

Practically, QMC wavefunctions therefore incorporate explicit short-range correlation structure (e.g. cusp-enforcing Jastrow factors), precisely to remove these singular residual pathologies and reduce the variance of local quantities [26, 58].

B.5 Implications for Stochastic Reconfiguration

SR performs a natural-gradient step using a (quantum) Fisher/metric matrix [24, 26, 57]. In one common formulation,

$$S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = 2\,\mathrm{Cov}_{p_\theta}[E_L, O_i], \qquad S\,\delta\theta = -g, \tag{B.19}$$

with $O_i = \partial_{\theta_i} \log \Psi_\theta$.

Coulomb-induced cusp mismatch produces E_L spikes of the form (B.15)–(B.16), which has two consequences:

1. **Heavy-tailed Monte Carlo forces.** The estimator of $g_i = \mathrm{Cov}[E_L, O_i]$ becomes noisy when E_L has rare but extreme spikes (short-range events). This increases variance and can destabilize the linear solve unless one uses damping/regularization, as standard in SR practice [26].

2. **Coupling to curvature sensitivity.** Even if SR corrects anisotropy in parameter space (natural gradient) [24, 57], it does not remove the underlying operator-driven stiffness: the spikes originate from imperfect kinetic–potential cancellation enforced by Coulomb cusp physics.

Summary. Laplacian dominance creates stiffness through k^4-type amplification of curvature errors; Coulomb creates stiffness by enforcing cusp constraints (in 2D: 1 vs 1/3 depending on spin channel) and cancellation against the Laplacian. Both mechanisms worsen conditioning of strong residual objectives and increase statistical difficulty of SR through heavy-tailed local-energy fluctuations [26, 40, 58].

Chapter C

The collocation–backflow catch-22

This appendix formalizes a structural difficulty encountered when training backflow transformations under the *strong-form* collocation objective—the one that minimises the variance of the pointwise local energy E_L—and presents extensive numerical evidence for the N=6, ω=0.5 quantum dot. The core observation is a *catch-22*: backflow's learning signal is concentrated near the nodal surface, but the local energy diverges there for any imperfect wavefunction, and every sampling strategy we tried that includes those points produced gradients whose variance we could not control. The result, across every intervention below, is an accuracy floor at roughly 0.3%—set by the Jastrow alone—that is absent under VMC + SR training of the same ansatz family. We call the floor irreducible *within this pipeline*: it is an empirical bound established by exhaustion over the interventions we tested, not a proof that no strong-form scheme can do better.

The scope of this result needs two qualifications. First, it is a property of the *strong form* specifically: the weak-form (score-function / REINFORCE) collocation objective used for the results in the main text never differentiates through the Laplacian and does not inherit this pathology, which is why it trains the same ansatz to far below the 0.3% floor. The value here is therefore a diagnosis of one objective's limits, not a measure of what MCMC-free training can achieve. Second, this analysis is what motivated the shift to the weak form, and is reported as a negative result rather than as a headline number.

C.1 Setup and notation

We consider a Slater–Jastrow–backflow ansätz of the form

$$\Psi_\theta(\mathbf{R}) = \det \tilde{S}_\uparrow(\mathbf{R};\theta_{\mathrm{BF}})\ \det \tilde{S}_\downarrow(\mathbf{R};\theta_{\mathrm{BF}})\ \exp\bigl(J(\mathbf{R};\theta_J)\bigr), \tag{C.1}$$

where $\tilde{S}_\sigma$ is the Slater matrix evaluated at backflow-shifted coordinates $\tilde{\mathbf{r}}_i = \mathbf{r}_i + \Delta\mathbf{x}_i(\mathbf{R};\theta_{\mathrm{BF}})$, J is a neural Jastrow correlator, and $\theta = (\theta_J, \theta_{\mathrm{BF}})$ are trainable parameters. A CTNN message-passing network produces the backflow shifts $\Delta\mathbf{x}_i$.

We train by minimizing the *screened collocation* objective

$$\mathcal{L}(\theta) = \mathrm{Var}_{\mathcal{S}}\bigl[E_L(\mathbf{R};\theta)\bigr], \qquad E_L = \frac{\hat{H}\Psi_\theta}{\Psi_\theta}, \tag{C.2}$$

where the collocation set $\mathcal{S}$ is obtained by importance-sampling from a Gaussian proposal q and retaining the top-K points ranked by $|\Psi_\theta|^2/q$ ("screened collocation"). This top-K selection naturally suppresses near-node configurations where $|\Psi|^2 \approx 0$.

C.2 The divergence of E_L near nodes

The local energy decomposes as

$$E_L = -\frac{1}{2}\frac{\nabla^2\Psi}{\Psi} + V = -\frac{1}{2}\Bigl[\nabla^2 J + |\nabla J|^2 + 2\,\nabla J\cdot\frac{\nabla D}{D} + \frac{\nabla^2 D}{D}\Bigr] + V, \tag{C.3}$$

where $D := \det \tilde{S}_\uparrow \det \tilde{S}_\downarrow$ and the last term $\nabla^2 D/D$ contains second derivatives of the backflow shifts $\nabla^2 \Delta\mathbf{x}_i$ amplified by the inverse determinant.

For the exact wavefunction, E_L is constant. If Ψ is an eigenstate, $E_L = E$ everywhere. The cancellation is a property of the *full* local energy $\hat{H}\Psi/\Psi$ with $\Psi = D\, e^J$, in which the Jastrow cross term $2\nabla J \cdot \nabla D/D$ of Eq. (C.3) participates alongside $\nabla^2 D/D$; we isolate the determinant terms below because they are the ones the backflow parameters move, not because the others are absent. For an exact eigenstate, the potentially singular contributions in the full expression $\nabla^2(De^J)/(De^J)$ cancel at a simple node. This is not a property of $\nabla^2 D/D$ in isolation: for a generic smooth determinant with $D \sim d$ near the nodal surface $\mathcal{N}$, $\nabla^2 D$ may remain $\mathcal{O}(1)$. The exact eigenstate satisfies

$$\frac{\nabla^2(De^J)}{De^J} \text{ finite on } \mathcal{N}, \tag{C.4}$$

through cancellation among the determinant, Jastrow-cross, and remaining Jastrow terms in Eq. (C.3).

For an approximate wavefunction, the cancellation generically fails. During training, the backflow parameters θ_{BF} are imperfect: the *learned* nodal surface $\mathcal{N}_\theta$ (where $D(\cdot;\theta) = 0$) does not coincide with the *true* nodal surface. Let d_θ denote the signed distance from the learned node and d^* that from the true node. Near a point where $d_\theta \approx 0$ but $d^* \neq 0$, the determinant $D \sim d_\theta \to 0$ while the kinetic contributions from the Jastrow and potential terms remain finite:

$$E_L(\mathbf{R}) \supset \frac{\nabla^2 D(\mathbf{R};\theta)}{D(\mathbf{R};\theta)} \sim \frac{c}{d_\theta} \to \pm\infty \qquad \text{as } d_\theta \to 0,\ d^* \neq 0, \tag{C.5}$$

for some configuration-dependent constant $c \neq 0$. Conversely, near a point on the true node ($d^* = 0$) but off the learned node ($d_\theta \neq 0$), $\Psi \sim d^* \to 0$ and the wavefunction assigns vanishing weight. The pathology (C.5) is therefore not an artifact of the nodal structure itself—it is a property of *mismatched* nodes, and is worst precisely when backflow still has something to learn.

C.3 The geometric mismatch argument

We now locate the divergence. It is *not* in the variance of the local energy, which the calculation below shows to be finite for a simple node; it is one derivative later, in the parameter gradient.

Consider the contribution to $\mathrm{Var}[E_L]$ from a neighborhood U_ε of the learned nodal surface:

$$\mathrm{Var}[E_L] \geq \int_{U_\varepsilon} |\Psi(\mathbf{R};\theta)|^2 \left(E_L(\mathbf{R};\theta) - \bar{E}\right)^2 d\mathbf{R}. \tag{C.6}$$

When the node is "simple" (D vanishes linearly), we have

$$|\Psi|^2 \sim d_\theta^2, \qquad\qquad E_L^2 \sim d_\theta^{-2\alpha}, \tag{C.7}$$

where α characterises the severity of the mismatch. For the generic case where the numerator $\nabla^2 D$ does *not* vanish on $\mathcal{N}_\theta$ (because $\mathcal{N}_\theta \neq \mathcal{N}^*$), one has $\alpha = 1$ so that $E_L^2 \sim d_\theta^{-2}$.

In D-dimensional configuration space, the volume element in a tubular neighborhood of a codimension-1 surface scales as $dV \sim d_\theta^0\, d\sigma\, dd_\theta$. Thus the integrand in (C.6) behaves as

$$|\Psi|^2 \cdot E_L^2 \cdot dV \sim d_\theta^2 \cdot d_\theta^{-2} \cdot dd_\theta = dd_\theta, \tag{C.8}$$

which integrates to $\mathcal{O}(\varepsilon)$ and is therefore *finite*: a simple node does not, by itself, make $\mathrm{Var}[E_L]$ diverge under the $|\Psi|^2$ measure. The instability enters one derivative later, in the *parameter gradient.* But the key observation is that this is the *best case.* In practice, the mismatch between $\mathcal{N}_\theta$ and $\mathcal{N}^*$ is not uniform: there exist regions where the surfaces cross at angles, creating pockets where $|d^*|/|d_\theta|$ is large (the true wavefunction is non-negligible while the learned determinant is near-zero), and there the E_L^2 spike is not compensated by $|\Psi|^2$ suppression.

More concretely, the gradient $\partial\mathcal{L}/\partial\theta_{\mathrm{BF}}$ involves terms of the form

$$\frac{\partial E_L}{\partial \theta_{\mathrm{BF}}} \supset \frac{1}{D}\frac{\partial(\nabla^2 D)}{\partial\theta_{\mathrm{BF}}} - \frac{\nabla^2 D}{D^2}\frac{\partial D}{\partial\theta_{\mathrm{BF}}}, \tag{C.9}$$

both of which diverge as $D \to 0$. Even in the marginal case where $\mathrm{Var}[E_L]$ is finite, the *gradient variance* (which involves these higher-order poles) generically diverges, making gradient-based optimisation impossible at points near the learned node.

C.4 The catch-22

The preceding analysis reveals a three-sided dilemma:

1. **Backflow learns at the node.** The purpose of backflow is to reshape the nodal surface of the Slater determinant. The gradient signal $\partial E_L/\partial\theta_{\mathrm{BF}}$ is largest where the backflow shifts actually change which side of the node a configuration point lies on, i.e., where $D \approx 0$.
2. **E_L diverges at the node for imperfect θ.** As shown in §C.2–C.3, the local energy and its parameter gradient contain poles of the form D^{-1} and D^{-2} whenever the learned node does not coincide with the exact node. The resulting gradient variance is unbounded.
3. **Good collocation sampling must avoid the node.** Under screened collocation, the top-K selection ranks points by $|\Psi|^2/q \propto D^2 e^{2J}/q$. Since $D^2 \to 0$ at the node, near-node points are systematically excluded from the training batch. This is *protective*—it prevents gradient explosion—but it also removes the points where backflow's learning signal lives.

A "catch-22" emerges:

> *Including near-node points gives divergent, noisy gradients and backflow shrinks defensively or diverges. Excluding them gives stable training but removes backflow's learning signal. In either regime, collocation cannot effectively train backflow beyond an accuracy floor set by the correlator quality alone.*

C.5 Why VMC + SR escapes

Under VMC, the sampling distribution *is* $|\Psi_\theta|^2$, which assigns vanishing measure to the nodal surface. Near-node configurations are strongly *suppressed*—for a simple node the probability of lying within a distance ϵ scales as ϵ^3—rather than strictly absent, which is enough to keep the D^{-1} singularity out of the gradient estimator in practice. However, VMC *still optimises the node position* through a subtle mechanism:

Indirect nodal information through bulk correlations. The variational energy $\langle E_L\rangle_{|\Psi|^2}$ is sensitive to nodal position because shifting the node changes the wavefunction *everywhere* (through normalisation and through the functional form of D). The SR force vector $g_i = \mathrm{Cov}_{|\Psi|^2}[E_L, O_i]$ (where $O_i = \partial_{\theta_i} \log\Psi$) receives contributions from the *bulk*—regions where both $|\Psi|^2$ and E_L are well-behaved—that are *correlated with* nodal position but do not require evaluating E_L at the node.

SR amplifies the signal. The Fisher metric $S_{ij} = \mathrm{Cov}[O_i, O_j]$ captures how sensitive $\log\Psi$ is to each parameter direction. By solving $(S + \varepsilon I)\,\delta\theta = -g$, SR amplifies exactly those subtle bulk signals that encode nodal information, compensating for the fact that the node itself has zero weight under $|\Psi|^2$.

Core asymmetry. The asymmetry needs stating carefully, because a power count that is too quick proves too much. Under $|\Psi|^2$ the SR force $g_i = \mathrm{Cov}[E_L, O_i]$ has a finite *mean*: both E_L and $O_i = \partial_{\theta_i}\log\Psi$ carry a d_θ^{-1} pole near a mismatched node, their product goes as d_θ^{-2}, and the measure supplies $|\Psi|^2 \sim d_\theta^2$, so the integral converges. Its *variance* does not, by the same counting—so VMC is not immune to near-node pathology either, and in practice this is what the damping and trust region of §4.4 are for.

The real distinction is one derivative higher. VMC and SR never differentiate E_L: the local energy enters only as a multiplicand in a covariance, so the D^{-2} pole of Eq. (C.9)—which comes from ∂_θ acting on $\nabla^2 D/D$—never appears. Strong-form collocation differentiates E_L by construction, and inherits it. That is why the same node is survivable in one scheme and not in the other, and it is the argument

§7.3.1 makes for the weak form as well. None of the sampling, loss, or preconditioning strategies we tested (§C.6) overcame this asymmetry within the strong-form collocation framework; we present it as a structural obstruction of the tested pipeline rather than a theorem that no collocation method can ever avoid it.

C.6 Experimental evidence

We present a systematic study for N=6 electrons in a 2D harmonic trap with ω=0.5 (DMC reference: E_{DMC} = 11.78484(6) [13]). The ansätz uses a neural Jastrow correlator (∼15,000 parameters) and a CTNN backflow network (∼8,600 parameters) with bf_scale = 0.7. All energies are evaluated via MCMC with 15 000 samples after burn-in.

C.6.1 Baseline: correlator alone vs. correlator + backflow

Table C.1: Baseline collocation results for N=6, ω=0.5. "Correlator only" trains without backflow; "joint" trains both networks from scratch for 300 epochs with bf_scale = 0.7. The backflow provides only a marginal improvement of ∼0.1 percentage points.

Configuration	E (Ha)	σ_E	err (%)
Correlator only (300 ep)	11.834	0.003	0.42
Correlator + BF, joint (300 ep)	11.823	0.003	0.33
Correlator + BF, separate opt	11.822	0.003	0.32
DMC reference	11.78484	—	0.00

The backflow reduces the error from 0.42% to 0.32–0.33%, far short of the <0.01% reached by VMC + SR on the same ansatz *family* (Table 6.2). The two are not matched in size: the networks here total ∼24k parameters against ∼236k for the production model, so the comparison bounds what this objective achieves at this capacity rather than isolating the objective alone. The floor is nonetheless a property of the objective, because it does not move under any of the interventions below, including ones that add capacity.

C.6.2 Sampling strategies

We tested multiple strategies to bring near-node information into the collocation batch, hoping to give the backflow a stronger learning signal.

Table C.2: Effect of sampling strategy on collocation accuracy. All experiments use 50-epoch fine-tuning from the best collocation checkpoint (0.35%), except "hard-smooth (scratch)" which trains 300 epochs from random initialisation.

Method	Description	err (%)	Outcome
Top-K baseline	standard screened collocation	0.35	stable
Hard mining + Huber	top-K of residuals	0.41	neutral
Hard mining + smoothness	+ $\|\nabla^2 \Delta x\|^2$ penalty	0.38	slight help
SIR (single proposal)	$\|\Psi\|^2$-reweighted resampling	0.47	worse
SIR (mixture)	Gaussian mixture proposal	0.44	worse
Gumbel top-K [59]	stochastic top-K selection	5.00	catastrophic

Every strategy that includes more near-node points (SIR, Gumbel) produced *worse* results—consistent with the catch-22: the additional points carry divergent E_L values that corrupt the gradient. Gumbel top-K (which replaces deterministic selection with stochastic Gumbel-softmax) was catastrophic, with early stopping triggered by rapid energy divergence.

C.6.3 Loss modifications

Table C.3: Effect of loss function modifications. The hybrid loss adds a mean-energy (variational) term; the det filter discards points with small $|\det \tilde{S}|$.

Method	Description	err (%)	Outcome
Variance loss (baseline)	Huber on $E_L - \bar{E}$	0.33	best collocation
Hybrid anneal	$\beta \cdot \mathrm{Var} + (1{-}\beta) \cdot \mathrm{Mean}$, $\beta{:}0.7{\to}0.15$	2.59	catastrophic
Gumbel + hybrid	stochastic top-K + hybrid	5.00	catastrophic
Hybrid + det filter	β=0.2 fixed + $|\det|$ filter	24.75	broken

The mean-energy gradient $\nabla_\theta \langle E_L \rangle_{\mathcal{S}}$ under biased collocation sampling does *not* equal the VMC gradient—the sampling distribution is not $|\Psi|^2$—so the variational principle does not hold, and the backflow drifts to large, unphysical shifts ($|\Delta \mathbf{x}| \approx 0.83$, versus $\sim$0.23 under variance-only training). The variance loss's implicit constraint on backflow magnitude is *protective*, not pathological.

C.6.4 Gradient regularisation

We attacked the divergence sources in E_L directly:

Table C.4: Regularisation of the E_L singularity (50-epoch fine-tune from 0.35% checkpoint). Each modification smooths or clips one of the divergent contributions to E_L.

Method	Modification	err (%)	Outcome
Soft Coulomb	$1/r \to 1/\sqrt{r^2+\varepsilon^2}$, ε=0.2	1.41	much worse
Regularised det	$\log|D| \to \mathrm{logaddexp}(\log|D|, \log\delta)$	0.49	worse
T clipping	$\max(|T|, T_{\max})$, $T_{\max}$=30	0.43	neutral

Soft Coulomb trains on wrong physics (V is no longer the true potential). Regularising the determinant distorts the $\log\Psi$ landscape. Kinetic energy clipping is neutral because the clipped points are exactly those that carried the (noisy) backflow signal. None of the regularisations improve upon the baseline.

C.6.5 Gradient preconditioning

Table C.5: Gradient preconditioning experiments (300-epoch scratch training unless noted). K-FAC [60] uses block-diagonal Kronecker-factored Fisher on the backflow network only.

Method	err (%)	Outcome
Vanilla Adam	0.33	baseline
K-FAC on CTNN	0.34	no improvement
K-FAC + smoothness	0.51	worse
Variance-minimising pretrain	0.54	worse
Adam fine-tune (30 ep, lr=1e-4)	0.34	no improvement
Adam burst (15 ep, lr=3e-4)	0.56	worse

K-FAC provides a block-diagonal approximation to the Fisher matrix and should, in principle, help with the ill-conditioned parameter-space geometry. In practice, it does not: the conditioning problem is *operator-level* (Appendices A–B), not merely parameter-space anisotropy, and K-FAC cannot address the fundamental divergence in E_L.

C.6.6 Summary of all interventions

The pattern is striking:

- Every intervention that *includes* more near-node points (SIR, Gumbel) *increases* the error—backflow shrinks defensively under gradient noise or diverges entirely.

Table C.6: Complete summary of all collocation-based backflow experiments for N=6, ω=0.5. No intervention breaks the ~0.32% accuracy floor. The final row is the same ansatz *family* under VMC + SR at production width (Table 6.2), an order of magnitude larger in parameter count; it is included to show the size of the gap, not as a matched comparison.

Experiment	err (%)	Category
Correlator + BF, sep opt (best)	0.32	baseline
Correlator + BF, joint 300 ep	0.33	baseline
K-FAC on CTNN	0.34	preconditioner
Adam fine-tune 30 ep	0.34	optimiser
Top-K baseline (retrained)	0.35	sampling
Hard mining + smoothness	0.38	sampling
Hard mining + Huber	0.41	sampling
Correlator only	0.42	no backflow
T clipping	0.43	regularisation
SIR (mixture)	0.44	sampling
SIR (single)	0.47	sampling
Regularised det	0.49	regularisation
K-FAC + smoothness	0.51	preconditioner
Var-min pretrain	0.54	pretrain
Adam burst lr=3e-4	0.56	optimiser
Soft Coulomb	1.41	regularisation
Hybrid loss (annealed)	2.59	loss
Gumbel top-K + hybrid	5.00	sampling + loss
Hybrid + det filter	24.75	loss + filter
Same ansatz, VMC + SR	**<0.01**	**variational**

- Every intervention that *changes* the loss to tolerate near-node contributions (hybrid mean-energy, det filter) produces catastrophic results because the collocation sampling is not $|\Psi|^2$ and the variational principle does not apply.
- The smoothness penalty $\lambda\|\nabla^2\Delta\mathbf{x}\|^2$ helps modestly (from 0.59% to 0.38% without it vs. with it) by damping the $\nabla^2 D/D$ poles, but cannot eliminate the fundamental divergence.
- Ironically, the only method that trains backflow effectively is MCMC-based VMC + SR—which sidesteps the entire collocation framework.

C.7 Connection to Appendices A and B

The catch-22 is a specific manifestation of the general conditioning difficulties analysed in Appendices A–B:

Laplacian stiffness (Appendix A). The k^4-type amplification of curvature errors (§A.3) is compounded by backflow, because $\nabla^2(\det\tilde{S})$ introduces second derivatives of $\Delta\mathbf{x}$ that are not present in a fixed-node ansatz. These additional high-frequency modes inflate the spectral spread of the training operator L^*L.

Coulomb singularity (Appendix B). The $1/r$ Coulomb potential (§B.1) adds a singular multiplication operator to $\hat{H}$. Near both the coalescence point ($r_{ij}\to 0$) and the node ($D\to 0$), E_L contains products of these two singularities, and the cusp cancellation (§B.2) must be maintained simultaneously with the nodal cancellation (C.4). In the strong residual squared $\|LE_L\|^2$, these compound singularities create an extremely stiff landscape.

The backflow-specific pathology. What distinguishes the backflow catch-22 from the general conditioning problem is the *self-referential* nature of the singularity: the backflow parameters θ_{BF} determine *where* the node is, so updating θ_{BF} moves the singularity in configuration space. Under the collocation loss, the gradient pushes θ_{BF} in a direction determined by E_L values at the current collocation points, but the resulting node shift invalidates those very points. VMC+SR avoids this because it samples from the *current* $|\Psi_\theta|^2$ at every step, so the sampling distribution co-adapts with the node position.

Chapter D

Additional collocation results

This appendix records the collocation results that Chapter 7 refers to but does not tabulate: the N=20 energies, and the variants that were tried and failed. Both bear on the same question—what limits the Markov-chain-free route—from the side of what does not work.

D.1 What was varied

Four axes were explored. The *sampling*: Gaussian and Gaussian-mixture proposals scaled by $1/\sqrt{\omega}$, oversample-then-resample with ESS diagnostics, ESS-adaptive oversampling, weight tempering and log-weight clipping. The *objective*: finite-difference collocation residuals, a hybrid weak form, and pure REINFORCE, which became the default, with the Laplacian kept in the reward path but never differentiated. The *preconditioner*: Adam, diagonal-Fisher natural gradient, and full SR in Woodbury and CG modes with damping anneals and trust-region caps. And the *training style*: single sweeps, wave-based cascades with checkpoint inheritance, and asynchronous campaigns relaunching from the best available checkpoint.

Two results held across the search. Optimisation quality dominated architectural novelty: moderately sized backflow+Jastrow models with stable training beat more ambitious variants with weaker optimisation geometry. And at low ω, regularising the resampling distribution (ESS control, top-mass reduction) helped where hard cusp gating did not—a controlled 2×2×3 matrix over gating, resampling regularisation and three seeds showed no measurable gain from gating and a consistent gain from regularisation.

Warm-starting along the confinement cascade ω : $1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001$ was necessary throughout; cold starts at low ω did not converge. Representative cascade results are N=6 at ω=1.0 ($E = 20.16167 \pm 0.00181$, $+0.012\%$), ω=0.5 ($E = 11.78466 \pm 0.00199$, -0.002%) and ω=0.1 ($E = 3.56525 \pm 0.00046$, $+0.32\%$), and N=12 at ω=1.0 ($E = 65.70699 \pm 0.00595$, $+0.010\%$). For N=2 the route covers the whole confinement range: $+0.012\%$, $+0.023\%$, -0.004%, $+0.002\%$ and $+0.154\%$ at $\omega = 1, 0.5, 0.1, 0.01, 0.001$, zero-variance extrapolated.

D.2 The $N = 20$ collocation energies

At N=20 the backflow's edge-feature tensor, which scales as $\mathcal{O}(N^2 h_{\rm bf})$, forces the hidden dimension down to $h_{\rm bf}$=48–64 to fit in memory. The best backflow run at ω=1.0 then reached only $\sim$+18% error, while a Jastrow-only ansatz—freed of that tensor, and so able to afford a larger collocation budget and higher oversampling—reached +1.32% ($E = 157.934 \pm 0.018$). The more expressive architecture is beaten by the simpler one, which is consistent with the memory-forced capacity cut degrading gradient quality more than its extra flexibility repays.

Polishing the Jastrow checkpoints at reduced learning rate improved the ω=1.0 error from $\sim$+2.6% to +1.32% over $\sim$400 epochs, and the ω=0.5 error from $\sim$+7.0% to +2.74% ($E = 96.447 \pm 0.015$); the best ω=0.1 result was +5.53% ($E = 31.637 \pm 0.008$). The gap to the sub-percent accuracy reached at $N \in \{6, 12\}$ tracks the capacity cut and the gradient noise it brings. A more memory-frugal backflow is the direct next intervention; these experiments do not isolate the collocation objective as the source of the remaining gap.

D.3 Negative results worth recording

Scalar final energy is not sufficient for method development on this route: ESS, top-mass concentration, rollback counts and the probe-to-final gap are what distinguish a run that converged from one that stopped moving. Several intuitively attractive interventions delivered nothing in the tested range—hard cusp gating most notably—and recording that was necessary to keep the search from circling. Stricter SR trust constraints, heavier damping, ESS-adaptive sampling and rollback controls improved operational stability without closing the accuracy gap at the hardest targets. What eventually closed it at low ω and N=6 was not a stabilisation measure but a change of proposal (§7.4).

Chapter E

The learned representation of one trained solution

This appendix records what the internal representation of *one* high-quality trained solution looks like. It is descriptive, it is measured on a single training run, and—as §7.5 sets out—most of the quantities in it are component-level and therefore gauge-like: the correlator and the backflow are partial substitutes, so how the labour is divided between them records the route the training took as much as the state it reached. Nothing in the architectural argument of Chapter 7 rests on anything here. It is reported because the regularities are striking and because they are ones the network was free *not* to adopt, not because they establish a property of the architecture.

Three findings survive that qualification well enough to be worth stating. Across $N \in \{2, 6, 12, 20\}$ and $\omega \in \{1, 0.5, 0.1, 10^{-2}, 10^{-3}\}$ the correlator feature space stays on one to three effective dimensions and never expands, even where the real-space density develops elaborate shell structure; what changes across the crossover is which branch the leading axis is built from, not how many axes there are. The displacement field is the opposite kind of object, occupying nearly all the spatial directions available to it. And the coupling between the two changes character near $\omega \approx 0.1$, the same place the energy partition crosses over (§6.2.3).

Every diagnostic below is evaluated on fresh samples drawn from $|\Psi_\theta|^2$, with the estimators and stability protocol of §5.2.

E.1 Correlator geometry: a low-rank representation that reorganises

We characterise the correlator feature space $Z = \overline{\phi}\|\overline{\psi}\|\mathbf{g}$ by its entropy-based effective rank $r_{\text{eff}}(Z)$, the correlation $|\rho(\text{head}, \text{PC1})|$ between the head output and its first principal component, and the power of PC1 decomposed across architectural branches. Table E.1 reports these quantities across the grid.

The manifold is low-dimensional everywhere. The effective rank stays below 3.1 across the entire grid, and the head output is almost perfectly reconstructed from PC1 alone in all but one analysed cell ($|\rho| \geq 0.94$ except N=20, ω=0.01, where $|\rho| = 0.79$). This holds even at strong correlation where the real-space density develops pronounced shell structure—the correlator does not respond by expanding its latent dimensionality. The modest peak at N=12, ω=0.1 ($r_{\text{eff}} = 3.08$) coincides with the intermediate-correlation regime where neither the shell picture nor the crystalline picture provides a clean single-mode description.

A $\phi \leftrightarrow \psi$ transition in the leading axis. What varies across regimes is *which branch dominates PC1.* At moderate-to-weak correlation ($\omega \geq 0.1$, $N \geq 6$), PC1 is predominantly ψ-loaded: the pair branch carries 60–73% of the variance, meaning the correlator's primary degree of freedom tracks pair correlations. At strong correlation the loading shifts toward ϕ (orbital) or $\mathbf{g}$ (auxiliary) branches for N=6 and N=12. The N=20 run does not follow the same branch-switching pattern: its pair branch remains dominant at both weakest confinements. Given that these are single-run, component-level diagnostics,

Table E.1: Correlator geometry (CTNN). r_{eff}: entropy-based effective rank of Z. $|\rho|$: absolute correlation between head output and PC1 reconstruction. $\phi/\psi/$g: fraction of PC1 variance from the orbital, pair, and auxiliary branches. The ψ-dominated entries ($\psi > 0.5$) are shown in bold.

N	ω	r_{eff}	$\|\rho\|$	ϕ	ψ	g
2	0.001	1.00	1.00	0.00	**1.00**	0.00
2	0.01	1.04	0.66	0.00	**1.00**	0.00
2	0.1	1.31	0.99	0.00	**0.53**	0.46
2	0.5	1.25	1.00	0.00	**0.58**	0.42
2	1.0	1.53	0.98	0.00	0.35	**0.65**
6	0.001	1.52	0.97	0.01	0.12	**0.87**
6	0.01	1.38	0.99	0.05	0.44	**0.50**
6	0.1	2.73	0.94	**0.66**	0.21	0.14
6	0.5	2.69	0.98	0.15	**0.63**	0.22
6	1.0	2.69	0.99	0.03	**0.62**	0.35
12	0.001	1.66	0.96	**0.54**	0.13	0.33
12	0.01	1.54	0.97	0.43	0.09	**0.48**
12	0.1	3.08	0.99	0.12	**0.63**	0.26
12	0.5	2.32	0.99	0.12	**0.69**	0.19
12	1.0	1.76	0.99	0.08	**0.73**	0.19
20	0.001	2.59	0.99	0.18	**0.74**	0.08
20	0.01	1.09	0.79	0.04	**0.94**	0.01
20	0.1	2.20	0.97	0.36	0.44	0.20
20	0.5	2.30	1.00	0.17	**0.70**	0.13
20	1.0	2.32	0.99	0.18	**0.71**	0.11

we do not interpret this discrepancy as a size-dependent physical transition. The transition is sharpest at N=12, where ψ goes from 0.09 at ω=0.01 to 0.63 at ω=0.1.

These loadings track the physics. When interactions are weak relative to the trap ($\omega \gtrsim 0.1$), the non-interacting shell structure provides a good zeroth-order description, and the correlator's primary role is refining pair correlations on top of that. When interactions dominate ($\omega \lesssim 0.01$), the leading correlator coordinate shifts away from pair features toward the single-particle and auxiliary branches. The network reorganises the features through which it represents the state, and that reorganisation—not any growth in dimension—is how it adapts. Nothing in the architecture imposes this behaviour: the N=6 and N=12 regime changes appear as a rotation of a low-dimensional axis across branches rather than as growth in latent dimensionality.

E.2 Backflow: a high-rank field with a low-rank footprint

Table E.2 characterises the message-passing displacement field. We report the normalised displacement magnitude $\|\Delta\mathbf{x}\|/\ell$, the effective rank of $\Delta\mathbf{x}$ (viewed as a $B \times 2N$ matrix of flattened displacements), and the CKA similarity between the correlator features with and without backflow.

Displacement magnitude scales with ω but saturates near $\|\Delta\mathbf{x}\|/\ell \sim 0.1$. At strong correlation ($\omega \leq 0.01$), displacements are small relative to the harmonic length, $\|\Delta\mathbf{x}\|/\ell \sim 0.02$–$0.05$. As the trap tightens and the system enters the moderate-to-weak correlation regime, the magnitude grows and saturates at $\sim$8–10% of ℓ for $\omega \gtrsim 0.1$. The message-passing backflow never produces large displacements—its tanh output bound and copresheaf message-passing structure act as implicit regularisers, in contrast to the conventional backflow, which can exhibit displacements comparable to the system scale in poorly converged regimes (e.g. $\|\Delta\mathbf{x}\|/\ell \sim 0.95$ for N=6, ω=0.001 with the conventional network).

Full-rank displacements, near-unity CKA. The effective rank of the displacement field is consistently $\approx 2N-2$, saturating the available spatial degrees of freedom modulo centre-of-mass conservation (confirmed by COM shifts of order 10^{-14}–10^{-17}). The backflow transformation is not a low-rank collective mode such as a breathing or rigid rotation; the displacement field spans essentially all COM-free

Table E.2: CTNN backflow geometry. $\|\Delta\mathbf{x}\|/\ell$: mean displacement magnitude normalised by harmonic length. Rank: effective rank of the displacement covariance. $2N-2$: maximum possible rank after centre-of-mass projection. CKA: linear centred kernel alignment between correlator features with and without backflow (1.0 = identical feature manifolds).

N	ω	$\|\Delta\mathbf{x}\|/\ell$	Rank	$2N-2$	CKA
2	0.001	0.003	1.9	2	1.000
2	1.0	0.029	2.0	2	1.000
6	0.001	0.017	10.0	10	0.999
6	0.01	0.035	9.9	10	0.998
6	0.1	0.096	10.0	10	0.995
6	0.5	0.093	10.0	10	0.991
6	1.0	0.089	10.0	10	0.979
12	0.001	0.017	21.9	22	1.000
12	0.01	0.030	21.9	22	0.998
12	0.1	0.096	21.9	22	0.993
12	0.5	0.096	21.9	22	0.993
12	1.0	0.091	21.9	22	0.995
20	0.001	0.023	37.7	38	0.999
20	0.01	0.046	37.4	38	1.000
20	0.1	0.075	37.8	38	0.997
20	0.5	0.099	37.6	38	0.990
20	1.0	0.099	37.6	38	0.991

spatial directions, so each particle receives a correction conditioned on the full configuration. The field stays near that ceiling across the whole grid, including the deepest Wigner points, where one might have expected the correction to become collective. Whether that is a property of *this* architecture or of trained backflows generally is not something the present measurement can decide; Chapter 7 settles it by holding everything else fixed (§7.1.2).

Yet the CKA between correlator features with and without backflow exceeds 0.99 in nearly all cases (and exceeds 0.979 everywhere). The correlator manifold is almost unperturbed by the displacement field. The correlator's feature geometry is therefore almost unaffected by the presence of the displacement field, which is a statement about that geometry and not about the subspace the backflow occupies. Whether the two components are genuinely complementary, rather than merely uncorrelated on this measure, is tested by ablation in §7.1.2.

E.3 Correlator–backflow coupling: from independence to cooperation

The previous subsection established that backflow leaves the correlator manifold intact. But are the correlator and the backflow *functionally* coupled—does the displacement depend on what the correlator computes? We measure this through three diagnostics: (i) the mean cosine similarity $\langle\cos(\nabla_{\mathbf{x}} f,\, \Delta\mathbf{x})\rangle$ between the correlator's gradient in configuration space and the displacement field (alignment implies the backflow pushes along directions the correlator is sensitive to); (ii) the fraction of configurations where $\Delta\mathbf{x}$ is aligned with ∇f; and (iii) $|\delta f|/\sigma_f$, where $\delta f = f(\tilde{R} + \Delta\mathbf{x}) - f(\tilde{R})$ is the change in the correlator output when it is evaluated on the backflow-shifted coordinates, in units of the correlator's own standard deviation σ_f. Two cautions attach to this last quantity. It is a *counterfactual sensitivity*: in the production ansatz the correlator sees the unshifted $\tilde{R}$ (only the determinant sees $\tilde{R} + \Delta\mathbf{x}$; §3.5), so δf measures how much the correlator *would* respond to the displacement, not an actual term in the wavefunction. And it is a typical relative change, not a variance decomposition: $|\delta f|/\sigma_f = 0.7$ means the counterfactual change is 0.7 standard deviations, not that 70% of the variance is attributable to backflow. Table E.3 reports these alongside the fraction of particles pushed toward their nearest neighbour.

Table E.3: Correlator–backflow coupling diagnostics (message-passing ansatz). $\langle\cos\rangle$: mean cosine between ∇f and $\Delta\mathbf{x}$. Aligned: fraction of configs with $\nabla f \cdot \Delta\mathbf{x} > 0$. $|\delta f|/\sigma_f$: backflow-induced change in the correlator output, relative to its standard deviation. $\to$NN: fraction of particles displaced toward their nearest neighbour (0.5 = isotropic). Only $N \geq 6$ shown; N=2 displacements are near-zero for most ω.

N	ω	$\langle\cos\rangle$	Aligned	$\lvert\delta f\rvert/\sigma_f$	$\to$NN
6	0.001	0.14	0.63	0.03	0.43
6	0.01	0.18	0.77	0.05	0.28
6	0.1	0.49	0.99	0.25	0.16
6	0.5	0.62	1.00	0.33	0.19
6	1.0	0.77	1.00	0.41	0.18
12	0.001	−0.48	0.01	0.03	0.61
12	0.01	0.22	0.82	0.05	0.29
12	0.1	0.56	1.00	0.39	0.12
12	0.5	0.65	1.00	0.49	0.20
12	1.0	0.75	1.00	0.55	0.21
20	0.001	0.32	0.97	0.06	0.44
20	0.01	0.22	0.89	0.07	0.45
20	0.1	0.39	1.00	0.25	0.17
20	0.5	0.65	1.00	0.58	0.22
20	1.0	0.78	1.00	0.69	0.25

A transition at $\omega \approx 0.1$. The data reveal a sharp regime change. At $\omega \leq 0.01$ the coupling is weak and inconsistent in sign: $\langle\cos\rangle$ runs from −0.48 to 0.32 across the six cells, the alignment fraction is erratic, and backflow barely affects the correlator output ($|\delta f|/\sigma_f < 0.07$ everywhere). The N=12, ω=0.001 cell is strongly *anti*-aligned ($\langle\cos\rangle = -0.48$, alignment fraction 0.01), which we record without an account of it: at that confinement the displacement field is small enough (Table E.2) that its direction relative to the correlator gradient carries little energy either way, but we cannot say why the sign is systematic rather than scattered. In this regime the backflow operates in a subspace largely orthogonal to the correlator's gradient—it modifies the wavefunction in directions that W_θ is insensitive to, consistent with its role being to adjust the Slater determinant's orbital positions rather than the correlator.

At $\omega \geq 0.1$, the picture changes qualitatively. The gradient cosine jumps to 0.5–0.8, the alignment fraction reaches 1.0 for $N \geq 6$, and the counterfactual output change reaches $|\delta f|/\sigma_f = 0.25$–$0.69$ (i.e. up to $\sim$0.7 standard deviations of the correlator). Here the backflow acts as a cooperative refinement: it pushes configurations along directions the correlator is maximally sensitive to, systematically correcting the correlator's residual errors. The transition is sharpest between $\omega = 0.01$ and $\omega = 0.1$, coinciding with the crossover in $\Gamma = V_{\text{int}}/T$ from $\Gamma > 10$ (interaction-dominated) to $\Gamma \sim 3$–6 (moderate correlation).

Directional structure: correlation-hole deepening. The nearest-neighbour direction fraction provides a complementary signature. At $\omega \geq 0.1$ for $N \geq 6$, only 12–25% of particles are displaced toward their nearest neighbour. The backflow preferentially pushes particles *apart*, actively deepening the correlation hole. This directional bias is strongest at N=12, ω=0.1 ($\to$NN = 0.12), exactly where the correlator effective rank peaks and the $\phi \leftrightarrow \psi$ transition occurs. At strong correlation ($\omega \leq 0.01$, large N), the fraction rises toward 0.5 (isotropic), because the large equilibrium inter-particle separations leave little room for further correlation-hole deepening.

E.4 Limits of component-level interpretation

Taken together these measurements describe two components of very different character. The correlator occupies one to three directions, whose leading axis tracks the dominant global coordinate of the cloud and changes identity rather than growing in number as the physics changes. The displacement field is high-rank and configuration-dependent, and its presence leaves the correlator's feature geometry almost unchanged (CKA > 0.98). The coupling between them is weak and inconsistent in sign at strong correlation, and strong and systematically aligned at moderate coupling, where the displacements follow the correlator's gradient; the change is sharp and coincides with the crossover in Γ.

These measurements describe one trained solution and do not by themselves establish which properties are architecture-specific. A different training route could divide the same labour differently and arrive at the same energy, which for two partial substitutes is what one should expect. Deciding the question requires holding everything else fixed and varying only the architecture.

Chapter F

Structural diagnostics in full

Chapter 6 reports the two radial quantities the argument turns on—r_{mode} and the width-to-scale ratio γ. The complete diagnostics are collected here: the mean radius, width, FWHM and decile quantiles for N=2, and the ring-angle spread and pair Lindemann ratio for N=6. Estimators and stability protocol are defined in §5.1. All lengths are in Bohr; angles in radians.

F.1 Fitted energy and radius exponents

Section 6.2.3 states two conclusions from the energy partition—that interactions come to dominate, and that the trap–interaction balance approaches its classical form—and defers the fitted exponents that support them to here.

Log–log fits $E(\omega) \propto \omega^{\alpha}$ over the grid $\omega \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$ give a kinetic exponent $\alpha_T \simeq 1.00$ at every N, an interaction exponent $\alpha_{V_{\text{int}}} \simeq 0.66$, and trap exponents $\alpha_{V_{\text{trap}}} \simeq 0.81, 0.73, 0.72$ for $N = 2, 6, 12$. The kinetic energy therefore scales essentially linearly with ω while both potential terms grow sublinearly with nearly N-independent exponents; the ratio $\Gamma \propto \omega^{\alpha_{V_{\text{int}}} - \alpha_T} \approx \omega^{-0.34}$ predicts a factor $10^{3\times 0.34} \approx 10.5$ across three decades, against the measured 9.7, 10.6 and 11.1.

From $V_{\text{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2 \rangle$ the rms radius is $r_{\text{rms}} = \sqrt{2V_{\text{trap}}/(N\omega^2)}$, and fitting $r_{\text{rms}} \propto \omega^{\alpha_r}$ gives $\alpha_r \simeq -0.60$ to -0.64, equivalently $\langle r^2 \rangle \propto \omega^{-\beta}$ with $\beta \simeq 1.19$–1.28. The cloud therefore expands faster than the non-interacting ω^{-1} law for $\langle r^2 \rangle$.

These are averaged quantities and should not be confused with the shell-resolved radius of §F.4, which is the one compared against the classical Wigner prefactor; the two differ because r_{rms} and r_{mode} mix the central electron of a $(1, n)$ topology with the ring, and the shell-resolved radius does not.

Table F.1: Radial diagnostics for N=2 across ω (Bohr units).

ω	r_{mode}	$\langle r \rangle$	σ_r	FWHM	q_{10}	q_{50}	q_{90}	γ
1.00	1.435	1.632	0.705	1.743	0.735	1.581	2.581	0.491
0.50	2.303	2.475	1.007	2.466	1.209	2.409	3.829	0.437
0.10	3.601	3.830	1.728	4.449	1.640	3.714	6.127	0.480
0.01	14.191	14.593	5.615	13.602	7.390	14.322	22.038	0.396
0.001	64.292	63.953	18.479	43.977	40.392	63.814	87.714	0.287

Table F.2: Radial and angular diagnostics for N=6 across ω (Bohr; angles in radians).

ω	r_{mode}	σ_r	γ	σ(ring angles)	$\gamma_{r_{ij}}$
1.00	2.0907	0.9249	0.4424	0.7046	0.3538
0.50	3.0702	1.3648	0.4445	0.6943	0.3397
0.10	8.1686	3.5025	0.4288	0.6842	0.3149
0.01	33.1538	13.9094	0.4195	0.6497	0.2889
0.001	113.5542	47.8076	0.4210	0.6368	0.2664

F.2 Shell-resolved structure across the whole grid

Chapter 6 quotes shell occupancies, radii, bond order and Lindemann numbers for individual (N, ω) cells in running prose. Table F.3 collects them for the whole grid from a single analysis pass, so that the trends can be read off rather than assembled from the text. Shell radii are given in trap units $\tilde{r} = \sqrt{\omega}\, r$; multiply by $\ell = \omega^{-1/2}$ for Bohr.

Reading $|\Phi_n|$ against its null. One column of that table needs a baseline to be interpretable, and its absence would make the bond-order numbers read far more favourably than they should. For n particles placed at *independent uniform* angles on a ring, $\Phi_n = n^{-1} \sum_k e^{in\phi_k}$ is a two-dimensional random walk of n unit steps, so

$$\big\langle |\Phi_n| \big\rangle_{\text{random}} = \frac{\sqrt{\pi}}{2\sqrt{n}} \approx \frac{0.886}{\sqrt{n}}, \qquad \big\langle |\Phi_n|^2 \big\rangle_{\text{random}} = \frac{1}{n}. \tag{F.1}$$

A finite $|\Phi_n|$ is therefore *not* by itself evidence of angular order: a five-particle ring with no order at all returns $|\Phi_5| \approx 0.40$. The ratio $|\Phi_n|/\langle|\Phi_n|\rangle_{\text{random}}$ is the quantity that carries information, and Table F.3 reports it.

Read that way the result is sharper than the raw values suggest, and it needs stating with the right threshold. From $\omega = 1$ down to $\omega = 0.1$ the ratio is 1.00–1.02 at every particle number and on every ring. At $\omega = 0.01$ it lifts slightly, to 1.10 at N=6, while N=12 and N=20 remain at 1.03 and 1.01. Only at $\omega = 10^{-3}$ does it become large: 1.65 on the N=6 ring, 1.19 and 1.56 on the two rings of the N=12 $(3, 9)$ molecule, and 1.19 and 1.47 on the outer two rings of the N=20 $(1, 7, 12)$ molecule.

Whether the small excesses are *resolved* is a separate question from whether they are *large*, and with these sample sizes the two come apart. Two caveats attach to the null before it is used quantitatively. The Rayleigh form is the large-n limit of the resultant of n unit steps; for the small rings here ($n = 3$ to 19) it is an approximation, accurate to well within the effect at 10^{-3} but of the same order as the few-percent excesses at $\omega \geq 0.1$. And the frames come from a Markov chain, so a naive standard error over K raw frames understates the uncertainty, whereas the block bootstrap of §5.2.9 is used for the reported errors elsewhere. Taking those at face value, the excess at N=6, $\omega = 0.1$ is of order tens of block standard errors and still only 2%.

The conclusion that survives both caveats is the one that matters: statistical detectability is not what separates the regimes here, magnitude is. On that criterion the ordering is flat through $\omega = 0.1$, weakly present at $\omega = 0.01$ and only at N=6, and unambiguous at 10^{-3}. A cleaner version of this comparison would generate the null numerically at each actual occupancy n, through the same block-bootstrap pipeline, rather than relying on the asymptotic form; that is a small addition we did not make.

The Lindemann numbers of the same table fall monotonically throughout, so radial stiffening precedes n-fold angular ordering rather than accompanying it—the separation between radial and angular melting anticipated in §1.4. Note that the values here are for the *modal* topology over all frames; the sector-conditioned values quoted in §6.2.5 are computed on the frames assigned to a given sector and run slightly higher.

Table F.3: Shell-resolved diagnostics at the modal topology, all frames, for every (N, ω). Occupancies are $(n_{\text{in}}, \dots, n_{\text{out}})$; $\tilde{r}_s$ are mean shell radii in trap units; $w_s = \sigma(r_s)/\langle r_s \rangle$ are per-shell radial relative widths; $|\Phi_{n_s}|$ is the rotation-registered bond order on shell s; and the final column is $|\Phi_{n_s}|$ divided by its random-angle expectation, Eq. (F.1). A value of 1.00 in that column means the ring is equal to the Rayleigh null approximation within quoted precision; see the text for why resolvability and magnitude come apart at these sample sizes. Single-particle inner shells are omitted from the angular columns, where $|\Phi_1| = 1$ trivially.

N	ω	occ.	$\tilde{r}_s$	w_s	$\|\Phi_{n_s}\|$	ratio to null
6	1.0	$(1,5)$	0.70, 1.66	0.43, 0.32	0.397	1.00
6	0.5	$(1,5)$	0.76, 1.78	0.42, 0.31	0.397	1.00
6	0.1	$(1,5)$	0.93, 2.15	0.41, 0.28	0.405	1.02
6	0.01	$(1,5)$	1.33, 2.96	0.42, 0.22	0.434	1.10
6	0.001	$(1,5)$	1.12, 4.29	0.63, 0.15	**0.654**	**1.65**
12	1.0	$(1,11)$	0.55, 2.00	0.47, 0.34	0.268	1.00
12	0.5	$(1,11)$	0.60, 2.18	0.47, 0.33	0.268	1.00
12	0.1	$(1,11)$	0.74, 2.68	0.45, 0.31	0.270	1.01
12	0.01	$(1,11)$	1.04, 3.77	0.41, 0.28	0.275	1.03
12	0.001	$(3,9)$	2.52, 6.04	0.61, 0.24	**0.611**, **0.462**	**1.19**, **1.56**
20	1.0	$(1,19)$	0.55, 2.32	0.44, 0.36	0.204	1.00
20	0.5	$(1,19)$	0.60, 2.52	0.44, 0.35	0.203	1.00
20	0.1	$(1,19)$	0.75, 3.14	0.42, 0.34	0.204	1.00
20	0.01	$(1,19)$	1.05, 4.47	0.49, 0.32	0.205	1.01
20	0.001	$(1,7,12)$	1.27, 4.19, 7.83	0.47, 0.24, 0.11	**0.398**, **0.375**	**1.19**, **1.47**

F.3 Shell-model fit to the pair distribution

The reconstruction quoted in §6.2.5 tests whether the measured pair distribution is explained by the detected shell geometry. Two versions of that test exist and they are not equally informative.

The weaker version decomposes the pairs of two-shell frames into inner–inner, inner–outer and outer–outer histograms and recombines them with the combinatorial weights of the ideal topology. Because the sector histograms and the total are accumulated from the same frames, this recombination is close to exact by construction—cosine similarities exceed 0.99999 at every (N, ω)—and it establishes only that the sector bookkeeping is self-consistent.

The stronger version is a *generative* one and is the test reported here. Shell radii and widths are fitted on one half of the frames; each shell is then modelled as a regular n_s-gon carrying a per-frame global rotation and a fitted angular jitter; and the resulting pair-distance density is compared with the empirical density on the *held-out* half. Against it we run an angle-randomised null with the same fitted radial distributions but uniform angles, which isolates what polygonal order contributes over and above shell radii alone. Table F.4 reports both.

The polygon model reproduces the held-out pair distribution to cosine similarity 0.987–0.999 across the whole grid. The null reaches 0.936–0.996, and the gap between them, $\Delta\cos$, grows monotonically as the trap weakens at every particle number—from 0.012 to 0.058 at N=6, from 0.006 to 0.026 at N=12, and from 0.003 to 0.010 at N=20. Shell radii alone account for most of the pair structure at strong confinement; angular order accounts for a growing remainder as the Wigner molecule forms. This is the same transition the bond-order ratios of Table F.3 record, measured on a different quantity and on held-out data.

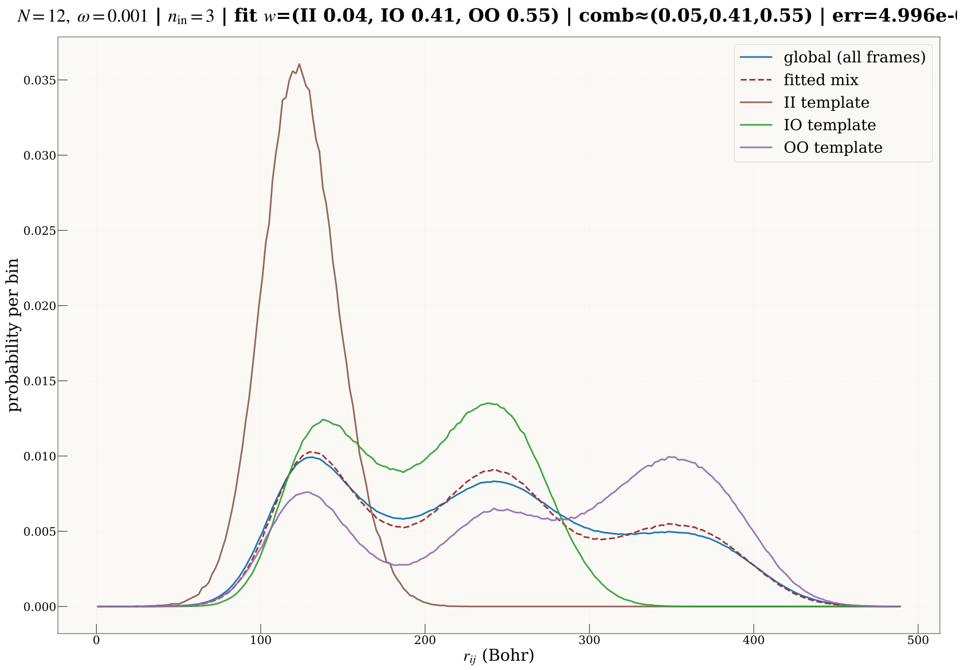


Figure F.1: Shell-resolved reconstruction of $g(r)$ at $\omega = 10^{-3}$ for N=12. The global pair distribution is reproduced by a convex mixture of inner–inner (II), inner–outer (IO) and outer–outer (OO) histograms with weights set by the observed $(n_{\text{in}}, n_{\text{out}})$ statistics.

Table F.4: Shell-model fit to the pair-distance distribution. The empirical $p_{\text{test}}(r_{ij})$ on held-out frames is compared with (i) a *polygon* model, in which each detected shell is a regular n_s-gon with a per-frame global rotation and fitted angular jitter, and (ii) a *uniform-angle* null with the same fitted radial distributions and no polygonal order. $\Delta\cos = \cos_{\text{poly}} - \cos_{\text{uni}}$ is what angular order buys; L^2_{poly} is the polygon model's residual.

N	ω	occ.	$\cos_{\text{poly}}$	$\cos_{\text{uni}}$	$\Delta\cos$	L^2_{poly}
6	1.0	$(1,5)$	0.9988	0.9866	0.0122	4.69×10^{-3}
6	0.5	$(1,5)$	0.9986	0.9836	0.0150	5.09×10^{-3}
6	0.1	$(1,5)$	0.9974	0.9741	0.0234	6.83×10^{-3}
6	0.01	$(1,5)$	0.9922	0.9536	0.0386	1.14×10^{-2}
6	0.001	$(1,5)$	0.9942	0.9358	**0.0585**	1.04×10^{-2}
12	1.0	$(1,11)$	0.9993	0.9937	0.0056	3.36×10^{-3}
12	0.5	$(1,11)$	0.9990	0.9927	0.0063	4.01×10^{-3}
12	0.1	$(1,11)$	0.9980	0.9896	0.0085	5.43×10^{-3}
12	0.01	$(1,11)$	0.9929	0.9805	0.0124	9.92×10^{-3}
12	0.001	$(3,9)$	0.9868	0.9604	**0.0264**	1.34×10^{-2}
20	1.0	$(1,19)$	0.9992	0.9964	0.0028	3.43×10^{-3}
20	0.5	$(1,19)$	0.9990	0.9958	0.0032	3.78×10^{-3}
20	0.1	$(1,19)$	0.9982	0.9942	0.0040	4.95×10^{-3}
20	0.01	$(1,19)$	0.9947	0.9900	0.0047	8.21×10^{-3}
20	0.001	$(1,7,12)$	0.9915	0.9820	**0.0095**	1.02×10^{-2}

F.4 The measured shell radius against the classical prediction

The classical minimum-energy configuration of six charges in a parabolic trap places five of them on a ring of radius $1.334\,\omega^{-2/3}$ [8, 9]. Nothing in the ansatz, the training objective or the sampler knows that number, so the fitted outer-shell radius of Table F.3 is an external check on the whole structural picture—and a sharper one than the exponent fits of §6.2.4, because it tests a prefactor as well as a power.

Table F.5: Fitted outer-shell radius of the N=6 $(1, 5)$ molecule against the classical Wigner ring $r_{\mathrm{cl}} = 1.334\,\omega^{-2/3}$. Radii in Bohr.

ω	r_{shell}	r_{cl}	ratio
1.0	1.66	1.33	1.247
0.5	2.52	2.12	1.188
0.1	6.80	6.19	1.098
0.01	29.65	28.74	1.032
0.001	135.67	133.40	1.017

The ratio falls monotonically toward unity, reaching 1.017 at $\omega = 10^{-3}$: the quantum ring sits 1.7% outside the classical one at the weakest trap, and the excess grows smoothly with confinement as zero-point motion pushes the shell outward. Fitting $r_{\mathrm{shell}} \propto \omega^{\alpha}$ gives $\alpha = -0.637$ over the full grid and -0.650 over the weakest three confinements, against the classical $-2/3$.

Two remarks on why this quantity, rather than the r_{mode} of Table 6.3, is the right one to compare. First, r_{mode} is the mode of the *single-particle* radial density, which for a $(1, 5)$ topology mixes the central electron with the ring and is therefore not the ring radius; fitting it gives $\alpha \approx -0.58$ at N=6 and -0.54 at N=2 over the full grid, and approaches $-2/3$ only on the weak-trap half. Second, the shell-resolved radius is the quantity the physics-informed proposal of §7.4 actually needs, so the agreement recorded here is what licenses building a sampler out of it.

Chapter G

Declaration on the Use of Artificial Intelligence

The collocation approach in this thesis was developed over several months of failed experiments. Its central difficulty was not merely that a network did not converge, but that the parts of the Schrödinger residual that made the method physically meaningful also made its gradients unstable near the wavefunction's nodes. During that period, AI systems were used as sounding boards: to test interpretations of unexpected results, to formulate possible explanations, and to decide which experiment should come next. The exchange below is a short sample from one longer conversation at a point where the negative evidence had begun to make abandoning collocation seem plausible.

Author

I am trying to develop a PINN to find ground-state solutions of the Schrödinger equation. I have tried multiple approaches, but I am not succeeding. ...What do you think about this? What should we do from here?

AI response 1

...Anything that injects near-node points makes gradients explode or become uselessly noisy. ...

⋮

AI response 2

...This asymmetry is fundamental and cannot be resolved by sampling tricks, loss modifications, or gradient preconditioning within the collocation framework. ...

⋮

AI response 3

...Stick with SR + energy (or energy+variance) VMC. ...

Artificial-intelligence systems were therefore used for discussion, planning, and review, rather than as a source of results. They helped turn observations into hypotheses that could be checked, suggested alternative diagnostics when a run failed, and reviewed the final prose for clarity and mistakes. The code, the experiments, the scientific judgments, and the text of this thesis were written by the author. In particular, the recommendation shown above was not accepted as a methodological conclusion: it became one claim among several to investigate. The work that followed led to the collocation analysis reported in this thesis.

Bibliography

[1] Giuseppe Carleo and Matthias Troyer. Solving the quantum many-body problem with artificial neural networks. *Science*, 355(6325):602–606, 2017. doi: 10.1126/science.aag2302.

[2] David Pfau, James S. Spencer, Alexander G. D. G. Matthews, and W. Matthew C. Foulkes. Ab initio solution of the many-electron schrödinger equation with deep neural networks. *Physical Review Research*, 2:033429, 2020.

[3] Jan Hermann, Zeno Schätzle, and Frank Noé. Deep-neural-network solution of the electronic schrödinger equation. *Nature Chemistry*, 12:891–897, 2020. doi: 10.1038/s41557-020-0544-y.

[4] Di Luo and Bryan K. Clark. Backflow transformations via neural networks for quantum many-body wave functions. *Physical Review Letters*, 122:226401, 2019. doi: 10.1103/PhysRevLett.122.226401.

[5] Gabriel Pescia, Jannes Nys, Jane Kim, Alessandro Lovato, and Giuseppe Carleo. Message-passing neural quantum states for the homogeneous electron gas. *Physical Review B*, 110(3):035108, 2024. doi: 10.1103/PhysRevB.110.035108.

[6] Jane Kim, Gabriel Pescia, Bryce Fore, Jannes Nys, Giuseppe Carleo, Stefano Gandolfi, Morten Hjorth-Jensen, and Alessandro Lovato. Neural-network quantum states for ultracold Fermi gases. *Communications Physics*, 7:148, 2024. doi: 10.1038/s42005-024-01613-w.

[7] R. Egger, W. Häusler, C. H. Mak, and H. Grabert. Crossover from fermi liquid to wigner molecule behavior in quantum dots. *Phys. Rev. Lett.*, 82:3320–3323, Apr 1999. doi: 10.1103/PhysRevLett.82.3320. URL `https://link.aps.org/doi/10.1103/PhysRevLett.82.3320`.

[8] Minghui Kong, B. Partoens, and F. M. Peeters. Transition between ground state and metastable states in classical two-dimensional atoms. *Phys. Rev. E*, 65:046602, Mar 2002. doi: 10.1103/PhysRevE.65.046602. URL `https://link.aps.org/doi/10.1103/PhysRevE.65.046602`.

[9] Vitaly A. Schweigert and François M. Peeters. Spectral properties of classical two-dimensional clusters. *Phys. Rev. B*, 51:7700–7713, Mar 1995. doi: 10.1103/PhysRevB.51.7700. URL `https://link.aps.org/doi/10.1103/PhysRevB.51.7700`.

[10] M. Manninen and S.M. Reimann. Chapter 12 - metal clusters, quantum dots, and trapped atoms: From single-particle models to correlation. In Purusottam Jena and A. Welford Castleman, editors, *Nanoclusters*, volume 1 of *Science and Technology of Atomic, Molecular, Condensed Matter and Biological Systems*, pages 437–484. Elsevier, 2010. doi: https://doi.org/10.1016/B978-0-444-53440-8.00012-4. URL `https://www.sciencedirect.com/science/article/pii/B9780444534408000124`.

[11] M. Mazars. Bond orientational order parameters in the crystalline phases of the classical yukawa-wigner bilayers. *Europhysics Letters*, 84(5):55002, dec 2008. doi: 10.1209/0295-5075/84/55002. URL `https://doi.org/10.1209/0295-5075/84/55002`.

[12] A. V. Filinov, M. Bonitz, and Yu. E. Lozovik. Wigner crystallization in mesoscopic 2d electron systems. *Phys. Rev. Lett.*, 86:3851–3854, Apr 2001. doi: 10.1103/PhysRevLett.86.3851. URL `https://link.aps.org/doi/10.1103/PhysRevLett.86.3851`.

[13] F. Pederiva, C. J. Umrigar, and E. Lipparini. Diffusion monte carlo study of circular quantum dots. *Phys. Rev. B*, 62:8120–8125, 2000. doi: 10.1103/PhysRevB.62.8120.

[14] Tosio Kato. On the eigenfunctions of many-particle systems in quantum mechanics. *Communications on Pure and Applied Mathematics*, 10:151, 1957.

[15] Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21:1087, 1953.

[16] Malvin H. Kalos, Dominique Levesque, and Loup Verlet. Helium at zero temperature with hard-sphere and other forces. *Physical Review A*, 9:2178, 1974.

[17] Gaopei Pan and Zi Yang Meng. The sign problem in quantum Monte Carlo simulations. In *Encyclopedia of Condensed Matter Physics*, pages 879–893. Elsevier, Oxford, 2 edition, 2024. doi: 10.1016/B978-0-323-90800-9.00095-0.

[18] Robert Jastrow. Many-body problem with strong forces. *Physical Review*, 98(5):1479–1484, 1955. doi: 10.1103/PhysRev.98.1479.

[19] Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, volume 2. Springer, 2009.

[20] Julius Berner, Philipp Grohs, Gitta Kutyniok, and Philipp Petersen. *The Modern Mathematics of Deep Learning*, page 1–111. Cambridge University Press, 2022.

[21] Lei Wu, Mingze Wang, and Weijie Su. The alignment property of sgd noise and how it helps select flat minima: A stability analysis, 2022. URL `https://arxiv.org/abs/2207.02628`.

[22] Diederik Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, San Diego, CA, USA, 2015.

[23] Yushun Zhang, Congliang Chen, Naichen Shi, Ruoyu Sun, and Zhi-Quan Luo. Adam can converge without any modification on update rules. In *Advances in Neural Information Processing Systems*, page 35:28386, 2022.

[24] Shun ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10:251, 1998.

[25] S. Amari and S.C. Douglas. Why natural gradient? In *Proceedings of the 1998 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP '98 (Cat. No.98CH36181)*, volume 2, pages 1213–1216 vol.2, 1998. doi: 10.1109/ICASSP.1998.675489.

[26] Federico Becca and Sandro Sorella. *Quantum Monte Carlo approaches for correlated systems.* Cambridge University Press, 2017.

[27] James Stokes, Josh Izaac, Nathan Killoran, and Giuseppe Carleo. Quantum natural gradient. *Quantum*, 4:269, 2020.

[28] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. URL `https://arxiv.org/abs/1806.07572`.

[29] Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 9, pages 249–256, 2010.

[30] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1026–1034, 2015. doi: 10.1109/ICCV.2015.123.

[31] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Russ R. Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[32] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *PMLR*, pages 1263–1272, 2017. URL `https://arxiv.org/abs/1704.01212`.

[33] Peter W. Battaglia, Jessica B. Hamrick, Victor Bapst, et al. Relational inductive biases, deep learning, and graph networks. *arXiv preprint arXiv:1806.01261*, 2018. URL `https://arxiv.org/abs/1806.01261`.

[34] George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2:303–314, 1989. doi: 10.1007/BF02551274. URL `https://doi.org/10.1007/BF02551274`.

[35] Kurt Hornik. Approximation capabilities of multilayer feedforward networks. *Neural Networks*, 4(2):251–257, 1991. ISSN 0893-6080. doi: https://doi.org/10.1016/0893-6080(91)90009-T. URL `https://www.sciencedirect.com/science/article/pii/089360809190009T`.

[36] Moshe Leshno, Vladimir Ya. Lin, Allan Pinkus, and Shimon Schocken. Multilayer feedforward networks with a nonpolynomial activation function can approximate any function. *Neural Networks*, 6(6):861, 1993.

[37] Andrew R. Barron. Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information Theory*, 39(3):930–945, 1993. doi: 10.1109/18.256500.

[38] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.

[39] I. E. Lagaris, A. Likas, and D. I. Fotiadis. Artificial neural networks for solving ordinary and partial differential equations. *IEEE Transactions on Neural Networks*, 9(5):987–1000, 1998. doi: 10.1109/72.712178.

[40] Tim De Ryck and Siddhartha Mishra. Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning. *Acta Numerica*, 33:633–713, 2024. doi: 10.1017/S0962492923000089.

[41] R. P. Feynman and Michael Cohen. Energy spectrum of the excitations in liquid helium. *Physical Review*, 102(5):1189–1204, 1956. doi: 10.1103/PhysRev.102.1189.

[42] M. Holzmann, D. M. Ceperley, C. Pierleoni, and K. Esler. Backflow correlations for the electron gas and metallic hydrogen. *Physical Review E*, 68(4):046707, 2003.

[43] Sandro Sorella. Green function monte carlo with stochastic reconfiguration. *Physical Review Letters*, 80(20):4558–4561, 1998. doi: 10.1103/PhysRevLett.80.4558.

[44] Ao Chen and Markus Heyl. Empowering deep neural quantum states through efficient optimization. *Nature Physics*, 20:1476–1481, 2024. doi: 10.1038/s41567-024-02566-1.

[45] Riccardo Rende, Luciano Loris Viteritti, Lorenzo Bardone, Federico Becca, and Sebastian Goldt. A simple linear algebra identity to optimize large-scale neural network quantum states. *Communications Physics*, 7:260, 2024. doi: 10.1038/s42005-024-01732-4.

[46] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256, 1992. doi: 10.1007/BF00992696.

[47] Aki Vehtari, Daniel Simpson, Andrew Gelman, Yuling Yao, and Jonah Gabry. Pareto smoothed importance sampling. *Journal of Machine Learning Research*, 25, 2024. URL `https://arxiv.org/abs/1507.02646`.

[48] Nicholas Gao and Stephan Günnemann. Neural Pfaffians: Solving many many-electron Schrödinger equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[49] H. Flyvbjerg and H. G. Petersen. Error estimates on averages of correlated data. *The Journal of Chemical Physics*, 91(1):461–466, 1989.

[50] Marius Jonsson. Standard error estimation by an automated blocking method. *Physical Review E*, 98:043304, 2018.

[51] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 3519–3529, 2019.

[52] Jørgen Høgberget. Quantum Monte-Carlo studies of generalized many-body systems. Master's thesis, University of Oslo, 2013. Source of the diffusion Monte Carlo benchmark energies for $N \in \{6, 12, 20\}$ used here; tabulated in Ref. [53].

[53] Even M. Nordhagen, Jane M. Kim, Bryce Fore, Alessandro Lovato, and Morten Hjorth-Jensen. Efficient solutions of fermionic systems using artificial neural networks. *Frontiers in Physics*, 11: 1061580, 2023. doi: 10.3389/fphy.2023.1061580.

[54] M. Taut. Two electrons in a homogeneous magnetic field: particular analytical solutions. *J. Phys. A: Math. Gen.*, 27:1045–1055, 1994. doi: 10.1088/0305-4470/27/3/040.

[55] F. Pederiva, C. J. Umrigar, and E. Lipparini. Erratum: Diffusion Monte Carlo study of circular quantum dots [Phys. Rev. B **62**, 8120 (2000)]. *Phys. Rev. B*, 68:089901, 2003. doi: 10.1103/PhysRevB.68.089901.

[56] Daniel Haas. Deep learning methods for quantum many-body systems: A study on neural quantum states. Master's thesis, University of Oslo, September 2024. Compilation from which the reference energies used here were taken; their origin is attributed in Table 6.2 to Høgberget's diffusion Monte Carlo calculations and, for $N = 2$, to Taut's analytic solutions.

[57] Chae-Yeun Park and Michael J. Kastoryano. Geometry of learning neural quantum states. *Physical Review Research*, 2:023232, 2020.

[58] Sami Siljamäki. *Wave Function Methods for Quantum Dots in Magnetic Field.* PhD thesis, Helsinki University of Technology, 2003.

[59] Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with Gumbel-softmax. In *International Conference on Learning Representations (ICLR)*, 2017.

[60] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 2408–2417, 2015.